

Report Concerning Space Data System Standards

SIMULTANEOUS TRANSMISSION OF GMSK TELEMETRY AND PN RANGING

INFORMATIONAL REPORT

CCSDS 413.1-G-2

GREEN BOOK

November 2021

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1 INTRODUCTION

1.1 PURPOSE AND SCOPE

A system capable of simultaneously transmitting high-rate telemetry and ranging has been recommended for telemetry symbol rates higher than 2 Msymbol/s in the 8400–8500 MHz Space Research Service (SRS) bands. In this system the telemetry is transmitted through a CCSDS standard GMSK modulator with the CCSDS standard regenerative Pseudo-random Noise (PN) ranging sequence included as an additional phase shift. The receiver first estimates the transmitted telemetry symbols, regenerates and removes the estimated GMSK signal from the received signal, and then estimates the ranging chips and, through a bank of correlators, the round-trip delay of the received ranging signal. Ranging is an interfering signal which degrades the performance of the telemetry subsystem, while errors in the estimation of telemetry symbols compromise the correct detection of the ranging chips. A careful parameter selection is therefore necessary for the system to work with minimum losses.

Transmission of simultaneous telecommand and regenerative PN ranging signals on the Earth-to-space link follows the CCSDS standard transmission modes and is not covered in this Informational Report.

This Informational Report provides the background information for CCSDS recommendations 401(2.4.22A), and 401(2.4.22B) addressing the use of spacecraft telemetry bandwidth-efficient modulations with simultaneous ranging.

This Report provides a review of the technical specification for the modulation techniques approved in the above-mentioned recommendations, together with a description of their main performance characteristics for the applications covered by the recommendations. All figures are based on simulations unless noted otherwise. However, a crosscheck of the main configurations' performance was also performed by hardware measurements (reference [12]).

The simulations and the measurements of the complete space-to-Earth transmission system encompass realistic synchronization of the recovered carrier as well as of the telemetry and ranging signals, GMSK telemetry regeneration through Laurent OQPSK approximation or through a look-up table, the effects of perfect or lack of synchronization between the transmitted telemetry and ranging signals, etc.

1.2 APPLICABILITY

The modulation techniques described in this document are applicable to high symbol rate (> 2 Msymbol/s for 8 GHz space research service) telemetry transmissions with simultaneous ranging. Two classes of modulation techniques are identified:

- those dedicated to space research, Category A missions, specified in recommendation 401(2.4.22A) B-1, applicable to frequency band 8450–8500 MHz;
- those dedicated to space research, Category B missions, specified in recommendation 401(2.4.22B) B-1, applicable to frequency band 8400–8450 MHz.

It should be noted that, *sensu stricto*, the above recommendations are applicable only to the mentioned frequency bands. However, the user should take note that extension to other SRS frequency bands could be envisaged in the future.

In no event will CCSDS or its members be liable for any incidental, consequential, or indirect damages, including any lost profits, lost savings, or loss of data, or for any claim by another party related to errors or omissions in this report.

1.3 REFERENCES

The following publications are referenced in this document. At the time of publication, the editions indicated were valid. All publications are subject to revision, and users of this document are encouraged to investigate the possibility of applying the most recent editions of the publications indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS publications.

- [1] *Radio Frequency and Modulation Systems—Part 1: Earth Stations and Spacecraft*. Issue 26. Recommendation for Space Data System Standards (Blue Book), CCSDS 401.0-B-26. Washington, D.C.: CCSDS, October 2016.
- [2] *Pseudo-Noise (PN) Ranging Systems*. Issue 2. Recommendation for Space Data System Standards (Blue Book), CCSDS 414.1-B-2. Washington, D.C.: CCSDS, February 2014.
- [3] E. Vassallo and M. Visintin. “Analysis of GMSK for Simultaneous Transmission of Ranging and Telemetry.” SLS-RFM_09-08. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, October 2009, Noordwijk, The Netherlands.
- [4] E. Vassallo and M. Visintin. “Analysis of UQPSK and GMSK/PN for Simultaneous Transmission of Ranging and Telemetry: Ranging Correlator Results.” SLS-RFM_10-03. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, May 2010, Portsmouth, Virginia.
- [5] E. Vassallo and M. Visintin. “GMSK and PN Ranging: Combining Option with Improved Spectral Performance.” SLS-RFM_10-09. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, May 2010, Portsmouth, Virginia.

- [6] D. Lee. "AI_09-16: Simulations of GMSK-PN and UQPSK." SLS-RFM_11-04. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, March 2011, Berlin, Germany.
- [7] E. Vassallo and M. Visintin. "Synchronization Analysis for GMSK/PN Modulation." SLS-RFM_11-01. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, March 2011, Berlin, Germany.
- [8] G. Sessler, M. Visintin, and E. Vassallo. "Analysis of GMSK/PN Modulation: Effects of Phase Jumps and Noise." SLS-RFM_11-16. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, November 2011, Boulder, Colorado.
- [9] *TM Synchronization and Channel Coding*. Issue 2. Recommendation for Space Data System Standards (Blue Book), CCSDS 131.0-B-2. Washington, D.C.: CCSDS, August 2011.
- [10] G. Sessler, E. Vassallo, and M. Visintin. "Lagrange/Mars Missions Link Budgets for GMSK and PN Ranging." SLS-RFM_11-17. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, November 2011, Boulder, Colorado.
- [11] "Performance of GMSK for Telemetry and PN Ranging under Realistic Conditions." In *Proceedings of TTC 2013: 6th International Workshop on Tracking, Telemetry, and Command Systems (10–13 September 2013, Darmstadt, Germany)*. Noordwijk, The Netherlands: ESA Conference Bureau, September 2013.
- [12] F. Winterstein. "Simultaneous Transmission of GMSK Telemetry and PN Ranging: Measurement Report in Support of the Draft CCSDS Recommendations 401(2.4.22A) and 401(2.4.22B)." SLS-RFM_14-11 rev.3. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, November 2014, London, England.
- [13] G. Sessler, E. Vassallo, and M. Visintin. "GMSK/PN RG: Interference of Telemetry on Ranging Chip Synchronization." SLS-RFM_12-13. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, October 2012, Cleveland, Ohio.
- [14] M. Visintin. *DTTL-Like Symbol Synchronizers for GMSK Signals*. Final Report of ESA/ESOC Contract 17136/03/D/CS(SC). N.p.: n.p., 2004.
- [15] M. Pent, et al. *End-to-End Study of GMSK Modulation*. Final Report of ESA/ESOC Contract 14295/00/D/CS. N.p.: n.p., 2002.
- [16] G. Sessler and E. Vassallo. "GMSK and PN Ranging: Improved Telemetry Signal Demodulation Using PN Ranging Signal Cancellation." SLS-RFM_17-02. Presented at CCSDS Radio Frequency and Modulation Working Group meeting, May 2017, San Antonio, Texas.

2 SCOPE OF GMSK+PN RANGING MODULATION

2.1 LIMITED SPECTRAL RESOURCES FOR SPACE TELEMETRY

In order to avoid a rapid saturation of the 2200–2290 MHz and 8450–8500 MHz bands with unsolvable interference conflicts, the CCSDS has issued a recommendation 401(2.4.17A) (reference [1]) for a limited set of common bandwidth-efficient modulation schemes to be used for high symbol rate transmissions, thus ensuring not only an optimum use of the band but also inter-agency cross-support capability.

Likewise, recommendation 401(2.4.17B) (reference [1]) addresses the Category B SRS bands 2290–2300 MHz and 8400–8450 MHz. These recommended modulations have been selected for their low loss and their bandwidth compactness.

The scope of GMSK+PN ranging modulation specified in the companion recommendations 401(2.4.22A) and 401(2.4.22B) (reference [1]) is to allow simultaneous space-to-Earth transmission of high-rate GMSK telemetry in accordance with 401(2.4.17A) or 401(2.4.17B) (reference [1]) and on-board regenerated PN ranging (reference [2]), limited to 8450–8500 MHz and 8400–8450 MHz, respectively, whenever the telemetry symbol rate exceeds 2 Msymbol/s.

The Earth-to-space transmission of simultaneous telecommand and regenerative PN ranging signals (see section 3 of reference [2]) follows the conventional transmission modes of references [1] and [2] and is not covered in this Informational Report.

2.2 REGULATIONS: THE SFCG SPECTRAL MASK

The SFCG maintains a spectral mask under recommendation 21-2 applicable to Category A bands 2200–2290 MHz, 8025–8400 MHz, and 8450–8500 MHz. SFCG Recommendation 23-1 provides guidance on the maximum allowable bandwidth as a function of data rates for space-to-Earth links in the Category B bands 2290–2300 MHz and 8400–8450 MHz. The SFCG Recommendations currently in-force can be found at the SFCG Web site <http://www.sfcgonline.org/>.

While the above recommendations do not address the ranging signal, they are used to provide an indication of how good the selected simultaneous telemetry and ranging transmission can be from a spectral management point of view.

2.3 A SELECTION OF BANDWIDTH-EFFICIENT TELEMETRY MODULATION METHODS

2.3.1 GENERAL

The selection of modulations schemes is the result of compromises on a number of criteria:

- bandwidth efficiency;
- link performances (in terms of BER or SER);
- implementation complexity and cost: onboard transmitter, ground receiver;
- robustness: susceptibility to interferers;
- programmatic aspects: cross-compatibility.

The telemetry GMSK modulation scheme to be used with simultaneous PN ranging is the same as for the case of telemetry-only transmission. This gives the advantage that no telemetry-modulation-scheme change is necessary when switching ranging on and off. The relevant telemetry recommendations are indicated in 2.3.2 and 2.3.3.

2.3.2 GMSK MODULATION FOR TELEMETRY TRANSMISSION OF SRS, CATEGORY A

For SRS Category A missions (2×10^6 km), only one GMSK modulation is specified in recommendation 401(2.4.17A) B-1 (reference [1]):

- GMSK ($BT_s=0.25$) with precoding.

2.3.3 GMSK MODULATION FOR TELEMETRY TRANSMISSION OF SRS, CATEGORY B

For SRS Category B missions, only one GMSK modulation is specified in recommendation 401(2.4.17B) B-1 (reference [1]):

- GMSK ($BT_s=0.5$) with precoding.

2.4 BIT AND SYMBOL RATE TERMINOLOGY

In the literature, the notations used for bit rate and symbol rate sometimes have different meanings. For this Informational Report, R_b refers to the information bit rate, R_s refers the coded symbol rate measured at the input of the modulator, and R_{CHS} refers to the channel symbol rate after the modulator. In this Informational Report (telemetry) symbol rate, R_{TM} and R_s are used interchangeably. The telemetry symbol interval is therefore given by $T_s = 1/R_s = 1/R_{TM}$.

Figure 2-1 shows the relationship between the different terms.

For the PN ranging signal, the chip rate is defined as R_{RG} so that the chip interval is given by $T_c = 1/R_{RG}$.

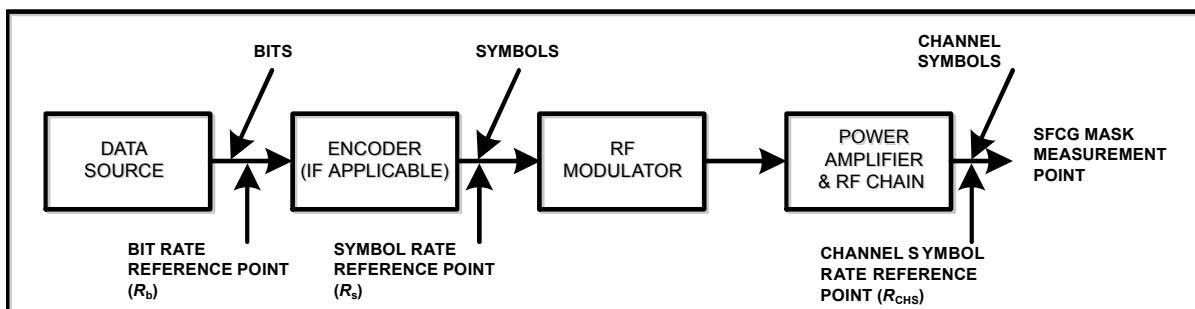


Figure 2-1: Bit and Symbol Rate Terminology

3 TECHNICAL DEFINITIONS

3.1 SIGNAL MODEL

A schematic of the transmitter under consideration is shown in figures 3-1 and 3-2.

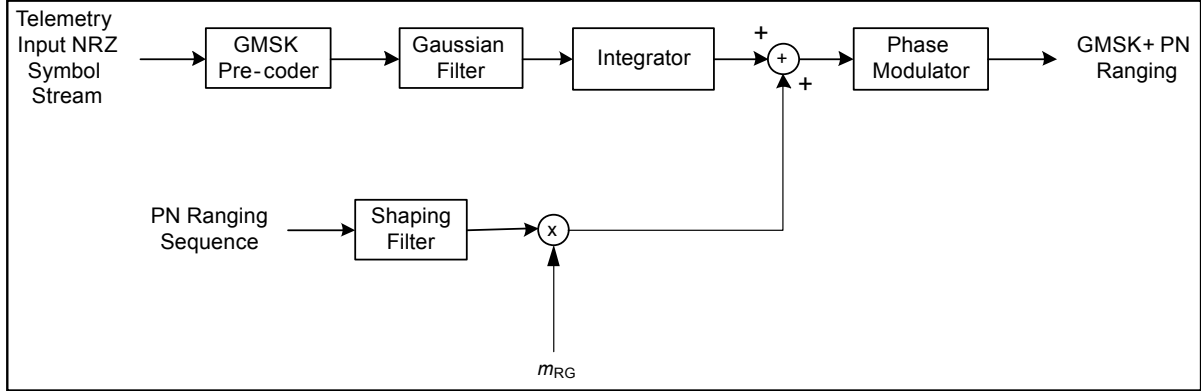


Figure 3-1: GMSK+PN Ranging Modulation Schematics

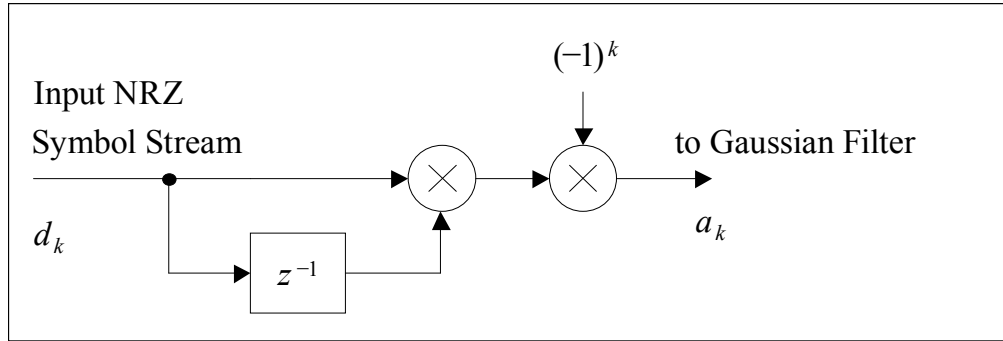


Figure 3-2: GMSK Precoder

Mathematically, the modulated RF carrier at the output of the phase modulator in figure 3-1 is expressed as:

$$x(t) = \sqrt{2P_T} \cos[2\pi f_c t + \phi_{TM}(t - \tau_{TM}) + \phi_{RG}(t - \tau_{RG})]; \quad (1)$$

where P_T is the transmit power,

f_c is the carrier frequency,

$\phi_{TM}(t)$ is the phase of the precoded GMSK signal (reference [1]) with symbol interval $T_s = 1/R_s$ given by:

$$\phi_{TM}(t) = \pi \sum_k a_k q(t - kT_s), \quad (2)$$

being

$$q(t) = \int_{-\infty}^t \frac{1}{4T_s} \left[\operatorname{erfc}\left(\frac{\tau - T_s}{\sqrt{2}\sigma}\right) - \operatorname{erfc}\left(\frac{\tau}{\sqrt{2}\sigma}\right) \right] d\tau$$

with $\lim_{t \rightarrow \infty} q(t) = 1/2$,

$$\sigma^2 = \ln(2)/(4\pi^2 B^2);$$

B is the single-sided 3-dB bandwidth of the Gaussian filter in figure 3-1:

$$BT_s = 0.25 \text{ or } BT_s = 0.5;$$

a_k are the precoded symbols to be transmitted obtained from the ± 1 level telemetry symbols by $a_k = (-1)^{k+1} d_k d_{k-1}$ (see figure 3-2),

and

$$\phi_{\text{RG}}(t) = m_{\text{RG}} \sum_k c_k h_{\text{sin}}(t - kT_c) \quad (3)$$

is the phase of the PN ranging signal;

m_{RG} is the peak PN ranging modulation index in radians;

$c_k = \pm 1$ the k th chip of the on-board regenerated PN ranging sequence (either a T2B or T4B sequence) (reference [2]);

$T_c = 1/R_{\text{RG}}$ the PN ranging chip interval;

$h(t) = h_{\text{sin}}(t) = \begin{cases} \sin(\pi t / T_c) & \text{for } t = [0, T_c] \\ 0 & \text{otherwise} \end{cases}$ the PN ranging shaping filter impulse response (a sinewave);

and τ_{TM} and τ_{RG} are random variables that model the absence of synchronization between the telemetry and the ranging signal.

Other PN ranging shaping filters were initially considered (rectangular and Square Root Raised Cosine [SRRC]), but were then discarded (see subsection A1).

Figure 3-3 (from reference [7]) shows a possible implementation of the receiver based on the complex envelope representation and the Laurent decomposition (filter coefficients $C_0(t)$) for GMSK with $BT_s = 0.5$ (an additional Wiener filter is needed for $BT_s = 0.25$).

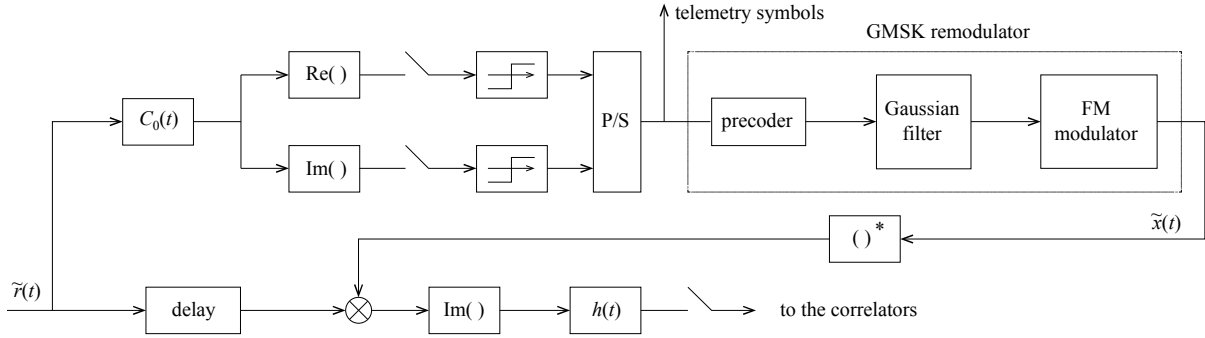


Figure 3-3: GMSK ($BT_s=0.5$)+PN RNG: Possible Demodulator Schematic

In other documents, the phase of the ranging signal was defined through a parameter h (dimensionless); to prevent confusion with the ranging shaping filter impulse response $h(t)$, the peak phase modulation index m_{RG} (rad) is instead used in this Informational Report. The relationship between h and m_{RG} is the following:

$$m_{RG} = \frac{\pi h}{\sqrt{2}}.$$

Therefore $h=0.1$ corresponds to $m_{RG} = 0.222$ rad peak while $h=0.2$ corresponds to $m_{RG} = 0.444$ rad peak.

In this document the considered values of m_{RG} are 0.111, 0.222, 0.444, 0.666 rad peak and it is shown that values from 0.222 to 0.444 rad can be used; for sake of simplicity the recommendations 2.4.22A and 2.4.22B round these values to 0.2 and 0.45 radians, but throughout this book 0.111, 0.222, 0.444, 0.666 rad peak are used.

3.2 RECOMMENDED STANDARD CONFIGURATION

3.2.1 GENERAL

After several tradeoffs, the choices were made to:

- have one recommendation for Category A missions (2.4.22A) and one for Category B missions (2.4.22B) and use the BT_s value for GMSK consistent with recommendation 2.4.17A and 2.4.17B (0.25 for Category A missions and 0.5 for Category B missions);
- allow a ranging modulation index m_{RG} between 0.222 and 0.444 rad (these values were later truncated to 0.2 and 0.45 rad);
- constrain the PN ranging chip-rate-to-telemetry-symbol-rate ratio to be a non-integer number higher than 1;¹

¹ A PN ranging chip-rate-to-telemetry-symbol-rate ratio slightly smaller than 1 may be used in case of mission need and provided that the link margins are adequate.

- set the telemetry signal level such that the resulting symbol error rate at the receiver is better than 0.1.²

The reasons for these choices are explained in the following subsections.

3.2.2 PN RANGING SIGNAL SHAPING

In light of the results given in 3.3, A2.1, and A2.2, *only sinewave shaping* is retained. Squarewave shaping is too bandwidth hungry while SRRC shaping has a significant power loss.

3.2.3 PN RANGING SIGNAL CODES

The regenerative PN ranging standard (section 3 of reference [2]) allows both options to be selected. T2B is recommended for low signal-to-noise links since it yields shorter acquisition times, while T4B is recommended for high ranging accuracy cases.

When combined with GMSK via sinewave shaping, T4B has a better spectral roll-off than T2B whereas the 99-percent power bandwidth is comparable.

The telemetry degradation does not depend on the PN code selection (T4B losses are only 0.1 dB smaller than T2B), as can be seen in table A-3 and for more general cases from figures A-26 to A-29.

The degradation of PN ranging acquisition time due to telemetry is some 0.3–0.9 dB smaller for T4B than for T2B, as can be seen in table A-4 and for more general cases from figures A-38 to A-41. However, as for conventional residual carrier modulation of the PN ranging signal, the acquisition time of T2B is much shorter (equivalent to 12 dB) than T4B (reference [2]).

The ranging jitter is obviously better for T4B than for T2B for standard modulation (equivalent to 3.5 dB) of the PN signal with a residual carrier. The telemetry GMSK modulation does not alter the difference in performance.

In light of these results (T4B is marginally better than T2B but the different characteristics of the two—optimal jitter for T4B and optimal acquisition time for T2B—are maintained), both T2B and T4B codes are allowed depending on mission characteristics: T4B is used when the ranging accuracy is of primary concern and T2B when the acquisition time is of primary concern.

² This value allows a simpler receiver implementation with telemetry cancellation (subtraction) at symbol level. The scheme is able to operate with a symbol error rate as high as 0.2 but may require a more complex cancellation or additional system margins.

3.2.4 PN RANGING MODULATION INDEX

Simulations (reference [7]) with real telemetry and ranging synchronizers have shown that the system is able to work as long as m_{RG} is kept between 0.222 (favoring telemetry) and 0.444 (favoring ranging). This can be seen in figures A-5, A-6, A-7, and A-8. Subsections A3.1, A3.3, and A3.4 confirm this result for non-synchronous telemetry and ranging signals as well as for several symbol-rate-to-chip-rate ratios.

Therefore it was agreed to select m_{RG} in the range $[0.222, 0.444]$ rad peak to allow some flexibility in the link design. For sake of simplicity, the recommendations give the allowable range of m_{RG} as $[0.2, 0.45]$ rad peak.

3.2.5 CHIP-RATE-TO-TELEMETRY-SYMBOL-RATE RATIO

In light of the results obtained, it is recommended that the ratio R_{RG}/R_{TM} be greater than one. Since the ranging signal is generated at the ground station, while telemetry is generated onboard the spacecraft, the possibility that R_{RG} will be exactly equal to R_{TM} is highly unlikely in practice; however, the ranging chip synchronizer loop bandwidth must be chosen so that potential periodic components of the ranging jitter are removed.

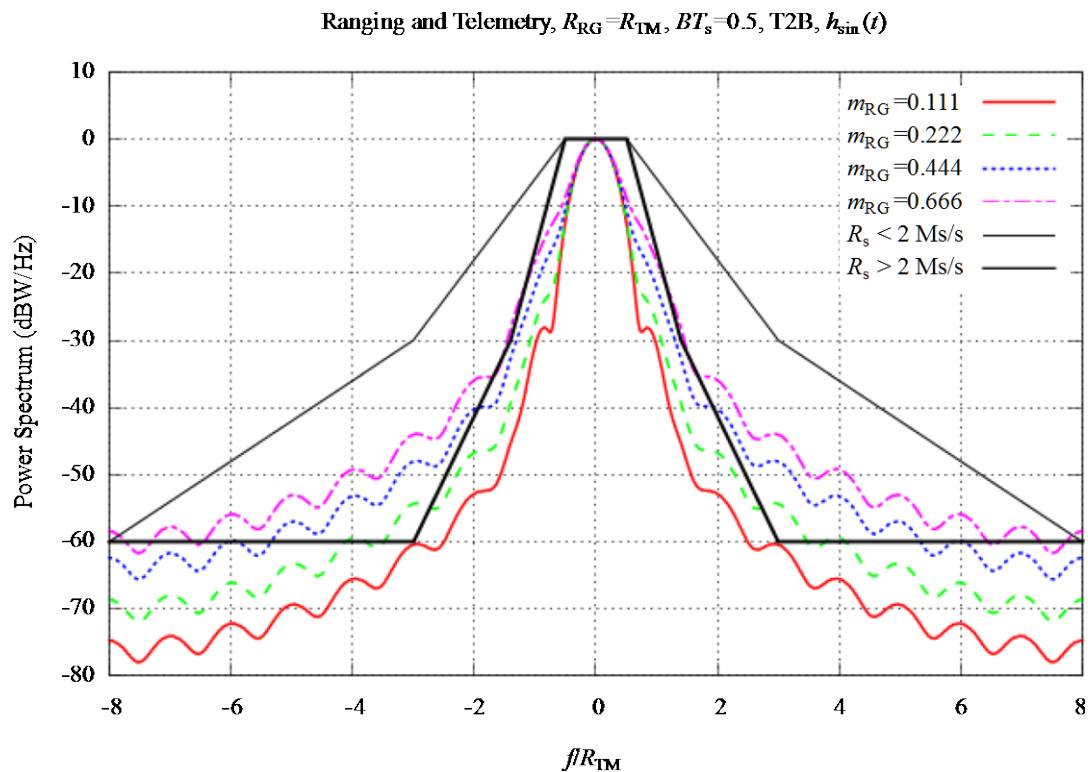
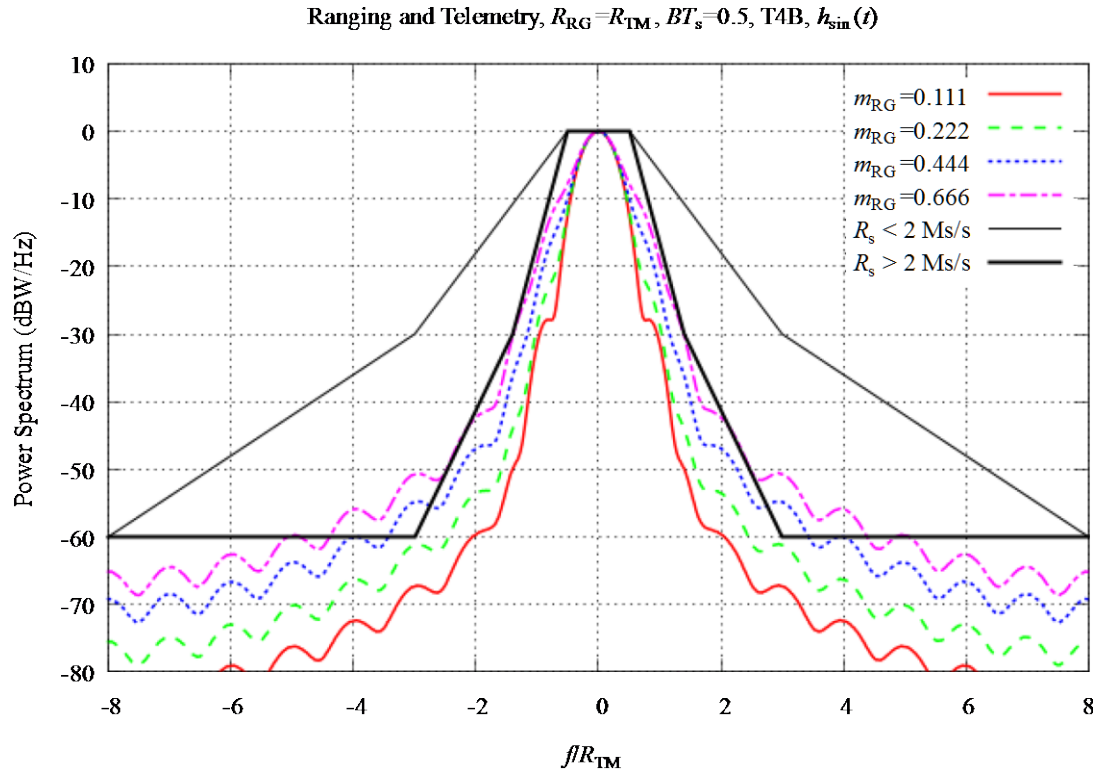
To reflect the general unsynchronized case, many simulations have a slightly offset telemetry rate, given by $R_{TM}=0.999$ or 0.99 instead of $R_{TM}=1$.

It must be noted that, when $R_{RG} \approx R_{TM}$ a periodic component at fundamental frequency $f_p = \frac{1}{2}|R_{RG} - R_{TM}|$ is present in the error signal which drives the chip synchronizer: if the chip synchronizer single-sided loop bandwidth is larger than f_p , then the ranging jitter standard deviation plotted versus $P_T T_s / N_0$ shows a floor, a minimum value which cannot be reduced by increasing the signal-to-noise ratio (reference [11]). Similarly, when $R_{RG} \approx 2R_{TM}$, the fundamental frequency is $f_p = \frac{1}{2}|R_{RG} - 2R_{TM}|$. For higher R_{RG}/R_{TM} ratios, the effect becomes negligible (reference [13]). (See A3.1 for more details.)

3.3 SPECTRAL PERFORMANCE

The SFCG spectral mask contained in recommendation 21-2R3 was not intended to be applicable to ranging or telemetry plus ranging, but to telemetry-only transmission. Therefore it is used only to check how good the selected modulation can be in the spectrum-management domain.

For $R_{RG}=R_{TM}$ the specified sinewave spectra can meet the SFCG mask if T4B is selected within a certain modulation index boundary. T2B is slightly less bandwidth efficient than T4B and, under the same conditions (sinewave shaping, same modulation index), exceeds the mask. This can be seen in figures 3-4, 3-5, 3-6, and 3-7 (sinewave), which refer to the case $R_{RG}=R_{TM}$. Figures 3-8, 3-9, 3-10, and 3-11 show the same simulations repeated for the case $R_{RG}=3R_{TM}$.

Figure 3-4: GMSK($BT_s=0.5$)/PN(Sine) Spectral Plots for Code T2B, $R_{RG}=R_{TM}$ Figure 3-5: GMSK($BT_s=0.5$)/PN(Sine) Spectral Plots for Code T4B, $R_{RG}=R_{TM}$

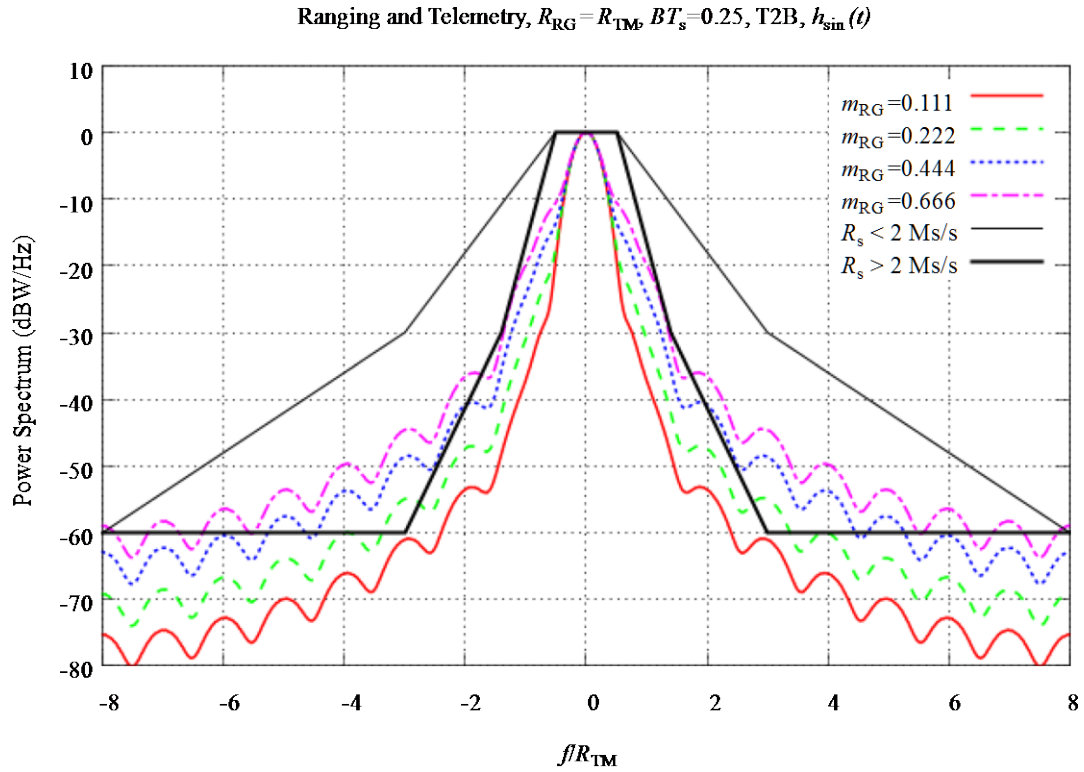


Figure 3-6: GMSK($BT_s=0.25$)/PN(Sine) Spectral Plots for Code T2B, $R_{RG}=R_{TM}$

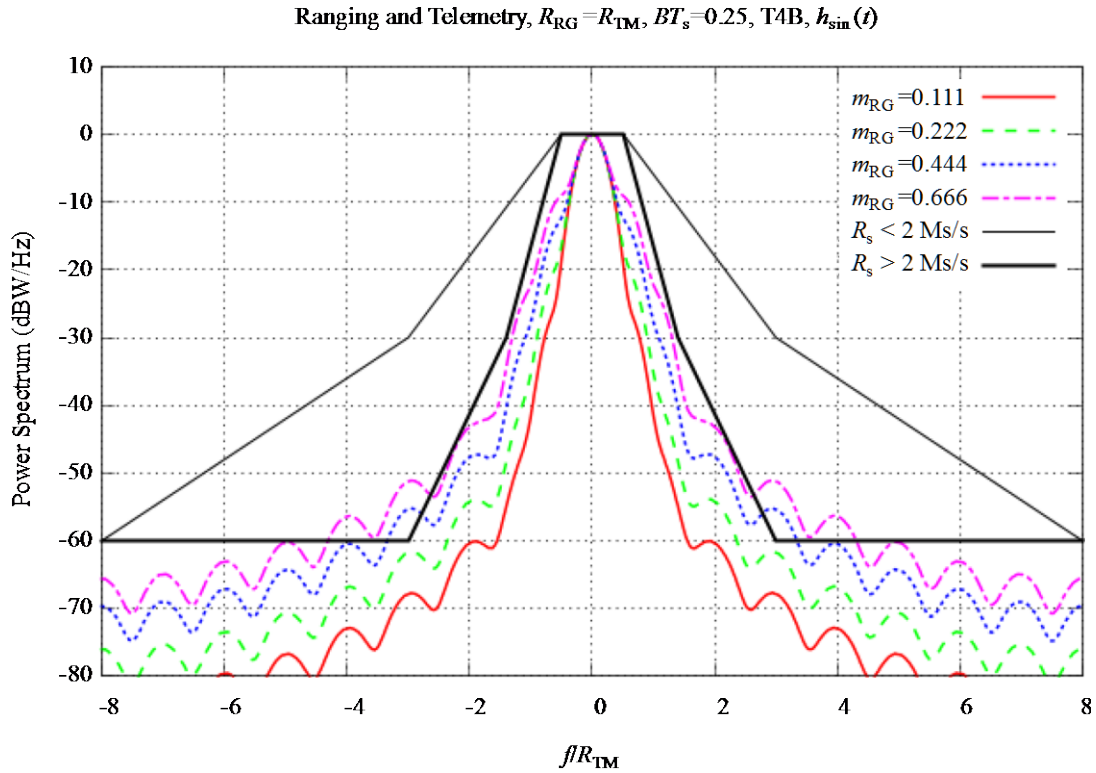
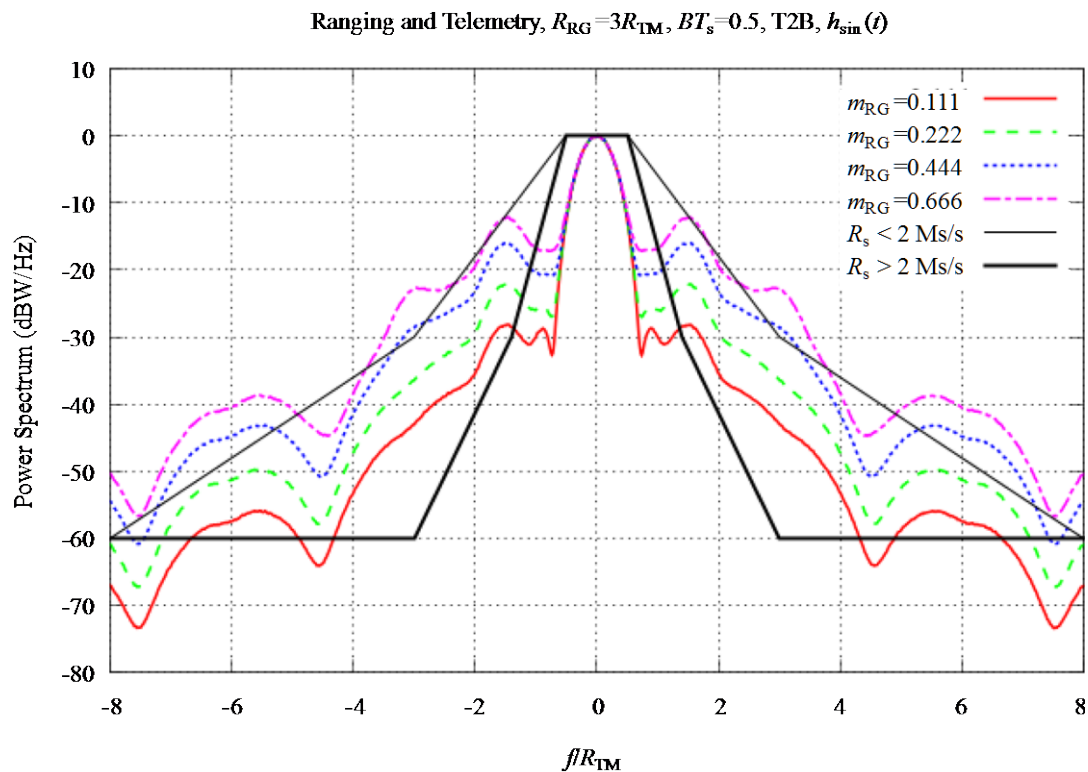
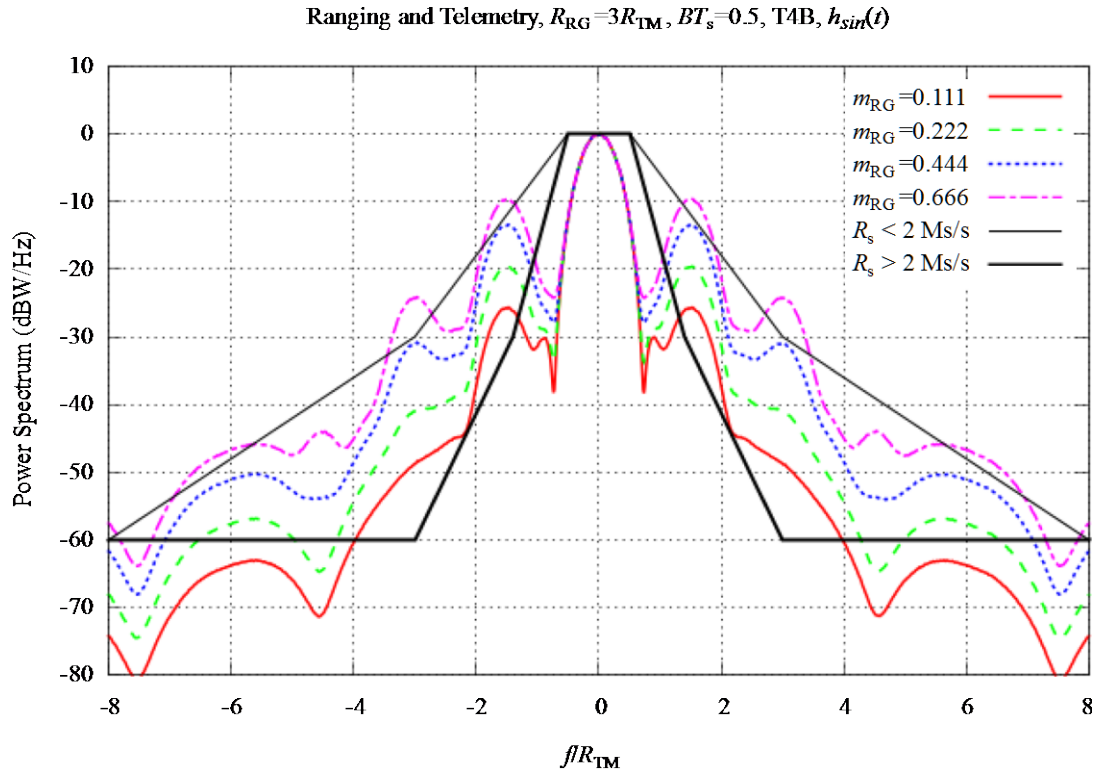


Figure 3-7: GMSK($BT_s=0.25$)/PN(Sine) Spectral Plots for Code T4B, $R_{RG}=R_{TM}$

Figure 3-8: GMSK($BT_s=0.5$)/PN(Sine) Spectral Plots for Code T2B, $R_{RG}=3R_{TM}$ Figure 3-9: GMSK($BT_s=0.5$)/PN(Sine) Spectral Plots for Code T4B, $R_{RG}=3R_{TM}$

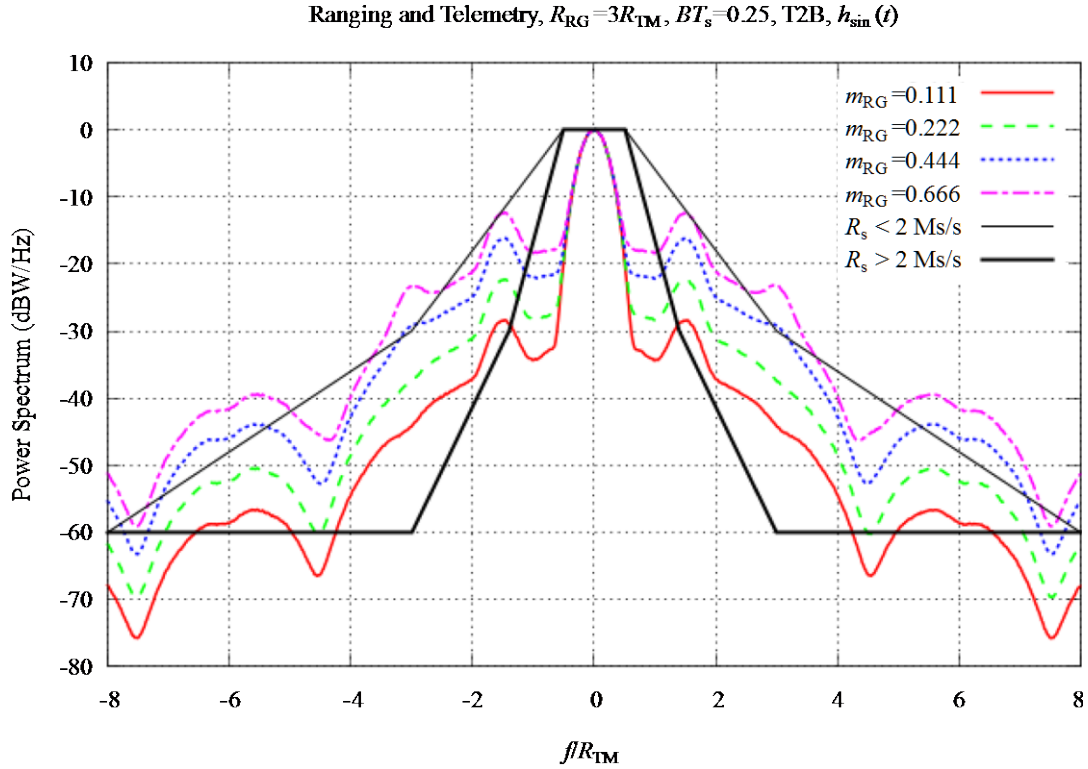


Figure 3-10: GMSK($BT_s=0.25$)/PN(Sine) Spectral Plots for Code T2B, $R_{RG}=3R_{TM}$

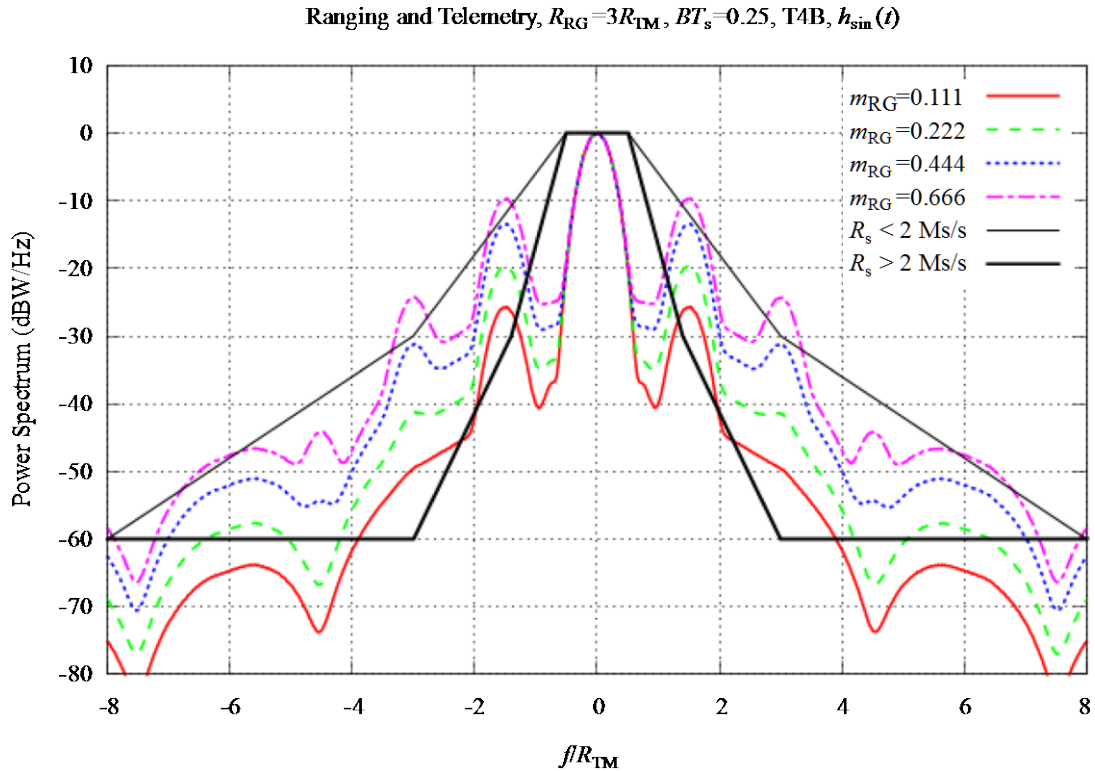


Figure 3-11: GMSK($BT_s=0.25$)/PN(Sine) Spectral Plots for Code T4B, $R_{RG}=3R_{TM}$

The occupied bandwidth or 99-percent power bandwidth B_x is defined by ITU as the width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to 0.5 percent of the total mean power of a given emission.

Tables 3-1 and 3-2 shows the measured normalized bandwidth B_x/R_{TM} for the specified sinewave cases with $R_{RG}=R_{TM}$ and $R_{RG}=3R_{TM}$.

Table 3-1: Normalized Occupied Bandwidth (99-Percent Power) for $R_{RG}=R_{TM}$, PN(Sine)

system	$R_{RG} = R_{TM}, h_{\sin}(t)$			
	GMSK $BT_s = 0.5$		GMSK $BT_s = 0.25$	
	T2B	T4B	T2B	T4B
$m_{RG} = 0.111$	1.070	1.075	0.902	0.911
$m_{RG} = 0.222$	1.183	1.206	1.062	1.106
$m_{RG} = 0.444$	1.656	1.592	1.559	1.479
$m_{RG} = 0.666$	1.979	1.829	1.911	1.711

Table 3-2: Normalized Occupied Bandwidth (99-Percent Power) for $R_{RG} = 3R_{TM}$, PN(Sine)

system	$R_{RG} = 3R_{TM}, h_{\sin}(t)$			
	GMSK $BT_s = 0.5$		GMSK $BT_s = 0.25$	
	T2B	T4B	T2B	T4B
$m_{RG} = 0.111$	1.119	1.137	0.939	0.952
$m_{RG} = 0.222$	2.945	3.086	2.953	3.072
$m_{RG} = 0.444$	4.073	3.603	4.042	3.498
$m_{RG} = 0.666$	5.515	4.042	5.526	3.913

For m_{RG} large (0.444 or 0.666) and sinewave shaping, T4B has a slightly narrower bandwidth than T2B.

3.4 TELEMETRY AND RANGING PERFORMANCE

3.4.1 INTRODUCTION

Unless otherwise specified, all telemetry symbol rates and ranging chip rates are unitless rates normalized with respect to R_{TM} . This choice is used for convenience since the relative performance of the system is set by the ratio R_{RG}/R_{TM} . In sections where the absolute performance is computed as example of achievable results, the actually used units are given.

3.4.2 OVERVIEW

In the following subsections, tables are given with telemetry and ranging performance, not only at $\text{SER}_{\text{TM}}=0.1$ but also at $\text{SER}_{\text{TM}}=0.2$ and 0.3 (useful when more powerful channel encoders are used) and $\text{SER}_{\text{TM}}=0.01$ (useful when a margin is present in the link budget); results are given only for $R_{\text{RG}}/R_{\text{TM}} \approx 1, 3, 1/3$. The case $R_{\text{RG}}/R_{\text{TM}} \approx 1/3$, which is not recommended, was actually analyzed only for GMSK with $BT_s=0.5$.

3.4.3 TELEMETRY PERFORMANCE

3.4.3.1 General

In the absence of ranging, the GMSK modulation has a loss L_{GMSK} with respect to ideal BPSK, because of the intersymbol interference generated by the Gaussian filter of the modulator; this loss depends on the receiver structure and on the target SER_{TM} value. For a simple OQPSK-like receiver (Laurent decomposition with 3-taps equalization in the case of $BT_s=0.25$), these losses are listed in table 3-3. The E_s/N_0 ratios necessary to provide SER_{TM} from 0.01 to 0.3 for ideal BPSK are listed in table 3-4.

Table 3-3: GMSK Modulation Losses L_{GMSK} (dB) with Respect to Ideal BPSK

SER_{TM}	$BT_s=0.5$	$BT_s=0.25$
0.3	0.042 dB	0.168 dB
0.2	0.010 dB	0.133 dB
0.1	0.012 dB	0.162 dB
0.01	0.021 dB	0.280 dB

Table 3-4: E_s/N_0 (dB) at Given SER_{TM} for Ideal BPSK

SER_{TM}	E_s/N_0
0.3	−8.62 dB
0.2	−4.51 dB
0.1	−0.86 dB
0.01	4.32 dB

The modulation loss of GMSK telemetry due to the power sharing with sinewave shaped ranging is given by:

$$L_{\text{TM-mod}} = J_0^2(m_{\text{RG}}), \quad (4)$$

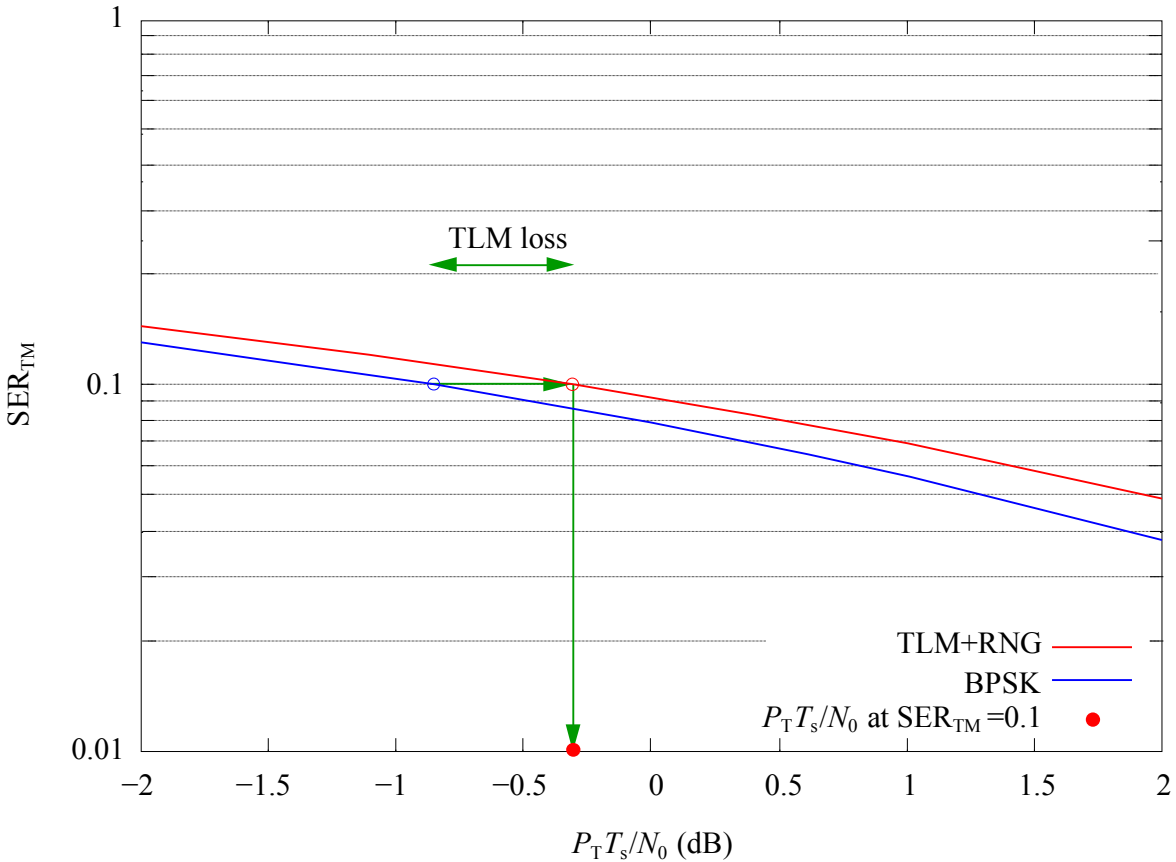
so that $L_{\text{TM-mod,dB}} = -10 \log_{10}[J_0^2(m_{\text{RG}})]$. Table 3-5 lists the telemetry modulation losses for some values of the ranging modulation index.

Table 3-5: Telemetry Modulation Loss $L_{\text{TM-mod}}$ (dB) versus Ranging Modulation Index

m_{RG} (rad)	0.111	0.222	0.444	0.666
$L_{\text{TM-mod}}$ (dB)	0.027	0.107	0.434	0.992

Because of the interference of ranging with telemetry, an additional degradation $L_{\text{TM-int}}$ has to be accounted for.

Tables 3-6 to 3-15 list the overall telemetry losses L_{TM} (the sum of $L_{\text{GMSK}} + L_{\text{TM-mod}} + L_{\text{TM-int}}$) with respect to ideal BPSK at the target SER_{TM} values and some ranging modulation indexes, for $R_{\text{RG}} \approx R_{\text{TM}}$, $R_{\text{RG}} \approx 3R_{\text{TM}}$, and $R_{\text{RG}} \approx R_{\text{TM}}/3$. As shown in figure 3-12, the value L_{TM} (dB) was obtained by subtracting the values E_s/N_0 (dB) for ideal BPSK listed in table 3-4 from the measured values of $P_{\text{T}}T_s/N_0$ (dB) that correspond to $\text{SER}_{\text{TM}}=0.01, 0.1, 0.2, 0.3$ in the complete simulation (telemetry and ranging, realistic synchronization, etc.). Values of L_{TM} (dB) versus m_{RG} are also plotted in figures A-26 to A-29 for $\text{SER}_{\text{TM}}=0.1$. The losses at higher SER_{TM} values (0.1 or 0.2), which correspond to lower E_s/N_0 values, are smaller than the losses at lower SER_{TM} values (0.01), since noise is the dominating impairment, and it masks the effects of ranging interference. At very high SER_{TM} values (0.3), the loss is again relatively large because the slope of the function SER_{TM} versus E_s/N_0 is almost zero and a very small difference in the SER_{TM} value corresponds to a large offset in terms of E_s/N_0 .

**Figure 3-12: Definition of Telemetry Loss L_{TM}**

NOTE – In figure 3-12, the telemetry loss L_{TM} is the difference between $(P_T T_s / N_0)_{TLM+RNG}$ (dB) necessary to obtain the target SER_{TM} with the complete system (telemetry and ranging both present) and the value of $(P_T T_s / N_0)_{BPSK}$ that gives the same SER_{TM} value for ideal BPSK. Example: $SER_{TM}=0.1$, $(P_T T_s / N_0)_{TLM+RNG}$ identified by a filled red circle, $(P_T T_s / N_0)_{BPSK} = -0.86$ dB.

Table 3-6: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod} for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.061	0.145	0.480	1.055
0.2	0.035	0.126	0.485	1.101
0.1	0.032	0.134	0.544	1.267
0.01	0.062	0.209	0.834	2.024

Table 3-7: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod} for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.061	0.145	0.472	1.041
0.2	0.035	0.122	0.474	1.074
0.1	0.029	0.125	0.517	1.196
0.01	0.058	0.189	0.739	1.753

Table 3-8: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod} for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.191	0.272	0.612	1.193
0.2	0.157	0.249	0.620	1.259
0.1	0.188	0.294	0.735	1.516
0.01	0.337	0.514	1.278	2.806

Table 3-9: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.186	0.270	0.603	1.164
0.2	0.158	0.243	0.597	1.194
0.1	0.184	0.279	0.666	1.342
0.01	0.320	0.442	0.958	1.897

Table 3-10: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.051	0.131	0.454	1.005
0.2	0.024	0.103	0.434	0.994
0.1	0.029	0.113	0.456	1.043
0.01	0.046	0.140	0.530	1.223

Table 3-11: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.052	0.131	0.450	0.992
0.2	0.022	0.102	0.423	0.969
0.1	0.028	0.108	0.429	0.983
0.01	0.042	0.126	0.460	1.034

Table 3-12: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.173	0.254	0.574	1.121
0.2	0.146	0.229	0.560	1.127
0.1	0.184	0.270	0.618	1.220
0.01	0.314	0.414	0.843	1.609

Table 3-13: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.172	0.253	0.570	1.112
0.2	0.146	0.224	0.545	1.090
0.1	0.182	0.262	0.587	1.140
0.01	0.308	0.395	0.734	1.325

Table 3-14: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1 / 3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.070	0.159	0.518	1.140
0.2	0.043	0.140	0.546	1.259
0.1	0.044	0.168	0.681	1.635
0.01	0.085	0.313	1.325	3.545

Table 3-15: Overall Telemetry Loss L_{TM} (Including the Modulation Loss L_{TM-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1 / 3$

	Loss (dB) with respect to ideal BPSK			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.070	0.159	0.517	1.140
0.2	0.041	0.139	0.545	1.256
0.1	0.044	0.168	0.678	1.625
0.01	0.085	0.310	1.310	3.511

3.4.3.2 Carrier Performance

The carrier phase jitter loss, as defined in figure A-21, is shown in tables 3-16 to 3-25 for the various considered cases. The measurements were performed using a closed loop second order phase synchronizer with normalized noise equivalent bandwidth $B_L T_s = 10^{-4}$.

Table 3-16: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.03	0.12	0.44	1.03
0.2	0.00	0.09	0.45	1.12
0.1	0.01	0.12	0.56	1.34
0.01	0.06	0.23	0.86	2.03

Table 3-17: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.03	0.11	0.37	0.85
0.2	0.00	0.07	0.36	0.89
0.1	0.00	0.07	0.40	0.99
0.01	0.05	0.12	0.50	1.27

Table 3-18: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.20	0.25	0.57	1.10
0.2	0.07	0.16	0.49	1.10
0.1	0.01	0.12	0.52	1.27
0.01	0.00	0.11	0.82	2.19

Table 3-19: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.19	0.26	0.47	0.90
0.2	0.06	0.13	0.39	0.85
0.1	0.00	0.06	0.35	0.88
0.01	0.00	0.03	0.41	1.08

Table 3-20: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.00	0.03	0.26	0.62
0.2	0.00	0.01	0.21	0.60
0.1	0.00	0.02	0.24	0.62
0.01	0.05	0.11	0.32	0.76

Table 3-21: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.00	0.03	0.22	0.58
0.2	0.00	0.01	0.19	0.51
0.1	0.00	0.02	0.19	0.47
0.01	0.00	0.06	0.19	0.44

Table 3-22: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.17	0.22	0.41	0.77
0.2	0.02	0.06	0.25	0.60
0.1	0.00	0.00	0.15	0.50
0.01	0.00	0.00	0.12	0.56

Table 3-23: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.17	0.22	0.41	0.73
0.2	0.02	0.06	0.23	0.54
0.1	0.00	0.00	0.11	0.38
0.01	0.00	0.00	0.03	0.21

Table 3-24: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1 / 3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.16	0.28	0.74	1.71
0.2	0.08	0.21	0.77	1.86
0.1	0.07	0.23	0.93	2.31
0.01	0.11	0.35	1.64	4.46

Table 3-25: Carrier Phase Jitter Loss (dB) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1 / 3$

	Carrier phase jitter loss (dB)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	0.16	0.24	0.74	1.67
0.2	0.08	0.20	0.75	1.80
0.1	0.06	0.21	0.86	2.17
0.01	0.06	0.29	1.31	3.68

3.4.4 RANGING PERFORMANCE

3.4.4.1 General

The modulation loss of the ranging signal is given by:

$$L_{\text{RG-mod}} = 2J_1^2(m_{\text{RG}}), \quad (5)$$

so that $L_{\text{RG-mod,dB}} = -10 \log_{10}[2J_1^2(m_{\text{RG}})]$

Table 3-26 lists the ranging modulation losses for some values of the ranging modulation index.

Table 3-26: Ranging Modulation Loss $L_{\text{RG-mod}}$ (dB) versus Ranging Modulation Index

m_{RG} (rad)	0.111	0.222	0.444	0.666
$L_{\text{RG-mod}}$ (dB)	22.11	16.13	10.27	7.02

3.4.4.2 Acquisition Time

As described in references [3] and [4], the number of correlation chips N_{chip} necessary to acquire the ranging sequence with an error of 10^{-3} , in the absence of telemetry, is equal to

$$N_{\text{chip}} = \frac{\eta}{E_{\text{RG}} / N_0} = \frac{\eta}{(\text{erf}^{-1}(1 - 2 \text{CER}))^2} \quad (6)$$

or

$$10 \log N_{\text{chip}} = \eta_{\text{dB}} - 10 \log_{10}(E_{\text{RG}} / N_0) = \eta_{\text{dB}} - 20 \log_{10}(\text{erf}^{-1}(1 - 2 \text{CER})), \quad (7)$$

where E_{RG}/N_0 is the chip energy-to-noise ratio, CER is the chip error rate, and η_{dB} is equal to 36.34 dB for T4B and 24.17 dB for T2B. The formula can be used to estimate the value of N_{chip} from the measured value of CER when telemetry is present, with an error of ± 1 dB (reference [4]).

In the ideal case (telemetry and ranging are both transmitted, but there is no interference between them; the two signals are orthogonal), the chip error rate CER can be derived simply as follows:

$$\text{CER} = \frac{1}{2} \text{erfc} \sqrt{\frac{E_{\text{RG}}}{N_0}} = \frac{1}{2} \text{erfc} \sqrt{\frac{P_{\text{RG}}}{N_0 R_{\text{RG}}}}, \quad (8)$$

where $P_{\text{RG}} = P_{\text{T}} L_{\text{RG-mod}} = P_{\text{T}} 2J_1^2(m_{\text{RG}})$. However, P_{T}/N_0 must be set so that the desired SER_{TM} is obtained, which is to say that the following must hold:

$$\frac{P_T}{N_0} J_0^2(m_{RG}) = \frac{E_s}{N_0} R_{TM}, \quad (9)$$

where E_s/N_0 is the signal-to-noise ratio necessary to obtain the specified SER_{TM} value using ideal BPSK (for example $E_s/N_0 = -0.86$ dB for $SER_{TM}=0.1$ (see table 3-4 for the other values of E_s/N_0). Therefore:

$$\frac{E_{RG}}{N_0} = \frac{P_{RG}}{N_0 R_{RG}} = \frac{P_T}{N_0 R_{RG}} 2J_1^2(m_{RG}) = \frac{E_s}{N_0} \frac{R_{TM}}{R_{RG}} \frac{2J_1^2(m_{RG})}{J_0^2(m_{RG})} = \frac{E_s}{N_0} \frac{R_{TM}}{R_{RG}} \frac{L_{RG-mod}}{L_{TM-mod}} \quad (10)$$

and

$$10\log_{10} N_{chip} = \eta_{dB} - \left(\frac{E_s}{N_0} \right)_{dB} + L_{RG-mod,dB} - L_{TM-mod,dB} + 10\log_{10} R_{RG} - 10\log_{10} R_{TM}. \quad (11)$$

The correlation time $t_{obs}=N_{chip}T_c$ can be thus expressed as

$$\begin{aligned} 10\log_{10} t_{obs} &= 10\log_{10} (N_{chip} T_c) = \\ &= \eta_{dB} - \left(\frac{E_s}{N_0} \right)_{dB} + L_{RG-mod,dB} - L_{TM-mod,dB} - 10\log_{10} R_{TM}, \end{aligned} \quad (12)$$

and it does not depend on R_{RG} since $T_c R_{RG}=1$. When GMSK telemetry is simultaneously transmitted, the above computed value has to be increased as per table A-4 or from figures A-38 to A-41, being the table valid only for ideal synchronization and R_{TM} exactly equal to R_{RG} .

The difference in dB between the value of N_{chip} estimated through the simulations (taking into consideration the telemetry interference and the imperfect synchronization) and the value obtained by the equation (11) is given in tables 3-27 to 3-36. It will be called $L_{RG-int-acq}$. For example, if the desired value of SER_{TM} is 0.1 and the selected system is GMSK ($BT_s=0.5$)/PN(T2B) with $m_{RG}=0.222$ rad and $R_{RG} \approx R_{TM}$, equation (11) is used with $E_s/N_0=-0.86$ dB (valid for ideal BPSK), and with $L_{RG-int-acq}=2.69$ dB, as given in table 3-27 to obtain the same value N_{chip} measured through the simulation.

Table 3-27: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	12.66	10.32	9.42	9.23
0.2	6.34	5.60	5.48	5.46
0.1	2.85	2.69	2.65	2.61
0.01	0.42	0.33	0.06	0.01

Table 3-28: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	11.24	9.42	9.04	8.92
0.2	5.62	5.25	5.19	5.20
0.1	2.54	2.48	2.50	2.49
0.01	0.37	0.31	0.11	0.01

Table 3-29: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	13.03	10.37	9.34	9.19
0.2	6.47	5.65	5.50	5.50
0.1	2.83	2.63	2.58	2.50
0.01	0.21	0.10	0.01	0.00

Table 3-30: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	10.68	8.83	8.53	8.37
0.2	5.12	4.85	4.85	4.90
0.1	2.17	2.15	2.18	2.18
0.01	0.08	0.04	-0.11	-0.36

Table 3-31: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	16.84	12.50	9.34	8.51
0.2	8.88	6.05	4.84	4.76
0.1	3.80	2.38	2.17	2.16
0.01	0.47	0.28	0.21	0.09

Table 3-32: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	14.12	10.62	8.34	8.10
0.2	7.69	5.08	4.57	4.58
0.1	2.86	2.09	2.05	2.07
0.01	0.32	0.27	0.25	0.25

Table 3-33: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	18.51	12.35	9.23	8.44
0.2	8.85	5.88	4.72	4.64
0.1	3.58	2.23	2.03	2.00
0.01	0.21	0.03	-0.08	-0.27

Table 3-34: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	16.38	10.86	8.26	7.93
0.2	7.31	4.88	4.37	4.40
0.1	2.58	1.91	1.84	1.87
0.01	0.07	0.00	0.00	-0.03

Table 3-35: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1/3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	17.67	13.59	11.22	10.76
0.2	8.83	6.97	6.58	6.83
0.1	3.89	3.37	3.47	3.85
0.01	0.47	0.34	-0.04	0.54

Table 3-36: Ranging Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1/3$

	Loss (dB) with respect to equation (11)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	17.29	12.39	10.58	10.59
0.2	8.10	6.60	6.53	6.80
0.1	3.52	3.31	3.46	3.84
0.01	0.45	0.34	-0.03	0.55

3.4.4.3 Jitter

For the proposed sinewave shaped ranging signal, the theoretical variance of the normalized³ (to the chip rate) clock jitter is given reference [7], for the ideal case, by:

$$\sigma_\varepsilon^2 = \frac{B_L T_c}{\pi^2 E_{cl,sin} / N_0}, \quad (13)$$

where B_L is the single-sided loop bandwidth (second order loop assumed), and $E_{cl,sin}$ is the energy per chip of the ranging clock (period $2T_c$, frequency $R_{RG,cl} = R_{RG}/2$) and can be obtained from the energy of the ranging chip by recalling that the clock is suppressed by $L_{ck} = 0.881$ ($L_{ck,dB} = 0.55$ dB) for T4B and by 0.3936 ($L_{ck,dB} = 4.05$ dB) for T2B:

$$E_{cl,sin} = L_{ck} E_{RG,sin} = L_{ck} P_{RG,sin} T_c = L_{ck} P_T L_{RG-mod} T_c = L_{ck} P_T 2J_1^2(m_{RG}) T_c, \quad (14)$$

so that the final formula is as per table 3-37, which also gives σ_ε^2 for the squarewave $h_{sq}(t)$. Again, P_T/N_0 is obtained from the specification of the target SER_{TM} value, assuming that telemetry and ranging signals are transmitted together but do not interfere:

$$\frac{P_T}{N_0} = \frac{E_s}{N_0} \frac{R_{TM}}{J_0^2(m_{RG})}, \quad (15)$$

$E_s/N_0 = -0.86$ dB being the signal-to-noise ratio necessary to provide $SER_{TM}=0.1$ for ideal BPSK (see table 3-4 for the values of E_s/N_0 to be used for other target SER_{TM} values). In the overall,

$$\sigma_\varepsilon^2 = \frac{B_L T_c}{\pi^2 L_{ck} \frac{P_T}{N_0} 2J_1^2(m_{RG}) T_c} = \frac{B_L T_c R_{RG} [J_0^2(m_{RG})]}{\pi^2 L_{ck} \frac{E_s}{N_0} R_{TM} [2J_1^2(m_{RG})]}. \quad (16)$$

³ The normalized jitter standard deviation σ_ε is equal to the jitter standard deviation σ_τ measured in seconds, divided by T_c .

The normalized jitter variance can also be expressed as:

$$\sigma_{\varepsilon}^2 = \frac{B_L T_c}{\pi^2 E_{\text{cl,sin}} / N_0} = \frac{B_L T_c}{\pi^2 P_{\text{cl,sin}} T_c / N_0} = \frac{B_L N_0}{\pi^2 P_{\text{cl,sin}}}, \quad (17)$$

while the (un-normalized) jitter variance is

$$\sigma_{\tau}^2 = (\sigma_{\varepsilon} T_c)^2 = \frac{\sigma_{\varepsilon}^2}{R_{\text{RG}}^2} = \frac{\sigma_{\varepsilon}^2}{4R_{\text{RG,cl}}^2}, \quad (18)$$

$R_{\text{RG,cl}} = R_{\text{RG}}/2$ being the frequency of the clock component of the PN code.

The actual variance taking into account the telemetry interference to the ranging signal is typically larger than the value predicted by the formula above; it should be noted that, because of the telemetry loss, the true signal-to-noise ratio at which the complete system works is larger than that of ideal BPSK, and therefore the ranging jitter loss can be negative, especially for high values of m_{RG} , which force large telemetry losses. The ratio (in dB) between the measured jitter variance and the theoretical one obtained through equation (16) can be computed from figures A-42, A-44, A-46, and A-48, and is listed in tables 3-38 to 3-47. The loss will be called $L_{\text{RG-int-jit}}$.

NOTE – In tables 3-38–3-47, losses with asterisks are not reliable, because of measurement errors or lack of synchronization.

Table 3-37: PN Ranging Normalized Jitter Variance

$h_{\text{sq}}(t)$	$\sigma_{\varepsilon}^2 = \frac{B_L}{4L_{\text{ck}} \frac{P_T}{N_0} \sin^2(m_{\text{RG}})}$
$h_{\text{sin}}(t)$	$\sigma_{\varepsilon}^2 = \frac{B_L}{\pi^2 L_{\text{ck}} \frac{P_T}{N_0} 2J_1^2(m_{\text{RG}})}$

Table 3-38: Ranging Jitter Loss (Excluding the Modulation Loss $L_{\text{RG-mod}}$) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{\text{TM}}=0.999$, $R_{\text{RG}}=1$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{\text{RG}}=0.111$	$m_{\text{RG}}=0.222$	$m_{\text{RG}}=0.444$	$m_{\text{RG}}=0.666$
0.3	7.33	5.25	5.13	5.00
0.2	2.82	2.60	2.52	5.21
0.1	1.47	1.39	1.21	0.94
0.01	0.28	0.21	0.00	0.00

Table 3-39: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	5.56	5.27	5.01	4.98
0.2	2.72	2.64	2.78	2.60
0.1	1.39	1.37	1.23	1.10
0.01	0.26	0.22	0.04	-0.41

Table 3-40: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	5.54	5.02	5.01	4.73
0.2	2.54	2.43	2.12	1.99
0.1	1.13	1.09	0.85	0.63
0.01	0.00	-0.13	-0.57	-1.61

Table 3-41: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	5.11	4.95	4.92	4.70
0.2	2.45	2.45	2.37	2.43
0.1	1.08	1.06	0.95	0.82
0.01	0.00	-0.03	-0.25	-0.65

Table 3-42: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	65.67*	4.99	4.50	4.38
0.2	-0.68*	2.32	2.31	2.26
0.1	0.99	0.83	0.77	0.69
0.01	-0.27	-0.32	-0.36	-0.52

Table 3-43: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	26.88*	4.62	4.42	4.38
0.2	2.48	2.33	2.30	2.34
0.1	0.88	0.78	0.77	0.75
0.01	-0.30	-0.34	-0.37	-0.37

Table 3-44: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	50.83*	5.15	4.58	4.56
0.2	8.33*	2.31	2.16	2.07
0.1		0.57	0.35	0.28
0.01		-0.67	-0.80	-0.99

Table 3-45: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	21.31*	4.78	4.64	4.71
0.2	2.52	2.18	2.14	2.09
0.1	0.62	0.57	0.44	0.42
0.01	-0.62	-0.66	-0.69	-0.71

Table 3-46: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1/3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	6.24	5.86	6.03	6.00
0.2	3.52	3.33	3.51	3.47
0.1	135	1.46	1.46	1.06
0.01	-0.13	-0.28	-0.87	-2.24

Table 3-47: Ranging Jitter Loss (Excluding the Modulation Loss L_{RG-mod}) for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, $R_{TM}=0.999$, $R_{RG}=1/3$

	Loss (dB) with respect to equation (16)			
Target SER_{TM}	$m_{RG}=0.111$	$m_{RG}=0.222$	$m_{RG}=0.444$	$m_{RG}=0.666$
0.3	6.05	5.71	5.97	6.32
0.2	3.50	3.47	3.67	3.53
0.1	1.38	1.49	1.55	1.30
0.01	-0.09	-0.23	-0.81	-2.16

3.4.5 SUMMARY FOR THE REFERENCE CASE

For the reference case $m_{RG}=0.222$ rad, $R_{RG} \approx R_{TM}=1$ and $SER_{TM}=0.1$, the parameters to be used in the link budgets are summarized in table 3-48, where P_T/N_0 is the total power-to-noise spectral density.

If R_{TM} is not equal to 1, then it is sufficient to increase the P_T/N_0 values given in table 3-48 by $10\log_{10}(R_{TM})$; if the obtained value is smaller than the available one, then the system correctly works. The losses given in table 3-48 do not depend on the value of R_{TM} .

Table 3-48: Parameters for the Link Budget Reference Case

GMSK BT_s PN code	$BT_s=0.5$ T2B	$BT_s=0.5$ T4B	$BT_s=0.25$ T2B	$BT_s=0.25$ T4B
E_s/N_0 for BPSK	-0.86	-0.86	-0.86	-0.86
Overall TM loss L_{TM} (dB)	0.134	0.125	0.294	0.279
P_T/N_0 (dBHz)	-0.72	-0.73	-0.56	-0.58
RG acquisition loss $L_{RG-int-acq}$ (dB)	2.69	2.48	2.63	2.15
RG jitter loss $L_{RG-int-jit}$ (dB)	1.39	1.37	1.09	1.06

3.4.6 LINK BUDGET EXAMPLES

Examples of a link budget for a hypothetical mission to the second Lagrange point called L2-M and one to Mars called Mars-M, similarly to the two missions considered in reference [10], are used here to explain the computation algorithms. However, extension to other missions can easily be done by using the data provided in the tables in 3.4.3 and 3.4.4.

In the examples, the total power-to-noise spectral density P_T/N_0 received at the station (with a 35-m antenna) is:

- $P_T/N_0 = 69.3$ dBHz for L2-M;
- $P_T/N_0 = 57.1$ dBHz for Mars-M.

L2-M GMSK information bit rate is 1.5 Mbit/s, so that in case of telemetry modulation only the available E_b/N_0 is 7.54 dB (where E_b is the energy per information bit). Such bit rate can be received with 4.84 dB margin with respect to the required E_b/N_0 ratio of 2.7 dB (assuming $SER_{TM}=0.1$) by using the CCSDS standard concatenated (approximate rate 1/2.29) encoding, resulting in a symbol rate of 3.435 Msymbol/s. The nominal mission-required margin is 3 dB; thus an extra 1.84 dB is available to counter the ranging losses.

Mars-M information bit rate is 150 kbit/s, so that in case of telemetry-only the available E_b/N_0 is 5.34 dB (where E_b is again the energy per information bit). Such bit rate can be received with 3.6 dB margin with respect to the required E_b/N_0 ratio of 1.74 dB by using the CCSDS standard Turbo (approximate rate 1/6.02 with resulting symbol rate of 903 ksymbol/s) encoding. The nominal mission required margin is 3 dB; thus an extra 0.6 dB is available to counter the ranging losses.

The nominal missions' performance for telemetry-only transmission is summarized in table 3-49.

The considered telemetry and ranging case is summarized in table 3-50.

For this example, only GMSK ($BT_s=0.25$) with code T4B for L2-M and GMSK ($BT_s=0.5$) with code T2B for Mars-M will be considered. While in reference [10] the link budgets are analyzed assuming the case of perfect synchronization between ranging and telemetry ($R_{RG}=R_{TM}$), the more general case $R_{RG} \approx R_{TM}$ is considered here.

Starting from the theoretical system performance, the losses due to the reciprocal interference between telemetry and ranging will be added, so that it becomes easier to check if it is possible to include the ranging signal in these two example missions, while keeping a reasonable margin for telemetry.

The Mars-M case is outside the boundary of recommendation 2.4.22B since, because of the use of Turbo rate 1/6.02 encoding, the telemetry symbol rate is much higher than the recommended $SER_{TM}=0.1$ value.

Table 3-49: Telemetry Only Link Budget Example

Mission	L2-M	Mars-M
P_T/N_0	69.3 dBHz	57.1 dBHz
GMSK modulation	$BT_s=0.25$	$BT_s=0.5$
Information bit rate	1.5 Mbit/s	150 kbit/s
Coding rate	1/2.29 (concatenated encoding)	1/6.02 (Turbo encoding)
Symbol rate	3.435 Msymbol/s	903 ksymbol/s
Required symbol error rate	0.1	0.24
Required E_b/N_0	2.7 dB	1.74 dB
Available E_b/N_0	7.54 dB	5.34 dB
Telemetry margin	4.84 dB	3.6 dB
Available E_s/N_0	3.94 dB	-2.44 dB
SER _{TM} at available E_s/N_0	0.013	0.143

Table 3-50: Telemetry and PN Ranging Set-Up Example

Mission	L2-M	Mars-M
P_T/N_0	69.3 dBHz	57.1 dBHz
GMSK modulation	$BT_s=0.25$	$BT_s=0.5$
Telemetry symbol rate	3.435 Msymbol/s	0.903 Msymbol/s
PN Ranging chip rate	≈ 3.435 Mchip/s	≈ 0.903 Mchip/s
PN Ranging modulation index	0.222 rad	0.222 rad
PN Ranging code	T4B	T2B
PN Ranging acquisition time constant η_{dB}	36.34 dB	24.17 dB
PN Ranging clock suppression L_{ck}	0.881 (0.55 dB)	0.3936 (4.05 dB)
PN Ranging single-sided loop bandwidth B_L	5 Hz	5 Hz

3.4.6.1 Telemetry Performance

For the L2-M mission with $R_{RG} \approx R_{TM}$, the value of the signal-to-noise ratio is $E_s/N_0=3.94$ dB and SER_{TM} is 0.013 in the absence of ranging; when ranging is present, a telemetry loss must be included, which increases the value of SER_{TM}. From tables 3-6 to 3-15 it is possible to evaluate the telemetry loss at specified SER_{TM} values, but in this case it is necessary to find SER_{TM} from the signal-to-noise ratio. One possibility is to build the curve SER_{TM} versus $P_T T_s/N_0=E_s/N_0$ from the data in tables 3-6 to 3-15 and then interpolate at $E_s/N_0=3.94$ dB. Such curve is defined through the four points $[E_s/N_0, \text{SER}_{TM}] = [(E_s/N_0)_{\text{BPSK}} + L_{TM}, \text{SER}_{TM}]$:

$[-8.62+0.27, 0.3]$, $[-4.51+0.243, 0.2]$, $[-0.86+0.279, 0.1]$, $[4.32+0.442, 0.01]$, with $(E_s/N_0)_{\text{BPSK}}$ from table 3-4 and L_{TM} from table 3-9 for $m_{RG}=0.222$ rad.

It is possible, however, to avoid this interpolation process by considering that the overall telemetry loss is between 0.279 dB for target $SER_{TM}=0.1$ and 0.442 dB for target $SER_{TM}=0.01$ (see table 3-9), but it is not possible to say which loss should be applied, since the true value of SER_{TM} is not known. This means that the useful E_s/N_0 ranges from $3.94-0.279=3.661$ dB (corresponding to a true $SER_{TM}=0.016$) to $3.94-0.442=3.498$ (corresponding to a true $SER_{TM}=0.017$), and the true SER_{TM} can be assumed as 0.017 (worst case.) Performing linear interpolation on the telemetry losses, but with a logarithmic scale (i.e., interpolating between $\log_{10}(0.01)$ with loss 0.442 dB and $\log_{10}(0.1)$ with loss 0.279 dB), and evaluating the loss at the intermediate point $\log_{10}(0.017)$ yields a loss equal to:

$$(0.442-0.279)/(\log_{10}(0.01)-\log_{10}(0.1)) \times (\log_{10}(0.017)-\log_{10}(0.01)) + 0.442 = 0.404 \text{ dB.}$$

This choice is reasonable, since BER curves are plotted versus E_s/N_0 using a vertical logarithmic scale, which gives smooth readable curves. Therefore a telemetry loss of 0.404 dB is estimated, out of which 0.11 dB result from the modulation loss L_{TM-mod} (see table 3-5) and 0.25 dB are the loss L_{GMSK} of GMSK with $BT_s=0.25$ with respect to BPSK (interpolating the data of table 3-3), and 0.044 dB result from the ranging interference. A useful $E_b/N_0=7.54-0.404=7.135$ dB is obtained, with a margin of 4.435 dB (implementation losses are not considered) with respect to the required value $E_b/N_0=2.7$ dB. Thus ranging can be included without affecting the telemetry subsystem. The ranging losses will then be evaluated for $SER_{TM}=0.017$, interpolating between the values given in the tables.

As far as the carrier is concerned, the phase jitter standard deviation of this example can be evaluated as follows:

- at $SER_{TM}=0.017$, the carrier phase loss can be interpolated from the values in the last two lines of table 3-19 for $m_{RG}=0.222$ rad:

$$(0.03-0.06)/(\log_{10}(0.01)-\log_{10}(0.1)) \times (\log_{10}(0.017)-\log_{10}(0.01)) + 0.03 = 0.037;$$

- the signal-to-noise ratio for evaluating equation (19) is $P_T T_s/N_0=3.94-0.037=3.903$ dB;
- equation (19) gives $\sigma_\zeta=1.38 \cdot 10^{-3}$ with $B_L T_s=10^{-4}$, so that the ‘unnormalized’ standard deviation of the phase jitter is $8.71 \cdot 10^{-3}$ rad.

For the Mars-M mission the value of SER_{TM} at the available E_s/N_0 is 0.143 in the absence of ranging; when ranging is present with $m_{RG}=0.222$ rad with $R_{RG} \approx R_{TM}$, the overall loss L_{TM} ranges from 0.134 at $SER_{TM}=0.1$ to 0.126 at $SER_{TM}=0.2$ (see table 3-6). Then the real SER_{TM} ranges from 0.146 to 0.149, and can be assumed equal to 0.149 (worst case). Performing the same interpolation explained above, the telemetry loss can be estimated equal to 0.129 dB, made of 0.11 dB of modulation loss L_{TM-mod} (see table 3-5), 0.011 dB of GMSK loss with respect to BPSK (see table 3-3), and 0.008 dB loss due to ranging interference. Therefore a useful $E_b/N_0=5.34-0.129=5.210$ dB is obtained and the margin with respect to the required value 1.74 is 3.470 dB, larger than the 3-dB requirement. The ranging losses must be evaluated for $SER_{TM}=0.149$, interpolating between the values given in the tables.

3.4.6.2 Ranging Performance

3.4.6.2.1 Acquisition Time

The difference in dB between the value of N_{chip} estimated through simulation (taking into consideration the telemetry interference and the imperfect synchronization) and the value obtained by the equation (11) is called $L_{\text{RG-int-acq}}$ and is shown in table 3-30 for L2-M and table 3-27 for Mars-M for the various SER_{TM} values.

In the L2-M link budget, for PN code T4B, equation (7) allows finding the number of correlation chips in the absence of interference from telemetry as

$$10 \log_{10} N_{\text{chip}} = \eta_{\text{dB}} - 10 \log_{10} (E_{\text{RG}} / N_0) = \eta_{\text{dB}} - 10 \log_{10} (P_{\text{T}} / N_0) + L_{\text{RG-mod,dB}} + 10 \log_{10} R_{\text{RG}},$$

with $\eta_{\text{dB}}=36.34$ (T4B), ranging modulation loss $L_{\text{RG-mod,dB}}=16.13$ dB as in table 3-26, $P_{\text{T}}/N_0=69.3$ dB, $10 \log_{10} R_{\text{RG}}=65.36$ dB ($R_{\text{RG}}=3.435$ Mc/s), which yields the value

$$10 \log_{10} N_{\text{chip}} = 36.34 - 69.3 + 16.13 + 65.36 = 48.53 \text{ dB}.$$

The corresponding acquisition time is approximately 20.7 ms. When telemetry is present, a loss $L_{\text{RG-int-acq}}$ must be included, corresponding to $\text{SER}_{\text{TM}}=0.017$; the loss at $\text{SER}_{\text{TM}}=0.01$ and at $\text{SER}_{\text{TM}}=0.1$ can be interpolated from table 3-30.

Performing linear interpolation, but with a logarithmic scale (i.e., interpolating between $\log_{10}(0.01)$ with loss 0.04 dB and $\log_{10}(0.1)$ with loss 2.15 dB), and evaluating the loss at the intermediate point $\log_{10}(0.017)$ yields:

$$L_{\text{RG-int-acq}} = (0.04 - 2.15) / (\log_{10}(0.01) - \log_{10}(0.1)) * (\log_{10}(0.017) - \log_{10}(0.01)) + 0.01 = 0.526 \text{ dB}.$$

Therefore, for the simultaneous transmission of telemetry and ranging,

$10 \log_{10} N_{\text{chip}} = 48.53 + 0.526 = 49.056$ dB, which corresponds to an acquisition time $t_{\text{obs}} = N_{\text{chip}} T_{\text{c}}$ equal to approximately 23.4 ms.

In the Mars-M case, $\eta_{\text{dB}}=24.17$ (T2B), $P_{\text{T}}/N_0=57.1$ dB, $L_{\text{RG-mod,dB}}=16.13$ dB,

$10 \log_{10} R_{\text{RG}}=59.55$ dB ($R_{\text{RG}}=0.903$ Mc/s), so that in the absence of telemetry:

$$10 \log_{10} N_{\text{chip}} = 24.17 - 57.1 + 16.13 + 59.55 = 42.75 \text{ dB}.$$

The corresponding acquisition time is approximately 20.9 ms. The ranging acquisition loss corresponding to $\text{SER}_{\text{TM}}=0.149$ obtained through interpolation from table 3-27 is 4.36 dB.

Therefore, for the simultaneous transmission of telemetry and ranging,

$$10 \log_{10} N_{\text{chip}} = 42.75 + 4.36 = 47.11 \text{ dB},$$

which corresponds to an acquisition time t_{obs} equal to approximately 57 ms.

3.4.6.2.2 Jitter

The normalized ranging jitter standard deviation in the absence of telemetry can be obtained from equation (16) as

$$\sigma_\varepsilon^2 = \frac{B_L T_c}{\pi^2 L_{ck} \frac{P_T}{N_0} 2J_1^2(m_{RG}) T_c} = \frac{B_L}{\pi^2 L_{ck} \frac{P_T}{N_0} L_{RG-mod}}$$

or

$$10 \log_{10} \sigma_\varepsilon^2 = 10 \log_{10} B_L - 10 \log_{10} \pi^2 + L_{ck,dB} - 10 \log_{10} \frac{P_T}{N_0} + L_{RG-mod,dB}.$$

For the L2-M mission, $B_L=5$ Hz, $B_L T_c=1.45 \cdot 10^{-6}$, $L_{ck}=0.881$ (code T4B), $P_T/N_0=69.3$ dB, $L_{RG-mod,dB}=16.13$ dB. If no telemetry were present, then the ranging jitter variance could be computed as:

$$10 \log_{10} \sigma_\varepsilon^2 = 6.99 - 9.9430 + 0.55 - 69.3 + 16.131 = -55.572 \text{ dB},$$

and $\sigma_\varepsilon^2 = 2.77 \cdot 10^{-6}$.

The loss $L_{RG-int-jit}$ due to telemetry can be obtained by interpolating the losses given in table 3-41 ($m_{RG}=0.222$) for $SER_{TM}=0.017$.

Performing linear interpolation with a logarithmic scale (i.e., interpolating between $\log_{10}(0.01)$ with loss -0.03 dB and $\log_{10}(0.1)$ with loss 1.06 dB) and evaluating the loss at the intermediate point $\log_{10}(0.017)$ yields:

$$L_{RG-int-jit} = (-0.03 - 1.06) / (\log_{10}(0.01) - \log_{10}(0.1)) * (\log_{10}(0.017) - \log_{10}(0.01)) + 0.3 = 0.22 \text{ dB},$$

which corresponds to an increase of approximately 1.05 times.

For simultaneous transmission of ranging and telemetry, the normalized ranging jitter variance is therefore approximately $\sigma_\varepsilon^2 = 1.05 \times 2.77 \cdot 10^{-6} = 2.91 \cdot 10^{-6}$.

The normalized jitter standard deviation is then $\sigma_\varepsilon = 1.70 \cdot 10^{-3}$, and the jitter standard deviation is $\sigma_\tau = 0.311$ ns; finally, the estimated range standard deviation is $\sigma_d = \sigma_\tau c / 2 = 46$ mm.

For the Mars-M mission, $B_L=5$ Hz, $B_L T_c=5.53 \cdot 10^{-6}$, $L_{ck}=0.3936$ (code T2B), $P_T/N_0=57.1$ dB, $L_{RG-mod,dB}=16.13$ dB. If no telemetry were present than the ranging jitter variance could be computed as:

$$\begin{aligned} 10 \log_{10} \sigma_\varepsilon^2 &= 10 \log_{10} B_L - 10 \log_{10} \pi^2 + L_{ck,dB} - 10 \log_{10} \frac{P_T}{N_0} + L_{RG-mod,dB} = \\ &= 6.99 - 9.9430 + 4.05 - 57.1 + 16.131 = -39.872 \text{ dB}, \\ \text{and } \sigma_\varepsilon^2 &= 1.03 \cdot 10^{-4}. \end{aligned}$$

The loss due to telemetry is obtained from table 3-38 interpolating the loss at $m_{RG}=0.222$ for $SER_{TM}=0.149$; using the same interpolation as for the L2-M mission yields $L_{RG-int-jit}=2.086$ dB (factor 1.62).

For simultaneous transmission of ranging and telemetry, the normalized ranging jitter variance is $\sigma_\epsilon^2=1.62 \times 1.03 \cdot 10^{-4}=1.66 \cdot 10^{-4}$.

The normalized jitter standard deviation is then $\sigma_\epsilon=1.29 \cdot 10^{-2}$, and the jitter standard deviation is $\sigma_\tau=14$ ns; finally, the estimated range standard deviation is $\sigma_d=\sigma_\tau c/2= 2.15$ m.

3.4.7 IMPROVED PERFORMANCES

3.4.7.1 General

The performances of subsections 3.4.3–3.4.6 are based on a rather simple and therefore suboptimal demodulator as illustrated in figure 3-3. Such scheme just demodulates the telemetry symbol stream, remodulates, and removes it from the composite receive signal prior to ranging demodulation.

Significant improvements can be obtained by performing also symbol decoding down to bit level followed by encoding and remodulation. The major drawback of this scheme is the complexity and some delay in obtaining the ranging signal. The latter may not be important since normally telemetry transmission starts at the beginning of the pass, while ranging is initiated at a later stage

An alternative from a complexity and performance point of view is to iteratively remove the ranging signal prior to telemetry demodulation (see reference [16]). The telemetry degradation due to the combined TM and ranging approach is in the order of 0.2 to 1 dB, for the recommended use cases given in reference [1] (i.e., the RG modulation index is kept below 0.45 rad and the ranging-chip-rate-to-TM-rate ratio is larger than one). This subsection introduces an extended receiver design, which regenerates the RG signal and subtracts it before TM demodulation. This approach allows extending the usage of the GMSK + PN RG to higher modulation indices and to lower RG-chip-rate-to-TM-rate ratios, while significantly reducing the overall TM loss.

3.4.7.2 Proposed Approach for PN Ranging Cancellation

The GMSK and PN ranging modulator is shown in figure 3-1. In 3.1 the demodulator approach consists of demodulating the GMSK signal directly. The PN RG signal is correlated only after canceling the estimated GMSK signal from the received signal (figure 3-3).

Figure 3-13 suggests an alternative approach that removes the RG signal before TM demodulation. This can be done once the RG correlators have locked to the RG signal and the delay of the ranging sequence with respect to the TM signal can be determined. As the RG sequence is perfectly known a priori, only the delay needs to be taken into account when regenerating the RG signal. This delay from the ranging correlator is averaged/filtered and fed to the RG regenerator. The RG regenerator creates a time-varying phase offset, like the one generated for the RG signal in the spacecraft, only with the opposite sign. This phase offset is then used to cancel the RG signal from the GMSK signal before the GMSK demodulation is performed.

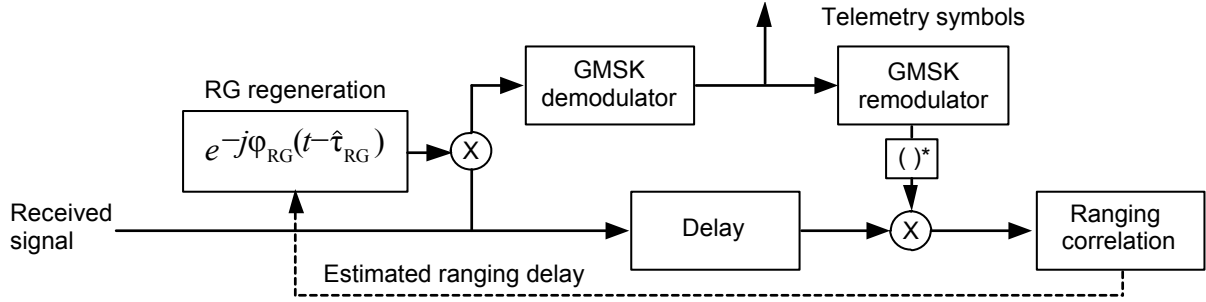


Figure 3-13: High-Level Diagram of GMSK+PN RG Receiver with Ranging Cancellation

3.4.7.3 Analytical Model of RG Cancellation

The GMSK + PN RG signal can be written as:

$$x(t) = \sqrt{2P_T} e^{j2\pi f_c t + j\phi_{TM}(t) + j\phi_{RG}(t - \tau_{RG})}$$

With P_T denoting the transmit power, f_c the carrier frequency, $\phi_{TM}(t)$ the phase of the GMSK modulated TM signal, and $\phi_{RG}(t - \tau_{TM})$ the phase of the RG signal with a time offset τ_{TM} with respect to the TM signal. The generation of the phase signals $\phi_{TM}(t)$ and $\phi_{RG}(t - \tau_{TM})$ is described in 3.1.

Using an Additive White Gaussian Noise (AWGN) channel, the received signal $y(t)$ is:

$$y(t) = \sqrt{2P_T} e^{j2\pi f_c t + j\phi_{TM}(t) + j\phi_{RG}(t - \tau_{RG})} + n(t)$$

The approach in 3.1 for TM demodulation at the receiver currently assumes that $\phi_{RG}(t - \tau_{TM})$ is negligible with respect to $\phi_{TM}(t)$ and therefore directly proceeds with the TM demodulation process.

Using instead the approach of RG regeneration and cancellation shown in figure 3-13, the signal becomes

$$\begin{aligned} y_{RG \text{ cancel}}(t) &= \left(\sqrt{2P_T} e^{j2\pi f_c t + j\phi_{TM}(t) + j\phi_{RG}(t - \tau_{RG})} + n(t) \right) \cdot e^{-j\phi_{RG}(t - \hat{\tau}_{RG})} \\ &= \sqrt{2P_T} e^{j2\pi f_c t + j\phi_{TM}(t) + j\phi_{RG}(t - \tau_{RG}) - j\phi_{RG}(t - \hat{\tau}_{RG})} + \underbrace{n(t) \cdot e^{-j\phi_{RG}(t - \hat{\tau}_{RG})}}_{\hat{n}(t)} \\ &= \sqrt{2P_T} e^{j2\pi f_c t + j\phi_{TM}(t) + j\phi_{RG}(t - \tau_{RG}) - j\phi_{RG}(t - \hat{\tau}_{RG})} + \hat{n}(t) \end{aligned}$$

With $\hat{\tau}_{\text{RG}}$ denoting the estimated ranging delay and $\hat{n}(t)$ representing the rotated noise. If the receiver achieves a perfect RG synchronization ($\hat{\tau}_{\text{RG}} = \tau_{\text{RG}}$), one obtains

$$y_{\text{RG cancel}}(t) = \sqrt{2P_T} e^{j2\pi f_c t + j\phi_{\text{TM}}(t)} + \hat{n}(t)$$

The *rotation* of the noise term does not affect the performance of the TM demodulation, as the mean and the variance of $\hat{n}(t)$ is identical to $n(t)$. Therefore, with this simple RG cancellation process, the TM degradation due to the simultaneous GMSK + PN RG approach is completely eliminated. Only in the case of imperfect RG synchronization during the cancellation process, some degradation could occur from the phase rotation introduced by $j\phi_{\text{RG}}(t - \tau_{\text{RG}}) - j\phi_{\text{RG}}(t - \hat{\tau}_{\text{RG}})$.

Subsection 3.4.7.4 presents some simulation results for typical values of the ranging cancellation timing error normalized to the RG chip interval $\hat{\tau}_{\text{RG}_n} = \frac{\hat{\tau}_{\text{RG}} - \tau_{\text{RG}}}{T_C}$.

3.4.7.4 Simulation Results

A GMSK + PN RG system was simulated including the RG cancellation approach presented in figure 3-13. The TM losses were calculated with respect to a perfect BPSK signal and (in brackets) with respect to a pure GMSK TM signal (i.e., without adding simultaneous PN RG). These losses therefore show the TM degradation for the simultaneous high-rate TM with PN RG, when using RG cancellation in the receiver. The difference of the two reported losses is due to the GMSK inherent losses as given in table 3-3.

Because of the variety of simulation cases, the following reference case was chosen (which, when not using RG cancellation, has non-negligible TM losses):

- $BT_s=0.25$ with T2B;
- $m_{\text{RG}} = 0.222, 0.444$, and 0.666 rad (for $m_{\text{RG}} = 0.111$ rad the TM losses are already very low, and therefore RG cancellation is typically not required, but if applied better values than for $m_{\text{RG}} = 0.222$ can be expected);
- ranging chip rate R_{RG} to TM rate R_{TM} ratios of $1/3$ [and ≈ 1];
- SER_{TM} (TM Symbol Error Rates) of $0.01, 0.1$, and 0.3 .

Furthermore, some simulations were performed with changing the parameters BT_s to 0.5 , using T4B and choosing $R_{\text{RG}} \approx 3 R_{\text{TM}}$ (all performed with the worst case modulation index $m_{\text{RG}} = 0.666$).

Simulations include the offset $\hat{\tau}_{\text{RG}_n}$ (ranging cancellation timing error normalized to the RG rate) in the RG cancellation, to simulate imperfect ranging cancellation. This parameter is

depending on the technical receiver implementation of the RG canceler. For the simulations this offset combines the errors that lead to an offset of the RG cancellation (e.g., due to sampling, jitter, filtering, Doppler, etc.) to a simple offset. The TM degradation resulting from this offset gives a good indication of how well the RG cancellation has to be aligned to the transmitted RG signal to maintain a low TM-degradation value. Apart from the perfect synchronization of the delay ($\hat{\tau}_{\text{RG}_n} = 0$), two realistic values ($\hat{\tau}_{\text{RG}_n} = 5$ percent and 10 percent) and a high-delay value were simulated ($\hat{\tau}_{\text{RG}_n} = 20$ percent).

Table 3-51: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{\text{TM}} = 0.99$, $R_{\text{RG}} = 1$, $m_{\text{RG}} = 0.222$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{\text{RG}_n} = 0$	$\hat{\tau}_{\text{RG}_n} = 5\%$	$\hat{\tau}_{\text{RG}_n} = 10\%$	$\hat{\tau}_{\text{RG}_n} = 20\%$
0.01	0.28 (0.00)	0.28 (0.00)	0.29 (0.01)	0.33 (0.05)
0.1	0.16 (-0.01)	0.16 (0.00)	0.17 (0.01)	0.20 (0.04)
0.3	0.18 (0.01)	0.18 (0.01)	0.19 (0.02)	0.22 (0.05)

Table 3-52: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{\text{TM}} = 0.99$, $R_{\text{RG}} = 1$, $m_{\text{RG}} = 0.444$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{\text{RG}_n} = 0$	$\hat{\tau}_{\text{RG}_n} = 5\%$	$\hat{\tau}_{\text{RG}_n} = 10\%$	$\hat{\tau}_{\text{RG}_n} = 20\%$
0.01	0.28 (0.00)	0.30 (0.02)	0.34 (0.06)	0.49 (0.21)
0.1	0.16 (-0.01)	0.17 (0.00)	0.20 (0.04)	0.33 (0.17)
0.3	0.18 (0.02)	0.20 (0.03)	0.23 (0.06)	0.35 (0.19)

Table 3-53: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{\text{TM}} = 0.99$, $R_{\text{RG}} = 1$, $m_{\text{RG}} = 0.666$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{\text{RG}_n} = 0$	$\hat{\tau}_{\text{RG}_n} = 5\%$	$\hat{\tau}_{\text{RG}_n} = 10\%$	$\hat{\tau}_{\text{RG}_n} = 20\%$
0.01	0.28 (0.00)	0.31 (0.03)	0.40 (0.12)	0.75 (0.47)
0.1	0.16 (-0.01)	0.18 (0.02)	0.26 (0.10)	0.55 (0.39)
0.3	0.18 (0.01)	0.21 (0.04)	0.28 (0.11)	0.56 (0.39)

Table 3-54: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 1 / 3$, $m_{RG} = 0.222$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.28 (0.00)	0.29 (0.01)	0.32 (0.04)	0.43 (0.15)
0.1	0.16 (0.00)	0.16 (0.00)	0.17 (0.01)	0.23 (0.07)
0.3	0.17 (0.00)	0.17 (0.01)	0.18 (0.02)	0.23 (0.06)

Table 3-55: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 1 / 3$, $m_{RG} = 0.444$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.29 (0.01)	0.33 (0.05)	0.45 (0.17)	0.87 (0.59)
0.1	0.16 (0.00)	0.17 (0.01)	0.23 (0.07)	0.44 (0.28)
0.3	0.16 (-0.01)	0.18 (0.01)	0.22 (0.05)	0.36 (0.19)

Table 3-56: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 1 / 3$, $m_{RG} = 0.666$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.28 (0.01)	0.38 (0.10)	0.64 (0.36)	1.53 (1.25)
0.1	0.16 (0.00)	0.20 (0.04)	0.33 (0.17)	0.79 (0.63)
0.3	0.17 (-0.01)	0.19 (0.03)	0.28 (0.12)	0.61 (0.44)

Table 3-57: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T4B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 1 / 3$, $m_{RG} = 0.666$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.29 (0.01)	0.39 (0.11)	0.69 (0.41)	1.70 (1.42)
0.1	0.15 (-0.01)	0.20 (0.04)	0.34 (0.18)	0.86 (0.70)
0.3	0.16 (-0.01)	0.19 (0.03)	0.28 (0.12)	0.64 (0.47)

Table 3-58: TM Loss When Using PN RG Cancellation; $BT_s = 0.5$, PN(T2B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 1 / 3$, $m_{RG} = 0.666$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.02 (0.00)	0.08 (0.06)	0.27 (0.25)	0.93 (0.91)
0.1	0.01 (0.00)	0.04 (0.03)	0.15 (0.14)	0.55 (0.54)
0.3	0.04 (0.00)	0.07 (0.03)	0.16 (0.11)	0.46 (0.42)

Table 3-59: TM Loss When Using PN RG Cancellation; $BT_s = 0.25$, PN(T2B), Sinusoidal Pulse, $R_{TM} = 0.99$, $R_{RG} = 3$, $m_{RG} = 0.666$ rad

	Loss (dB) with respect to ideal BPSK (ideal GMSK)			
Target SER_{TM}	$\hat{\tau}_{RG_n} = 0$	$\hat{\tau}_{RG_n} = 5\%$	$\hat{\tau}_{RG_n} = 10\%$	$\hat{\tau}_{RG_n} = 20\%$
0.01	0.28 (0.00)	0.30 (0.02)	0.37 (0.09)	0.63 (0.35)
0.1	0.17 (0.01)	0.19 (0.03)	0.26 (0.10)	0.52 (0.36)
0.3	0.18 (0.02)	0.20 (0.04)	0.27 (0.11)	0.55 (0.38)

For typical values of the normalized ranging cancellation error of 10 percent of the RG rate, the losses (in the following always with respect to ideal GMSK) are reduced from up to 1 dB down to < 0.07 dB for the cases recommended in the standard ($m_{RG} \leq 0.45$ rad and $R_{RG} / R_{TM} > 1$).

Because of low TM losses resulting from the RG cancellation approach, also high-rate TM with simultaneous PN RG using high modulation indices (simulated value 0.666 rad) and low PN-RG-rate-to-TM-rate ratios can be used (simulated value 1/3), while still obtaining low TM losses of < 0.41 dB for SER of 0.01 and < 0.19 dB for SER of 0.1 and 0.3 (all for 10 percent or less of normalized ranging cancellation error). For a technical implementation which does not align the RG cancellation well to the transmitted RG signal (simulated case: $\hat{\tau}_{RG_n} = 20$ percent), higher losses are observed. Nevertheless, also in this case the losses remain well below the losses observed when not using the RG cancellation approach.

As the improved TM performance is only available after TM synchronization and RG correlation has been performed, the TM acquisition margin has to be chosen large enough to acquire the TM signal before RG cancellation occurs.

3.4.7.5 Summary for the Improved Performances

Subsection 3.4.7 introduces a simple RG cancellation approach that regenerates the RG signal after RG correlation and subtracts it from the received signal before TM demodulation takes place. As the RG signal sequence is known a priori, the RG signal regeneration is very simple and just requires obtaining the RG offset with respect to the TM signal at the point of the RG cancellation. For an ideal RG cancellation, it is shown that the TM degradation is completely eliminated and the TM SER performance of the combined GMSK + PN RG approach is equivalent to a standard GMSK modulation (without RG).

Even for an imperfect RG cancellation, it is shown that the low TM losses introduced by using the combined GMSK + PN RG approach can be further reduced to a nearly negligible level for the cases recommended in the standard. Furthermore with the proposed RG cancellation, much lower RG-to-TM-rate ratios and higher RG modulation indices than recommended in the standard can be used, while maintaining a low TM degradation.

4 CONCLUSIONS

With more missions at high data rates demanding use of limited spectral resources for space-to-Earth telemetry in a power-limited scenario, the CCSDS approved recommendations 401 (2.4.22A) B-1, and 401 (2.14.22B) B-1 address the use of bandwidth-efficient modulations for high-data-rate missions (when the telemetry symbol rates exceed 2 Msymbol/s) and simultaneous ranging transmission in the SRS 8400–8500 MHz frequency allocations. Table 4-1 summarizes the CCSDS-recommended bandwidth-efficient modulations with simultaneous ranging and their respective frequency bands.

This Informational Report on bandwidth-efficient modulations supplements the two CCSDS recommendations on bandwidth-efficient modulations with ranging by providing technical descriptions of these modulations. The performance of these modulations is dependent on many factors, including channel coding, receiver type, channel characteristics, and transponder hardware distortions. Simulated end-to-end performance and spectral characteristics of most of the recommended modulations are provided for the selected channel model (AWGN given that the recommended modulations have constant envelope) and no hardware distortions.

Table 4-1: CCSDS Recommendations on Bandwidth-Efficient Modulations with Ranging

Frequency Band	Applicable CCSDS Recommendation(s)	Recommended Modulations
8450–8500 MHz	401 (2.4.22A) B-1	– Telemetry GMSK $BT_s=0.25$ with precoding; symbol error rate at the receiver better than 0.1. – Ranging – Regenerative PN T2B and T4B; Sinewave shaping; m_{RG} between 0.2 and 0.45 rad. – PN ranging chip rate to telemetry symbol rate ratio higher than 1 (non-integer).
8400–8450 MHz	401 (2.4.22B) B-1	– Telemetry GMSK $BT_s=0.5$ with precoding; symbol error rate at the receiver better than 0.1. – Ranging – Regenerative PN T2B and T4B; Sinewave shaping; m_{RG} between 0.2 and 0.45 rad. – PN ranging chip rate to telemetry symbol rate ratio higher than 1 (non-integer).

ANNEX A

SPECIAL CASES

A1 SQUAREWAVE AND SRRC SHAPING SPECTRAL PERFORMANCE

For $R_{RG}=R_{TM}$ the squarewave-shaped spectra are completely outside the SFCG mask (references [3] and [6]). As per figures A-1–A-4, the squarewave-shaped spectra do not have an appreciable roll-off even for a modulation index, m_{RG} , as small as 0.222.

The simulations (reference [5]) showed that with SRRC filtering of PN ranging, the composite spectrum would meet the SFCG mask at the expense of an additional degradation with respect to both sinewave and squarewave shaping in terms of telemetry and ranging performance.

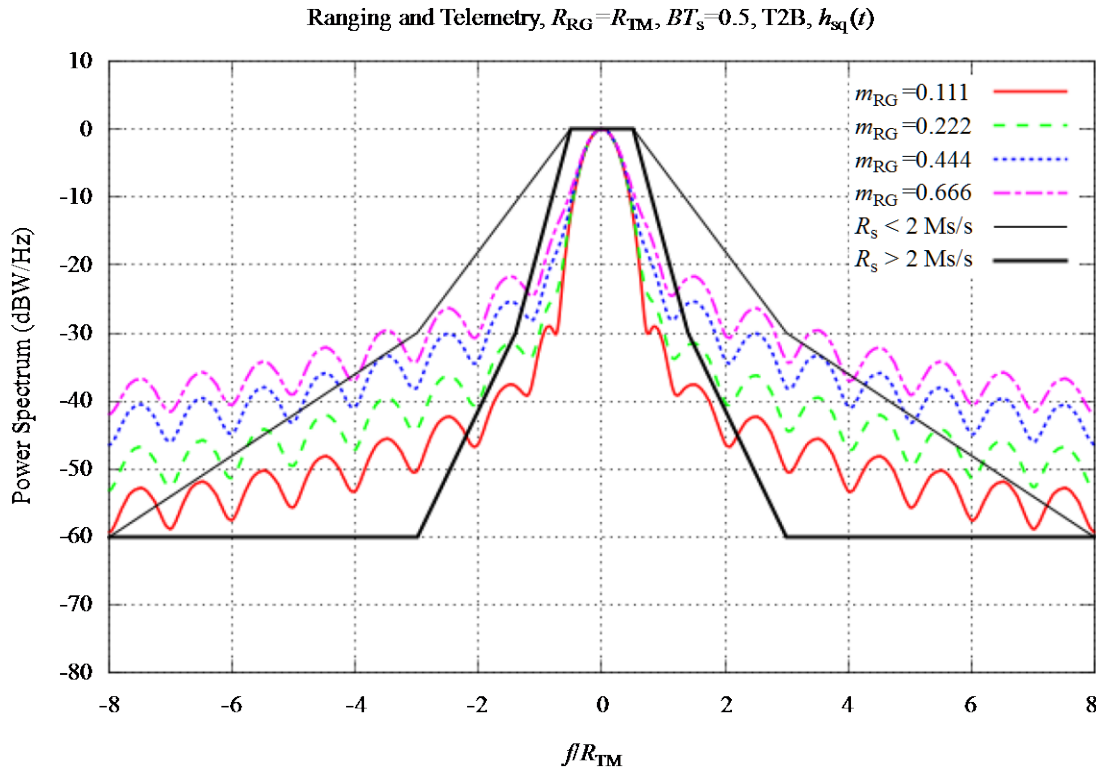
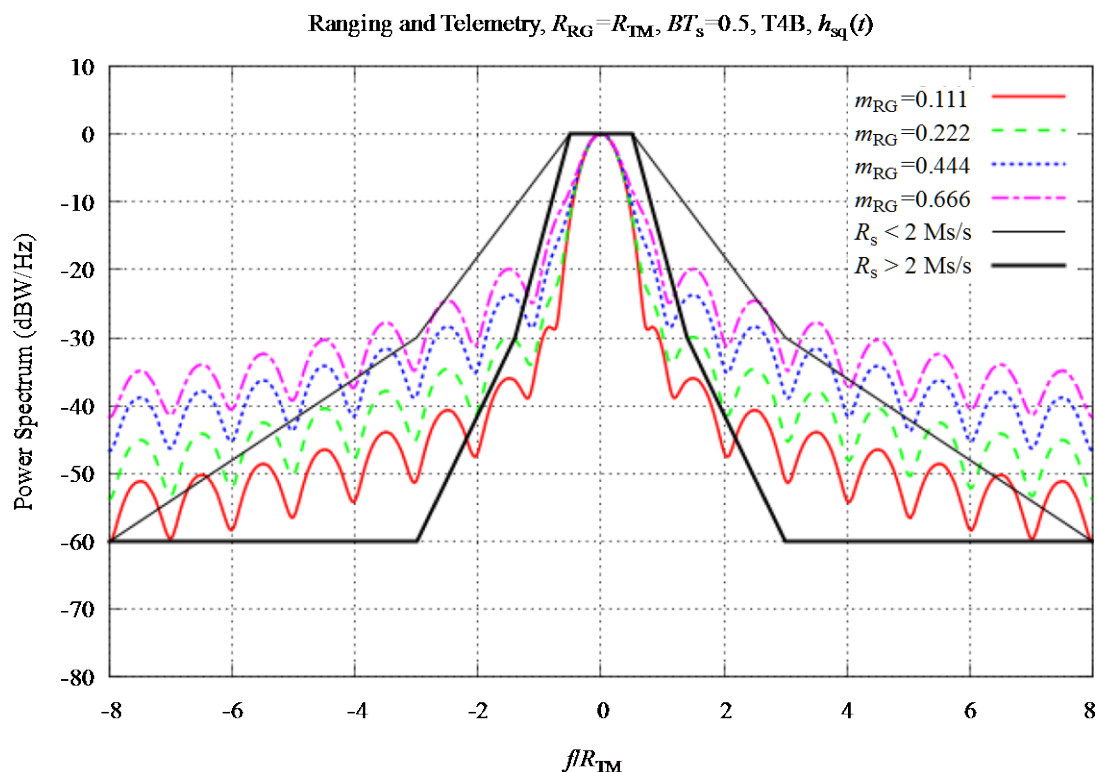
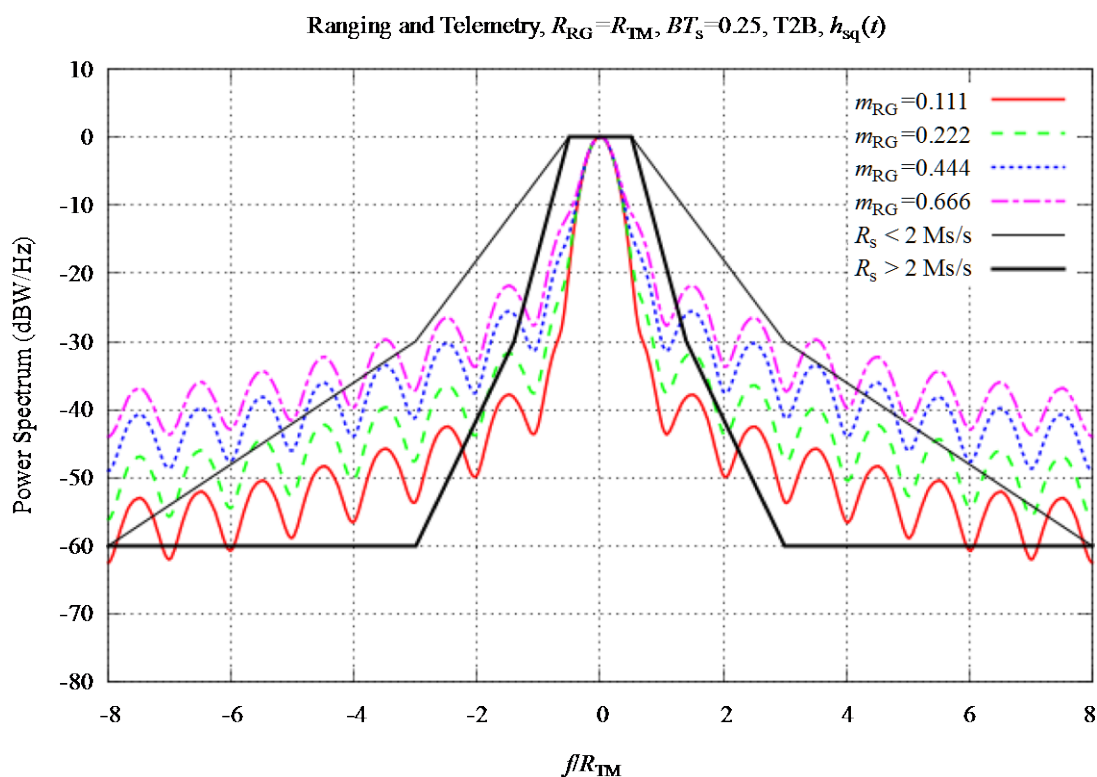


Figure A-1: GMSK($BT_s=0.5$)/PN(Square) Spectral Plots for Code T2B, $R_{RG}=R_{TM}$

Figure A-2: GMSK($BT_s=0.5$)/PN(Square) Spectral Plots for Code T4B, $R_{RG}=R_{TM}$ Figure A-3: GMSK($BT_s=0.25$)/PN(Square) Spectral Plots for Code T2B, $R_{RG}=R_{TM}$

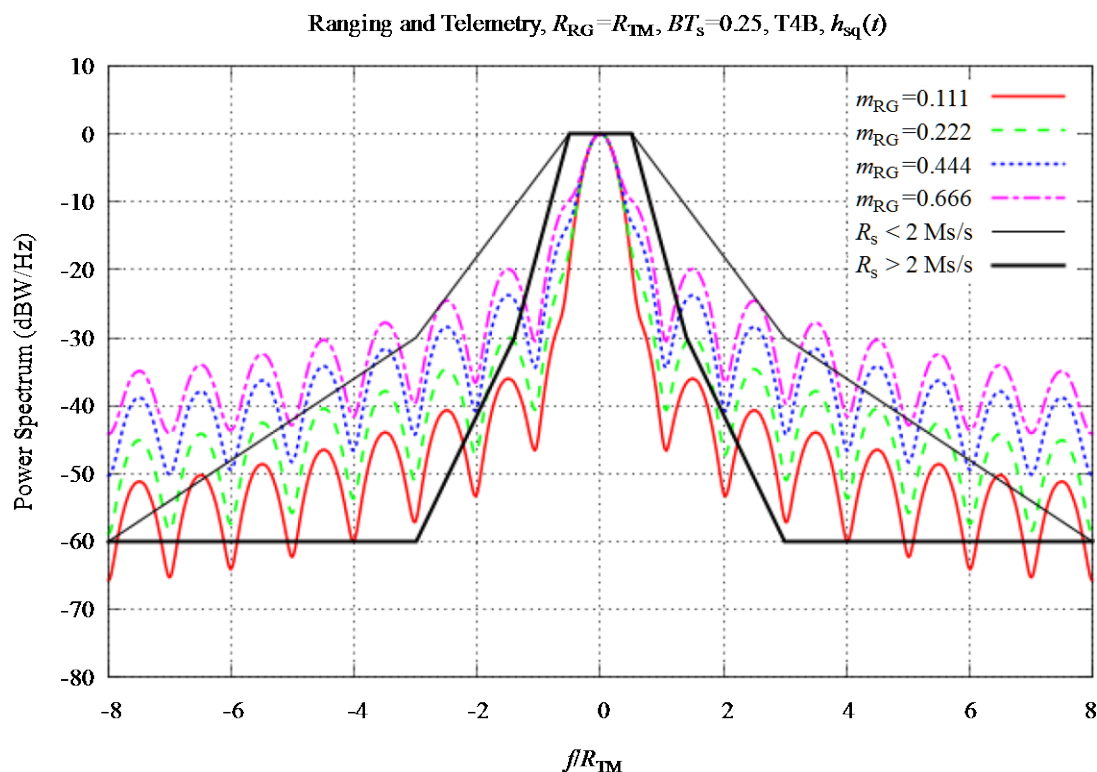


Figure A-4: GMSK($BT_s=0.25$)/PN(Square) Spectral Plots for Code T4B, $R_{RG}=R_{TM}$

Tables A-1 and A-2 show the measured normalized bandwidth B_x/R_{TM} for the squarewave cases with $R_{RG}=R_{TM}$ and $R_{RG}=3R_{TM}$.

It can be seen that the occupied bandwidth, when squarewave shaping is selected, can be much larger than that obtained with sinewave shaping if the ranging modulation index is large.

Table A-1: Normalized Occupied Bandwidth (99-Percent Power) for $R_{RG}=R_{TM}$, PN(square)

system	$R_{RG} = R_{TM}, h_{sq}(t)$			
	GMSK $BT_s = 0.5$		GMSK $BT_s = 0.25$	
	T2B	T4B	T2B	T4B
$m_{RG} = 0.111$	1.069	1.080	0.899	0.913
$m_{RG} = 0.222$	1.184	1.264	1.050	1.162
$m_{RG} = 0.444$	2.313	2.966	2.356	2.973
$m_{RG} = 0.666$	3.825	4.944	3.726	4.956

Table A-2: Normalized Occupied Bandwidth (99-Percent Power) for $R_{RG} = 3R_{TM}$, PN(square)

system	$R_{RG} = 3R_{TM}, h_{sq}(t)$			
	GMSK $BT_s = 0.5$		GMSK $BT_s = 0.25$	
	T2B	T4B	T2B	T4B
$m_{RG} = 0.111$	1.110	1.128	0.932	0.944
$m_{RG} = 0.222$	2.797	3.122	2.825	3.101
$m_{RG} = 0.444$	4.212	8.380	4.169	8.492
$m_{RG} = 0.666$	9.183	9.468	9.155	9.384

A2 SYNCHRONIZED TELEMETRY AND RANGING AT FIXED TELEMETRY SYMBOL ERROR RATE

A2.1 TELEMETRY PERFORMANCE

This subsection is devoted to a system designed to provide a telemetry symbol error rate equal to 0.1, which is consistent with the concatenated (rate 1/2.29) and LDPC (rate 1/2) coding schemes (reference [9]) that guarantee an actual bit error rate of better than 10^{-6} after decoding.

The telemetry degradation due to ranging interference and the PN ranging degradation due to telemetry interference were evaluated in reference [3] for the $R_{TM} = R_{RG}$ case and $\tau_{TM} = \tau_{RG} = 0$ (for $R_{TM} \approx R_{RG}$ and $R_{TM} \neq R_{RG}$, see A3).

It can be seen in reference [3] that the telemetry degradation does not depend on the shaping of the PN ranging up to a modulation index $m_{RG} = 0.444$, for both T2B and T4B. This is also visible in table A-3 where the telemetry loss is defined as the difference in dB between the signal-to-noise ratio $P_T T_s / N_0$ necessary to obtain $SER_{TM} = 0.1$ in the presence of ranging, and the value $E_s / N_0 = -0.84$ dB for $BT_s = 0.5$, $E_s / N_0 = -0.698$ dB for $BT_s = 0.25$, necessary to obtain the same SER_{TM} for a GMSK system in the absence of ranging (the telemetry loss thus includes the modulation loss because of the fact that part of the power is used to transmit ranging, which is equal to 0.107 dB for $m_{RG} = 0.222$ and 0.434 dB for $m_{RG} = 0.444$) (see equation (4) in 3.4.3). The difference between T2B and T4B is of the order of 0.1 dB (T4B has lower losses). The loss basically depends only on the selected value of the modulation index and is the same for both GMSK schemes considered.

It has to be noted that the results in reference [3] were obtained with ideal synchronization, as in reference [6].

The losses due to the synchronizers are negligible for normalized loop bandwidths of 10^{-4} when $R_{RG} = R_{TM}$ (synchronized telemetry and ranging signals).

Table A-3: Telemetry Losses (in dB) for $R_{RG}=R_{TM}$

system	code T2B $m_{RG} = 0.222$	code T2B $m_{RG} = 0.444$	code T4B $m_{RG} = 0.222$	code T4B $m_{RG} = 0.444$
GMSK $BT_s = 0.5, h_{sq}(t)$	0.12	0.50	0.10	0.45
GMSK $BT_s = 0.5, h_{sin}(t)$	0.11	0.48	0.10	0.44
GMSK $BT_s = 0.25, h_{sq}(t)$	0.13	0.56	0.10	0.45
GMSK $BT_s = 0.25, h_{sin}(t)$	0.12	0.53	0.10	0.44

A2.2 RANGING PERFORMANCE

A2.2.1 General

The ranging receiver input (see figure 3-3) is equal to the received signal whose phase has been modified by subtracting the estimated telemetry phase $\varphi_{TM}(t)$. Since the telemetry symbol error rate is equal to 0.1, a relative large loss must be expected for the performance of the ranging subsystem, both in terms of acquisition time and ranging jitter. If the telemetry SER is reduced, then the ranging performance improves, as shown in 3.4.

A2.2.2 Acquisition Time

As per reference [2], the number of correlation chips (proportional to the PN ranging acquisition time via the chip interval) necessary to provide an error probability in the detection of the ranging code phase equal to 10^{-3} has been used as the relevant criterion.

The increase of the acquisition time due to telemetry (reference [3]) is given in table A-4. In the worst case, the sinewave shaping loss is only 0.3 dB higher than squarewave shaping. Comparing T4B with T2B, T4B incurs in 0.3–0.6 dB smaller losses than T2B.

Table A-4: PN Ranging Acquisition Time Increase (in dB) for $R_{RG}=R_{TM}$

system	code T2B $m_{RG} = 0.222$	code T2B $m_{RG} = 0.444$	code T4B $m_{RG} = 0.222$	code T4B $m_{RG} = 0.444$
GMSK $BT_s = 0.5, h_{sq}(t)$	2.11	2.35	1.52	1.89
GMSK $BT_s = 0.5, h_{sin}(t)$	2.13	2.47	1.73	2.15
GMSK $BT_s = 0.25, h_{sq}(t)$	2.41	2.64	1.54	1.86
GMSK $BT_s = 0.25, h_{sin}(t)$	2.37	2.83	1.71	2.19

The ranging results in reference [3] were based on obtaining the chip error rate by simulations and computing the acquisition time by means of a formula with the chip error rate as input (see A3.4.1). Simulations (reference [4]) encompassing a bank of 76 correlators (reference [2]) were done to justify the simplified results. Such simulations indicated that the

correlation time estimated in reference [3] by the approximated formula are within ± 1 dB from the values obtained with a real correlator.

Additionally, the results in reference [3] were obtained with ideal synchronization as in reference [6]. However, simulations carried out with realistic synchronizers (reference [7]) have shown that the system is able to work as long as m_{RG} is kept between 0.222 (favoring telemetry) and 0.444 (favoring ranging). The losses due to the synchronizers are negligible for normalized (second order) loop bandwidths of 10^{-4} as far as $m_{RG} \geq 0.111$, while they are larger than 1.5 dB for $m_{RG} \leq 0.044$, which shows that such small values of m_{RG} should not be used in the real system.

A2.2.3 Jitter

As for the standard PN code scheme (reference [2]), code T4B has a stronger clock component than code T2B, so the ranging clock jitter variance for code T4B is 0.45 (3.5 dB) of the jitter variance for code T2B. Additionally, the squarewave shaping gives rise to a higher clock jitter variance with respect to the sinewave pulse (2.5 times higher, 4 dB) (see table 3-37).

The simulation results (reference [7]) confirmed the expected behavior, as can be seen in figures A-5, A-6, A-7, and A-8, obtained for $R_{RG} = R_{TM}$. The figures indicate the ranging jitter standard deviation normalized to the chip rate, for the case in which the chip rate is equal to the telemetry symbol rate and both rates are synchronous. The variances for sinewave, squarewave and two cases of SRRC filtering are shown in these figures. The clear advantage of sinewave shaping relative to squarewave shaping can be appreciated.

The normalized jitter standard deviation σ_ϵ is plotted versus the signal-to-noise ratio $P_T T_s / N_0$ necessary to guarantee telemetry SER equal to 0.1, for the considered values of m_{RG} ; in particular $m_{RG} = 0.111$ corresponds to the smallest $P_T T_s / N_0$, $m_{RG} = 0.666$ to the largest $P_T T_s / N_0$, and the case $m_{RG} = 0.222$ is identified by the filled circle. These results have been obtained with realistic second-order synchronizers with normalized (to the chip rate) loop bandwidths of 10^{-4} .

As m_{RG} increases, the ranging power increases and the ranging clock jitter variance decreases; when m_{RG} decreases below the threshold value $m_{RG} = 0.111$, the jitter variance dramatically increases, which reduces the interval of admissible values for m_{RG} .

In this case of perfect synchronization, the chip synchronizer is affected by the presence of a synchronized telemetry component which arises every time the detected telemetry symbol is wrong (which occurs with probability 0.1), and it shows a synchronization offset which depends on the relative delay between the chip and the telemetry symbol epochs ($\tau_{TM} - \tau_{RG}$ in equation (1)). As already stated in 3.2.5, it is, however, highly unlikely that R_{TM} and R_{RG} are exactly equal, and therefore the presence of the synchronization offset is not a real problem to be concerned about.

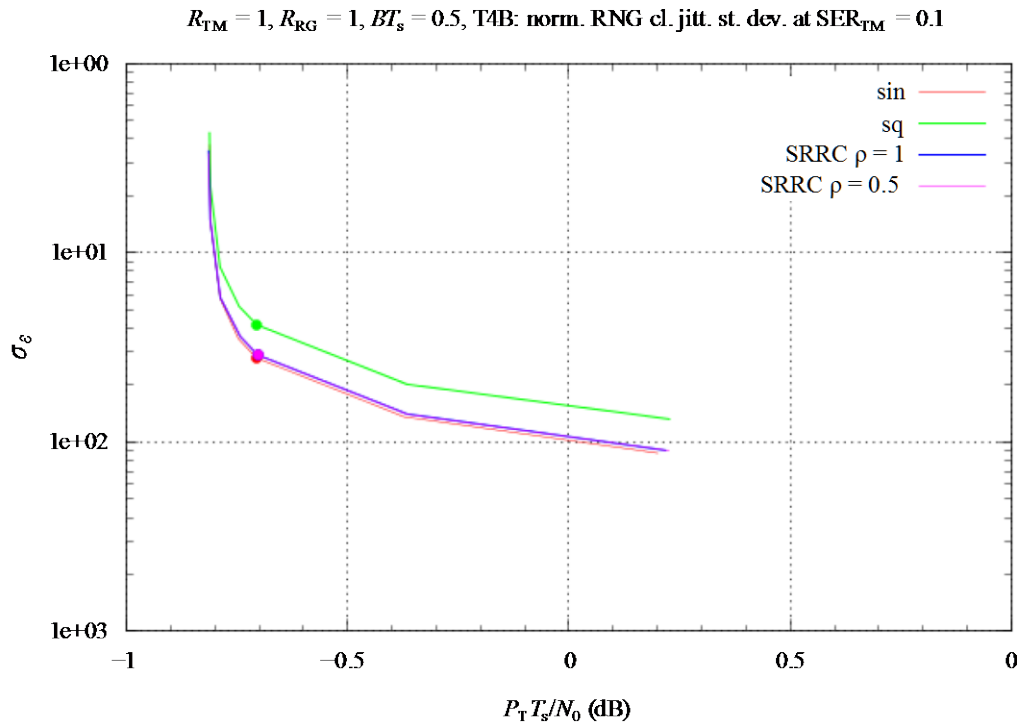


Figure A-5: Normalized Ranging Jitter Standard Deviation for GMSK($BT_s=0.5$)/PN(T4B)

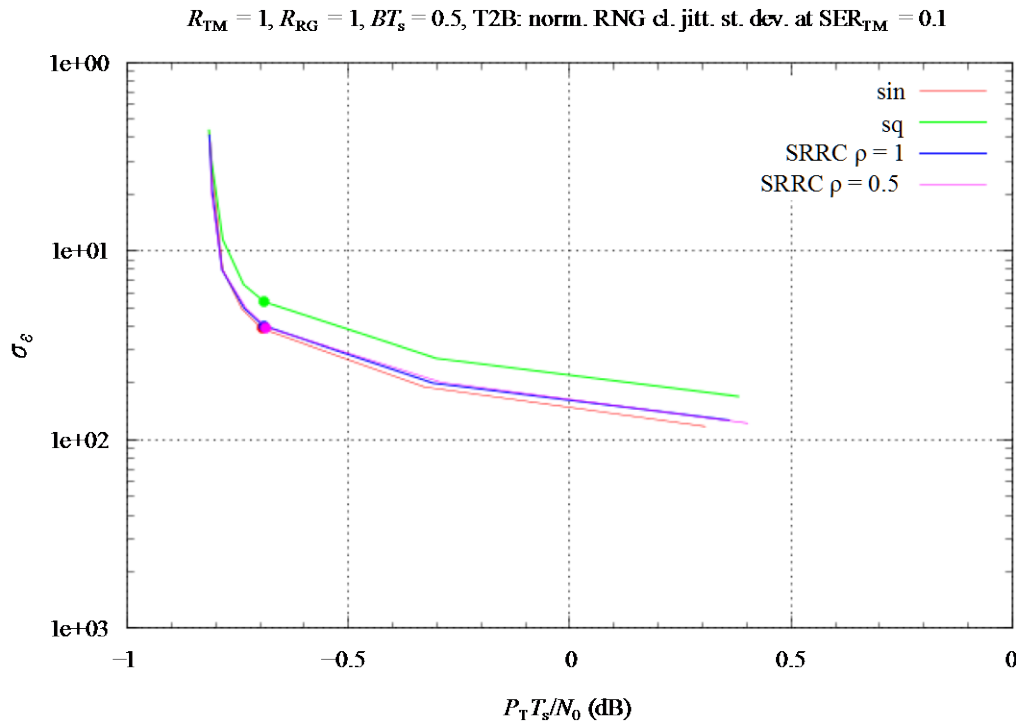


Figure A-6: Normalized Ranging Jitter Standard Deviation for GMSK($BT_s=0.5$)/PN(T2B)

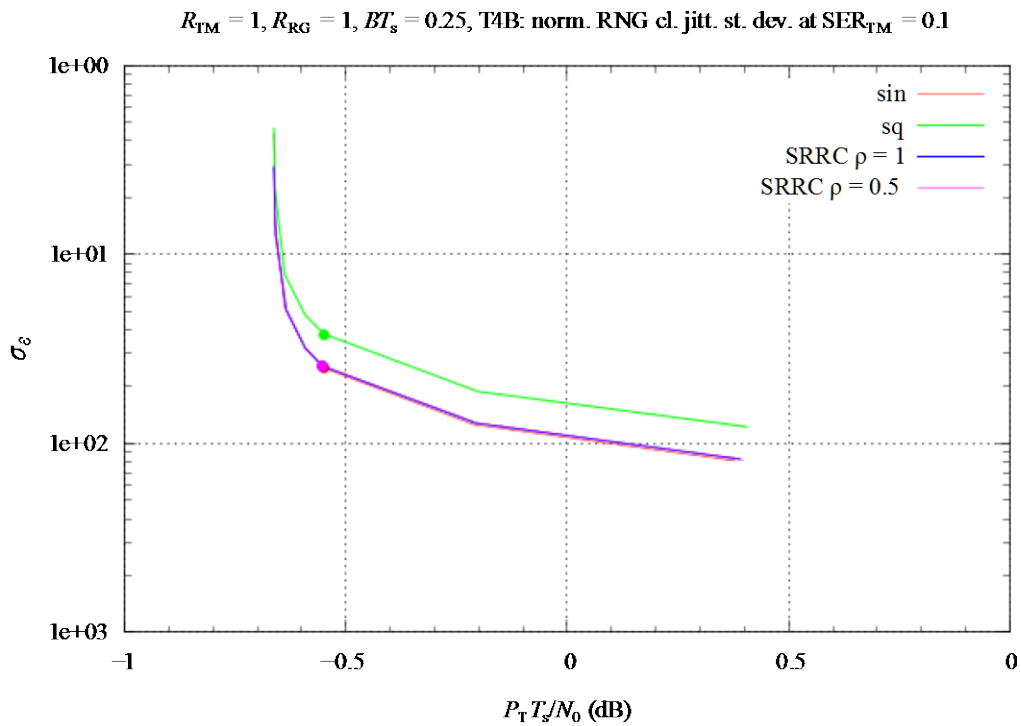


Figure A-7: Normalized Ranging Jitter Standard Deviation for GMSK($BT_s=0.25$)/PN(T4B)

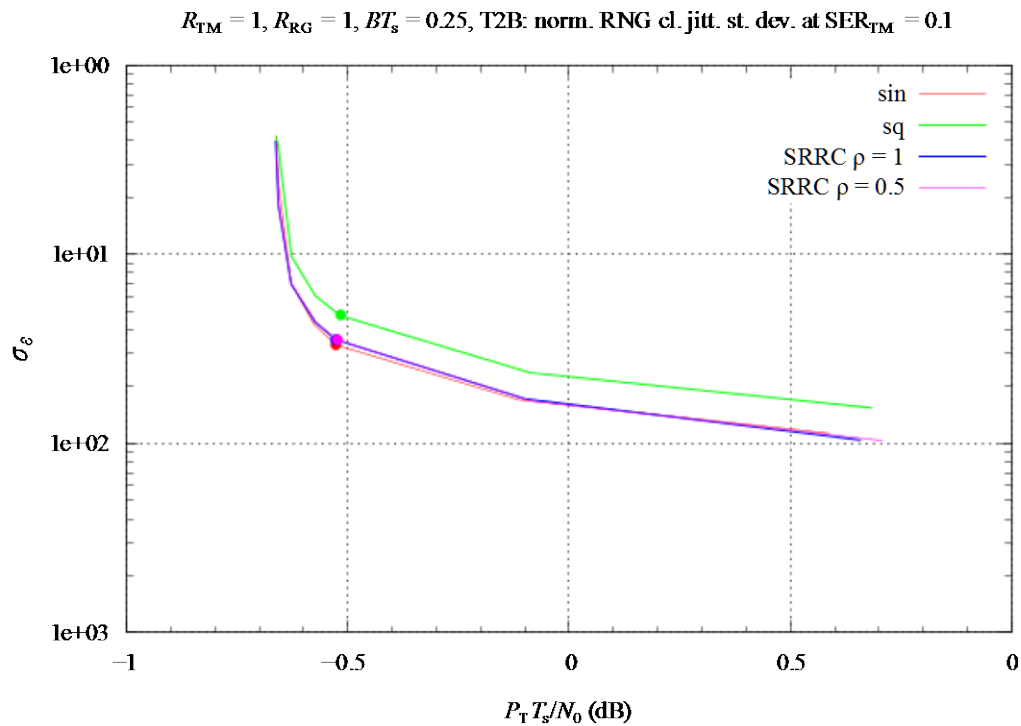


Figure A-8: Normalized Ranging Jitter Standard Deviation for GMSK($BT_s=0.25$)/PN(T2B)

A3 UNSYNCHRONIZED TELEMETRY AND RANGING WITH DIFFERENT CHIP RATE TO SYMBOL RATE RATIOS AT FIXED TELEMETRY SYMBOL ERROR RATE

A3.1 GENERAL CONSIDERATIONS

The results of A2.1 and A2.2 were related to the case in which the telemetry symbol rate R_{TM} is exactly equal to the ranging chip rate R_{RG} . The system was also analyzed for $R_{TM}=0.999$ (general unsynchronized case, defined by $R_{TM}=0.999$ instead of $R_{TM}=1$) with different values of R_{RG} , realistic synchronization, and sinewave shaping. In particular the telemetry clock and carrier phase synchronizers were simulated with loop noise equivalent bandwidths $B_L=10^{-4} R_{TM}$, while the ranging chip synchronizer was simulated with loop noise equivalent bandwidth $B_{LC}=10^{-5} R_{RG}$. In this subsection, a telemetry symbol rate equal to $R_{TM}=0.999$ Msymbol/s instead of a normalized value $R_{TM}=0.999$ is used to better appreciate ranging performance figures.

Figure A-9 shows the overall system performance of GMSK ($BT_s=0.5$)/PN (T2B): the x-axis is labeled with the value of signal-to-noise ratio $P_T T_s / N_0$ that guarantees an $SER_{TM} = 0.1$, while the y-axis is labeled with the estimated acquisition time (in seconds) necessary to obtain a 10^{-3} error probability on the PN code phase estimation. The considered values of m_{RG} are 0.111, 0.222, 0.444, and 0.666 rad. As an example, in figure A-9 each curve is defined by four points, where the leftmost corresponds to $m_{RG}=0.111$ rad and the rightmost corresponds to $m_{RG}=0.666$ rad. It can be seen that for higher values of m_{RG} , the value of $P_T T_s / N_0$ that guarantees $SER_{TM} = 0.1$ increases.

The ranging correlation (or acquisition, or observation) time is theoretically independent of the chip rate, when only ranging is transmitted. The curves shown in figure A-9 on the contrary reveal that, when telemetry is also present, the acquisition time is higher for smaller values of R_{RG} , for which also the value of $P_T T_s / N_0$ that guarantees as $SER_{TM} = 0.1$ is higher: therefore, for a similar interference from telemetry (SER_{TM} is always equal to 0.1), and in spite of an increased signal-to-noise ratio, lower values of R_{RG} correspond to longer acquisition times. For values $R_{RG}/R_{TM}>1$, however, the acquisition time becomes almost independent of the value of R_{RG} , as in the case of absence of telemetry. It can be concluded that, at least in terms of ranging acquisition times, the ranging component is almost orthogonal to the telemetry component when $R_{RG}/R_{TM}>1$. These considerations are valid for both codes T2B and T4B and for both BT_s values (see figures A-9, A-11, A-13, and A-15). Code T4B has a higher acquisition time than code T2B, and GMSK with $BT_s=0.25$ has higher acquisition times than GMSK with $BT_s=0.5$ for ratios $R_{RG}/R_{TM} < 1$. When m_{RG} decreases to 0.111 rad, then the ranging acquisition time increases, while telemetry shows a much smaller loss due to the reduced ranging interference, which accounts for the presence of a vertical asymptote on the left part of the plots at values $P_T T_s / N_0$ equal to -0.84 dB for GMSK with $BT_s=0.5$ (since $P_T T_s / N_0 = -0.84$ dB is the signal-to-noise ratio that corresponds to $SER_{TM} = 0.1$ when only telemetry is transmitted) and equal to -0.698 dB for GMSK with $BT_s=0.25$.

Figure A-10 shows the ranging jitter standard deviation σ_r (unit of measurement microseconds) for the case GMSK ($BT_b=0.5$)/PN (T2B) and for $R_{TM}=0.999$ Msymbol/s. For a given value of $P_T T_s/N_0$, the ranging jitter standard deviation obviously decreases as R_{RG} increases, being the normalized loop bandwidth $B_{LC} T_c$ constant. A vertical asymptote exists as in the corresponding plot of the acquisition time, because of the reduced power of the ranging component as m_{RG} decreases.

It must be noted that, when $R_{RG} \approx R_{TM}$ a periodic component is present in the error signal which drives the chip synchronizer at fundamental frequency $f_p = \frac{1}{2} |R_{RG} - R_{TM}|$: if the chip synchronizer single-sided loop bandwidth is larger than f_p , then the ranging jitter standard deviation plotted versus $P_T T_s/N_0$ shows a floor, a minimum value which cannot be reduced by increasing the signal-to-noise ratio (reference [11]). Similarly, when $R_{RG} \approx 2R_{TM}$, the fundamental frequency is $f_p = \frac{1}{2} |R_{RG} - 2R_{TM}|$. For higher R_{RG}/R_{TM} ratios, the effect becomes negligible (reference [13]).

With the parameters used in the simulations, being $B_{LC}=10^{-5}$ $R_{RG} < f_p = 10^{-3} R_{RG}$, the periodic component is removed and the floor is not present in the curve related to $R_{RG}/R_{TM} \approx 1$. Figures A-12, A-14, and A-16 extend the results of figure A-10 to the cases of GMSK ($BT_s=0.25$)/PN (T2B), GMSK ($BT_s=0.5$)/PN (T4B) and GMSK ($BT_s=0.25$)/PN (T4B). The results are very similar: code T4B, with a larger clock component, has a smaller jitter standard deviation with respect to code T2B, and GMSK with $BT_s=0.25$ introduces a larger penalty for $R_{RG}/R_{TM} < 1$ with respect to GMSK with $BT_s=0.5$.

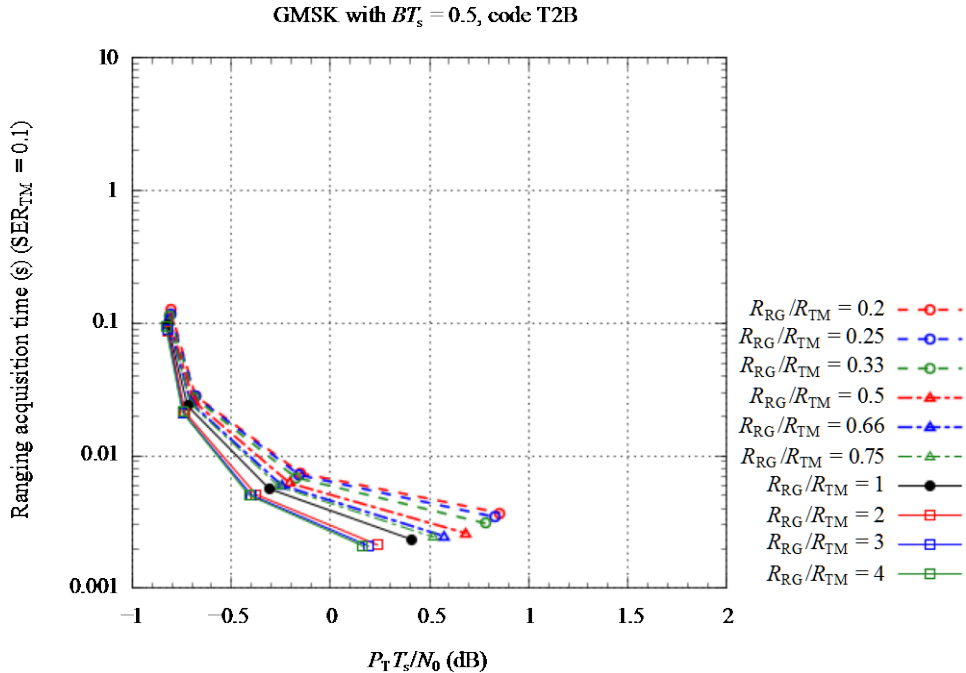


Figure A-9: Ranging Acquisition Time (s) versus $P_T T_s/N_0$ for GMSK ($BT_s=0.5$)/PN (T2B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, $B_{LC} T_c=10^{-5}$, Various R_{RG} Values

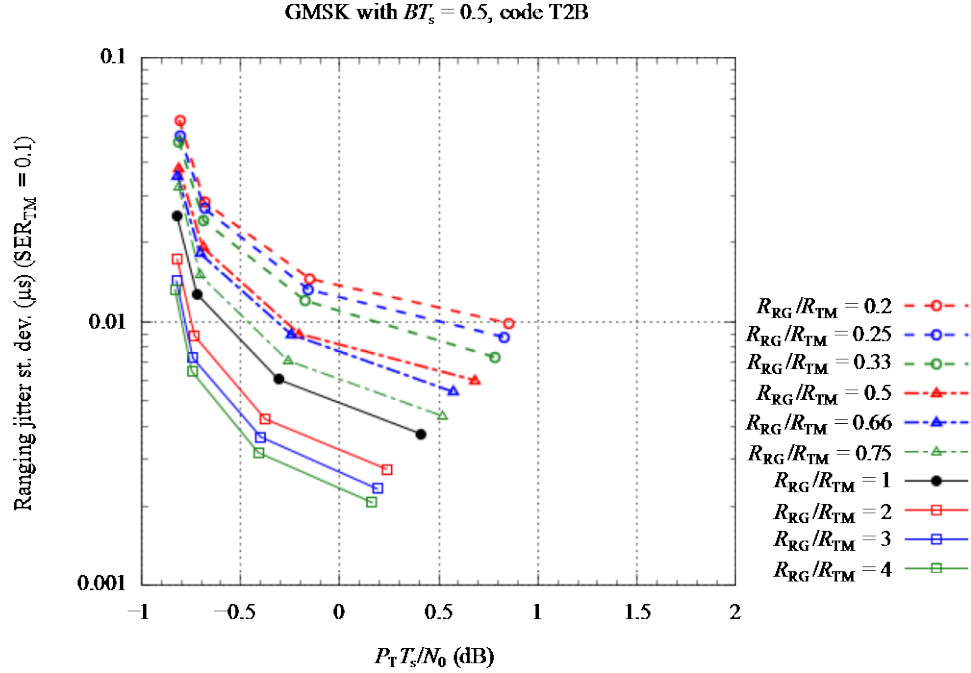


Figure A-10: Ranging Jitter Standard Deviation versus $P_T T_s / N_0$ for GMSK ($BT_s=0.5$)/PN (T2B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

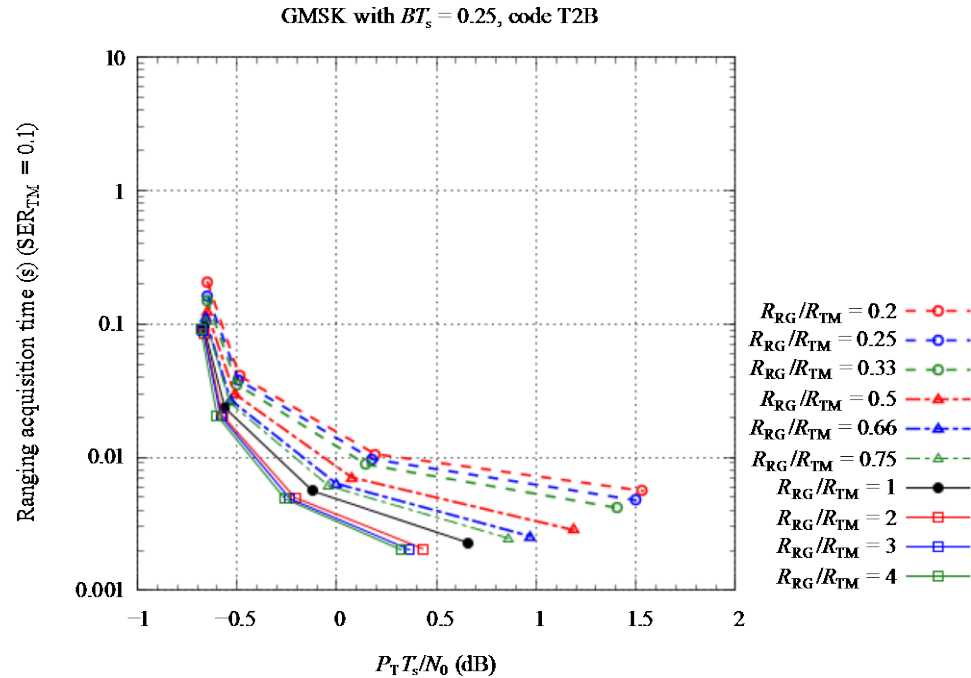


Figure A-11: Ranging Acquisition Time (s) versus $P_T T_s / N_0$ for GMSK ($BT_s=0.25$)/PN (T2B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, $B_{Lc} T_c=10^{-5}$, Various R_{RG} Values

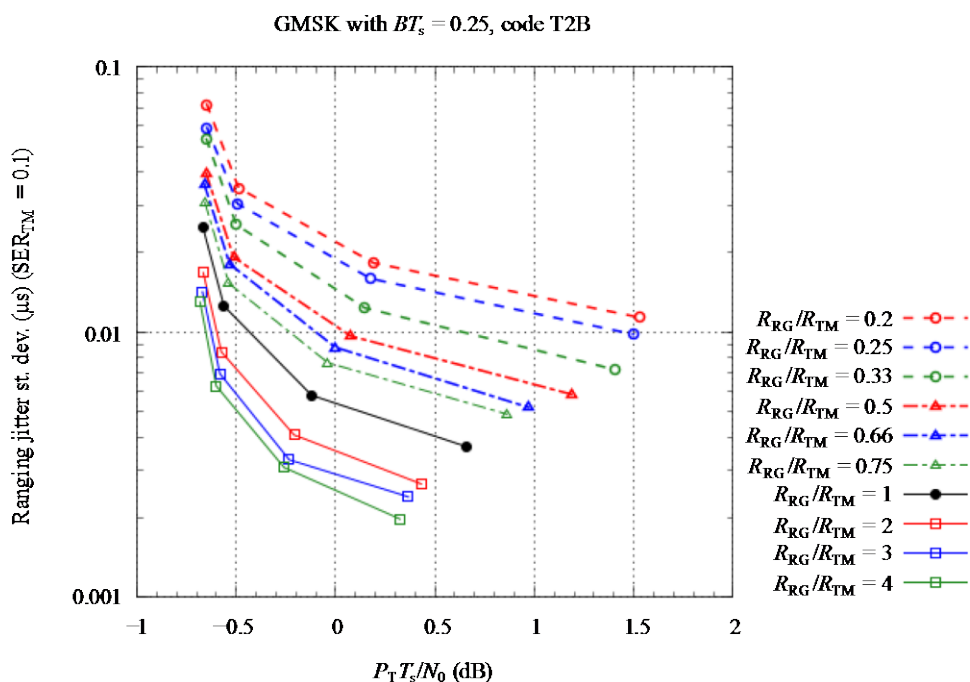


Figure A-12: Ranging Jitter Standard Deviation versus $P_T T_s / N_0$ for GMSK ($BT_s=0.25$)/PN (T2B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, Variable R_{RG} Values, $B_{Lc}T_c=10^{-5}$

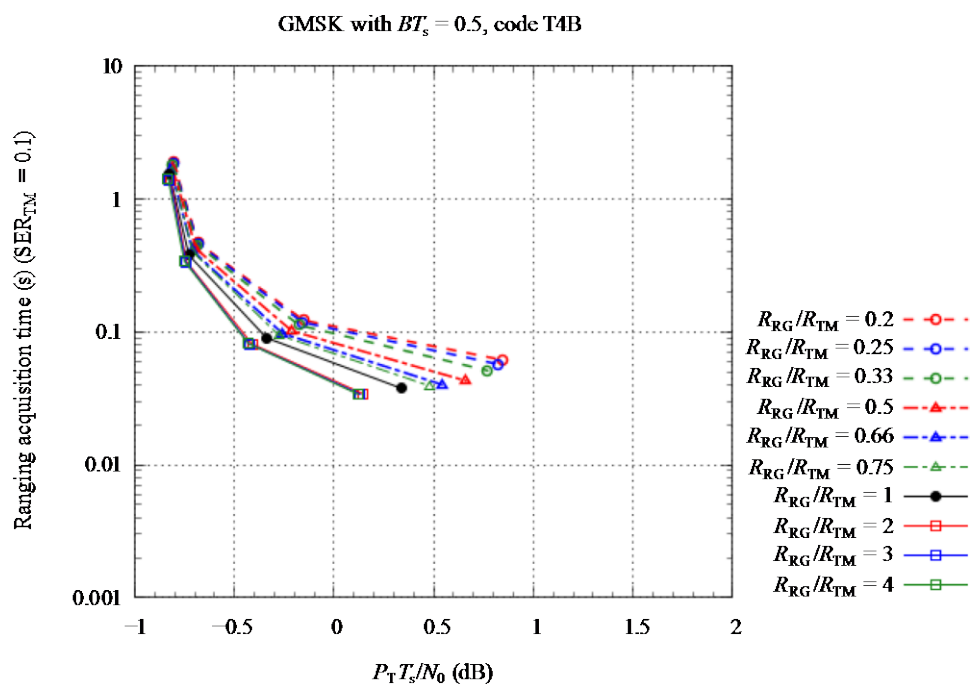


Figure A-13: Ranging Acquisition Time (s) versus $P_T T_s / N_0$ for GMSK ($BT_s=0.5$)/PN (T4B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, Various R_{RG} Values

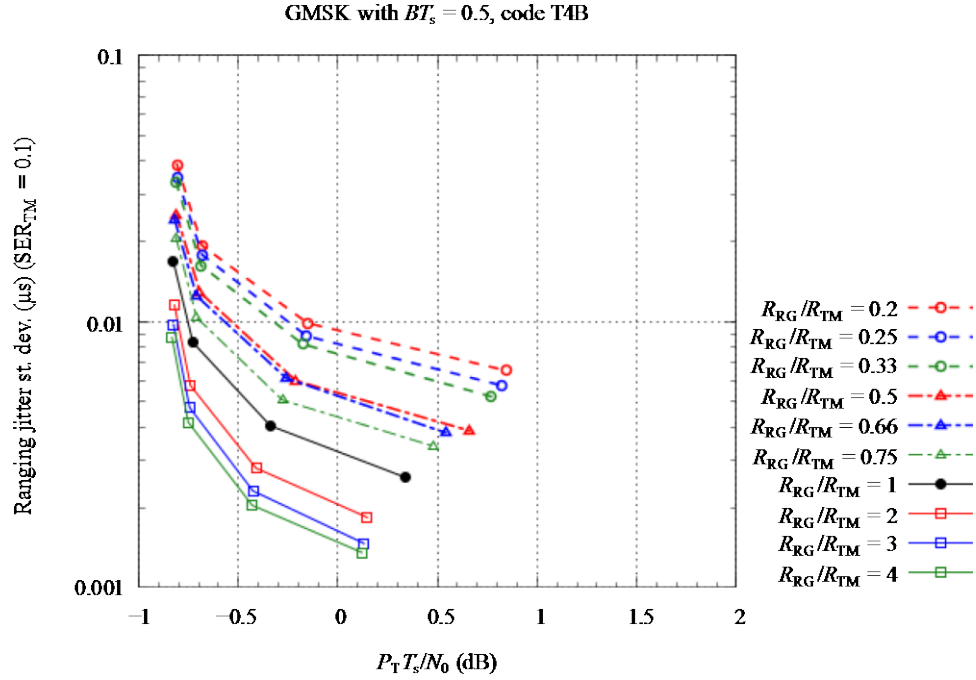


Figure A-14: Ranging Jitter Standard Deviation versus $P_T T_s/N_0$ for GMSK ($BT_s=0.5$)/PN (T4B) Sinusoidal Pulse, $R_{\text{TM}}=0.999$ Msymbol/s, Various R_{RG} Values, $B_{\text{LC}} T_c=10^{-5}$

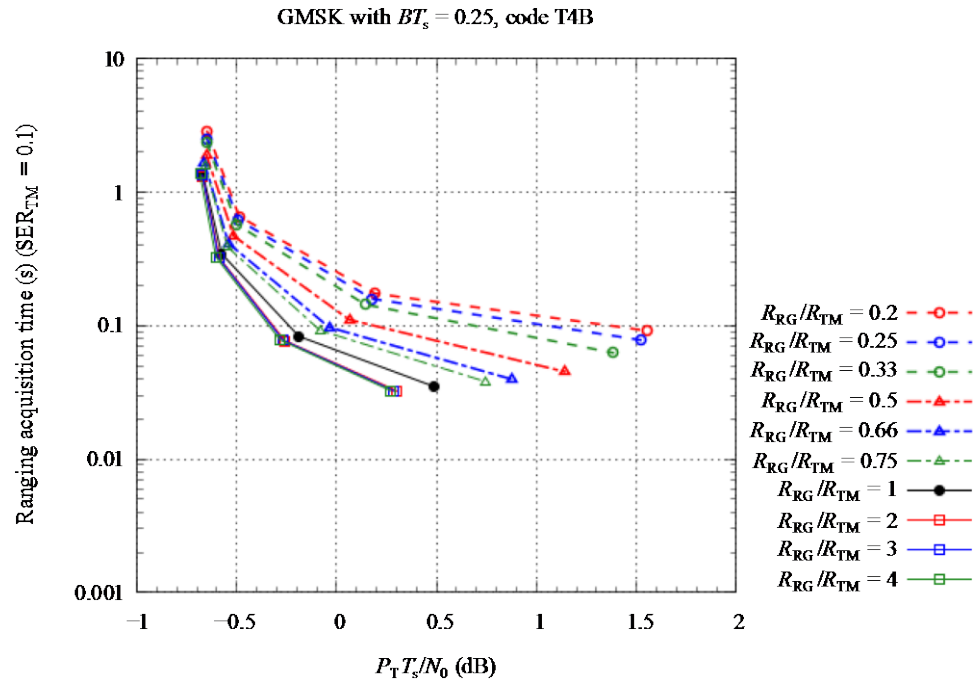


Figure A-15: Ranging Acquisition Time (s) versus $P_T T_s/N_0$ for GMSK ($BT_s=0.25$)/PN (T4B) Sinusoidal Pulse, $R_{\text{TM}}=0.999$ Msymbol/s, Various R_{RG} Values

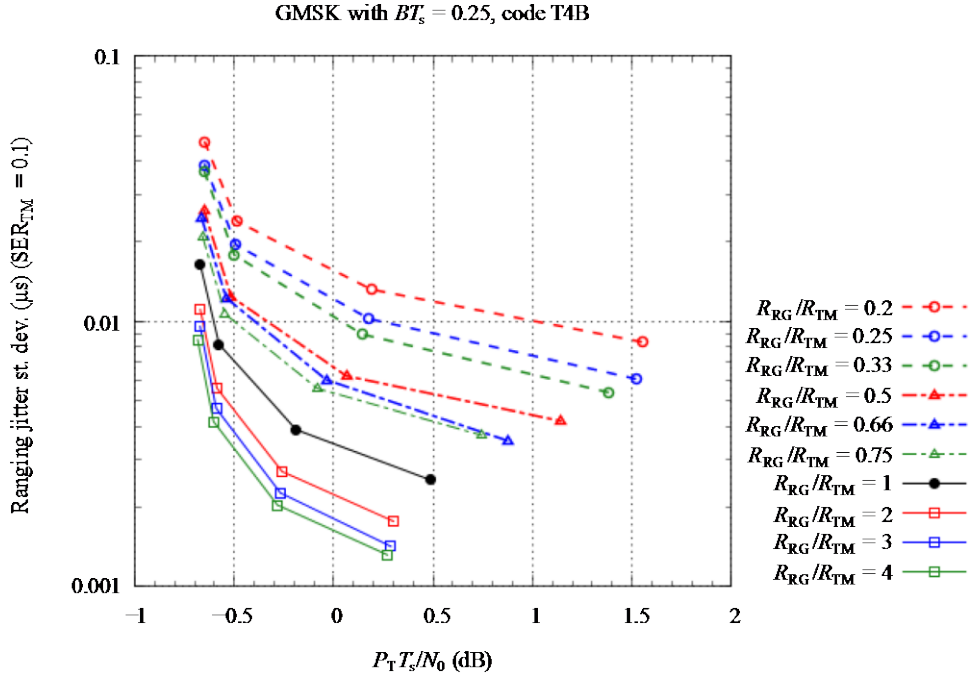


Figure A-16: Ranging Jitter Standard Deviation versus $P_T T_s / N_0$ for GMSK ($BT_s=0.25$)/PN (T4B) Sinusoidal Pulse, $R_{TM}=0.999$ Msymbol/s, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

The telemetry symbol rate R_{TM} and the ranging chip rate R_{RG} can take in principle any value. However, the channel models identified for this application in the 8 GHz space research service bands foresee telemetry transmission in the 1–10 Msymbol/s region. For such telemetry rates it is conceivable to transmit a chip rate equal to the telemetry symbol rate (at the low end) and smaller than the telemetry symbol rate (at the high end.) For missions with stringent ranging requirements, it may be possible for the chip rate to be higher than the telemetry symbol rate.

As far as the spectral mask and the occupied bandwidth value are concerned, the simulations carried out assume $R_{TM} = R_{RG}$ represent a worst case with respect to the cases where $R_{TM} > R_{RG}$. Obviously, if $R_{RG} > R_{TM}$, then the ranging signal drives the spectrum requirements.

The opposite result holds for what concerns the telemetry and ranging degradations. For a fixed value of R_{TM} , if R_{RG} increases, the orthogonality between telemetry and ranging components increases, and the reciprocal interference reduces. If, on the contrary, R_{RG} decreases below R_{TM} , then the probability that one telemetry symbol is wrong during one chip interval increases (being the telemetry SER constant is equal to 0.1), and the ranging subsystem performance deteriorates; at the same time, the telemetry signal is more affected by the ranging signal, especially for high values of the modulation index.

The next subsections specifically address the performance losses.

A3.2 CARRIER PERFORMANCE

The formula to evaluate the normalized standard deviation⁴ σ_ζ of the recovered carrier phase at the output of the phase synchronizer is given in equation (19); this is actually the formula valid for an unfiltered OQPSK modulation with a correction coefficient δ (see table A-5) to account for the different signal properties of the GMSK modulation (see reference [15]):

$$\sigma_\zeta = \delta \frac{1}{(2\pi)} \sqrt{\frac{B_L T_s (1 + 2P_T T_s / N_0)}{(P_T T_s / N_0)^2}}. \quad (19)$$

Table A-5: Parameter δ for the Evaluation of the Recovered Telemetry Carrier Phase Jitter Variance

BT_s	δ
0.5	0.88
0.25	1.15

Figures A-17 to A-20 show the normalized phase standard deviations versus the ratio R_{RG}/R_{TM} , with symbols identifying the measured values, continuous lines the best fit interpolating values, and dashed lines the theoretical values at the corresponding signal-to-noise values $P_T T_s / N_0$. The phase jitter losses were measured as described in figure A-21, and they are shown in figures A-22 to A-25. The conclusions are that the maximum phase jitter loss is 0.6 dB for m_{RG} in the range [0.222–0.444] and $R_{RG}/R_{TM} > 1$.

⁴ If σ is the phase standard deviation in radians, the normalized standard deviation is $\sigma_\zeta = \sigma / (2\pi)$

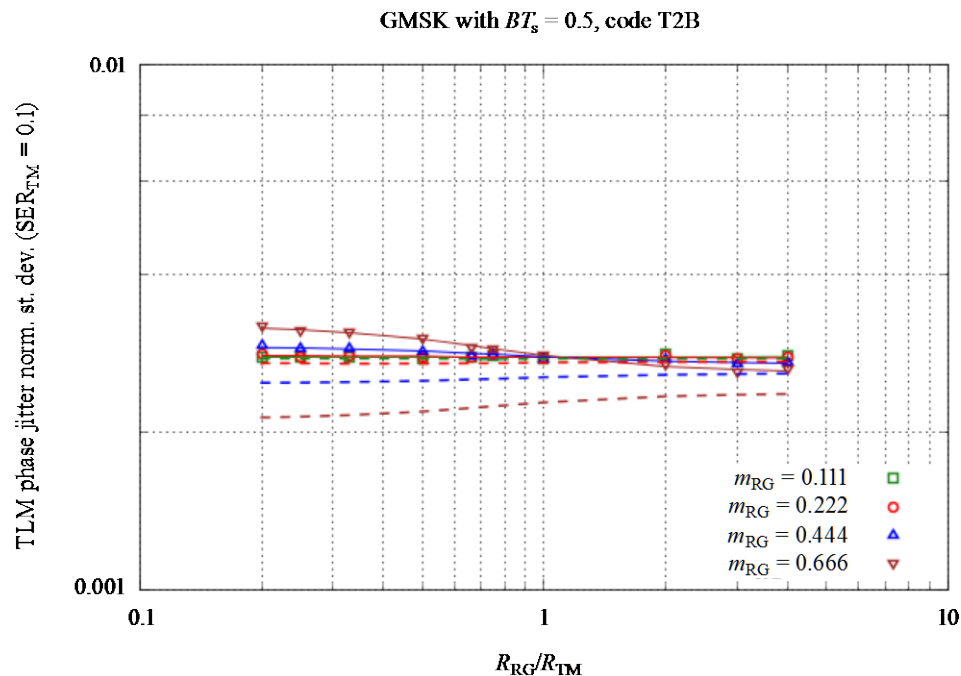


Figure A-17: Telemetry Carrier Phase Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

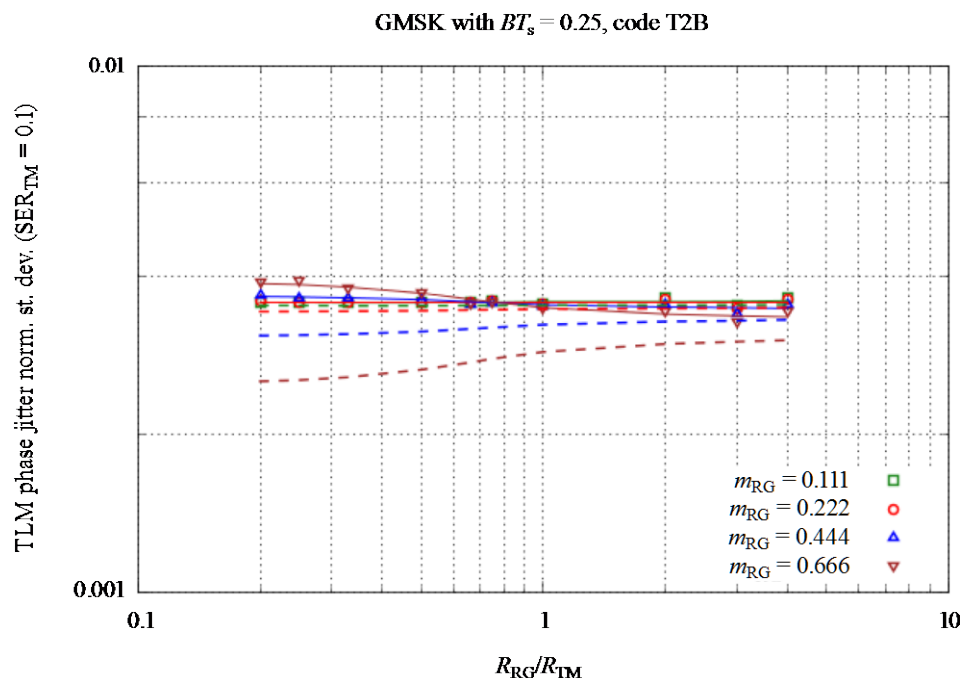


Figure A-18: Telemetry Carrier Phase Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

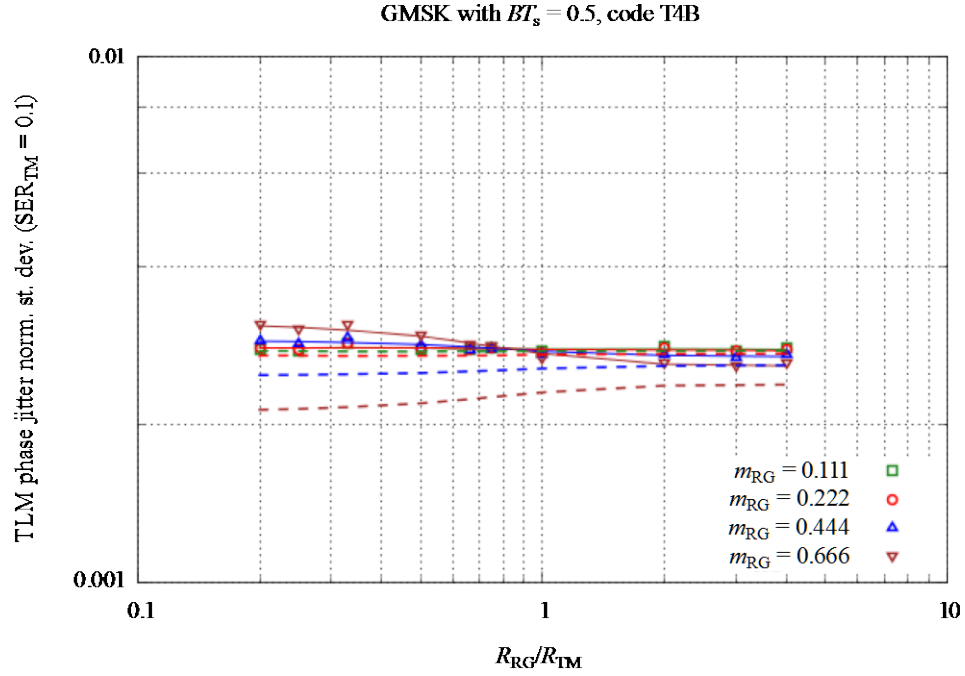


Figure A-19: Telemetry Carrier Phase Jitter Normalized Standard Deviation Versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

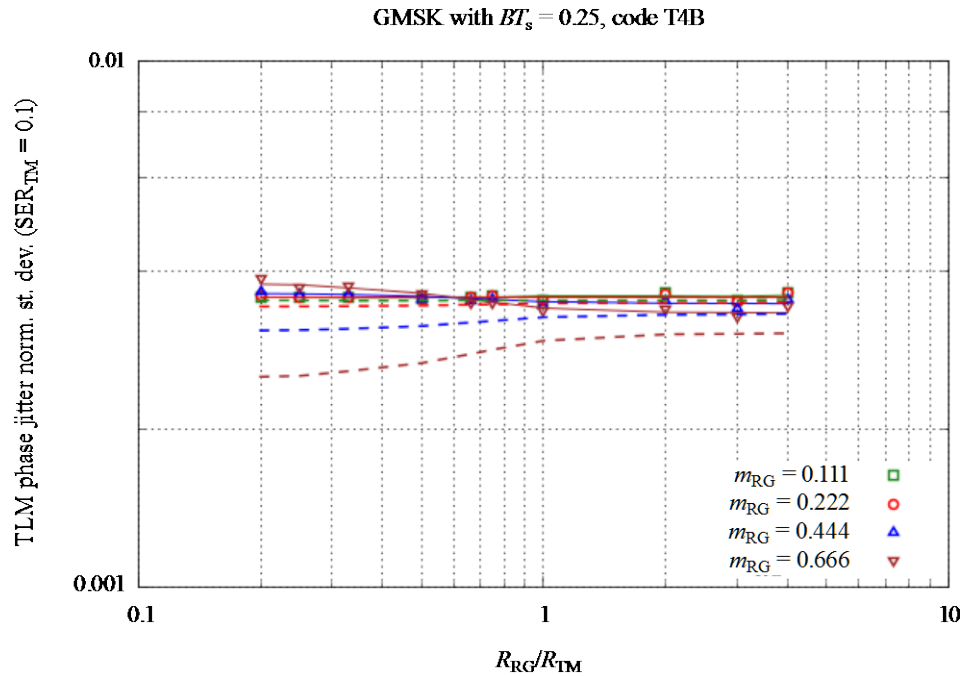


Figure A-20: Telemetry Carrier Phase Jitter Normalized Standard Deviation Versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

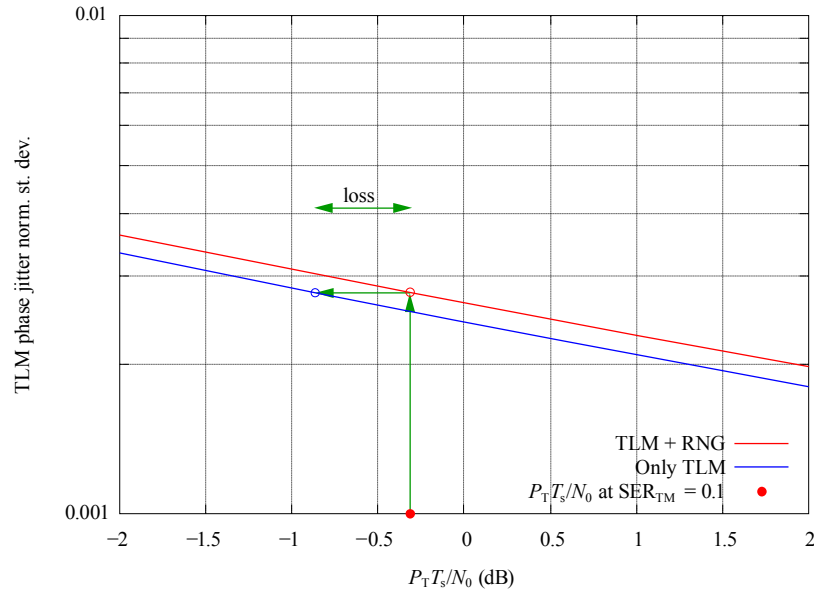


Figure A-21: Definition of Telemetry Carrier Phase Jitter Loss

NOTE – In figure A-21, the standard deviation of jitter $\sigma_{\zeta, \text{TML+RNG}}$ is measured for the complete system at the signal-to-noise ratio $(P_T T_s / N_0)_{\text{TLM+RNG}}$ (dB) at which $SER_{\text{TM}}=0.1$ (filled red dot in the picture); the signal-to-noise ratio $(P_T T_s / N_0)_{\text{TLM}}$ (dB) is found for which equation (19) gives the same standard deviation $\sigma_{\zeta, \text{TML+RNG}}$; the phase jitter loss is defined as $(P_T T_s / N_0)_{\text{TLM+RNG}} - (P_T T_s / N_0)_{\text{TLM}}$.

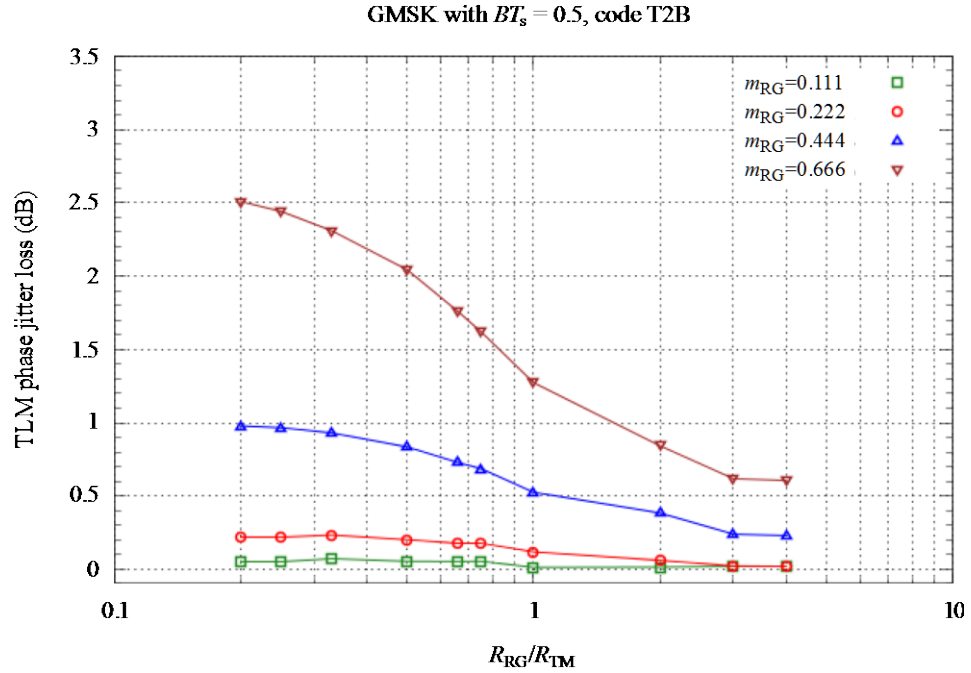


Figure A-22: Telemetry Carrier Phase Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

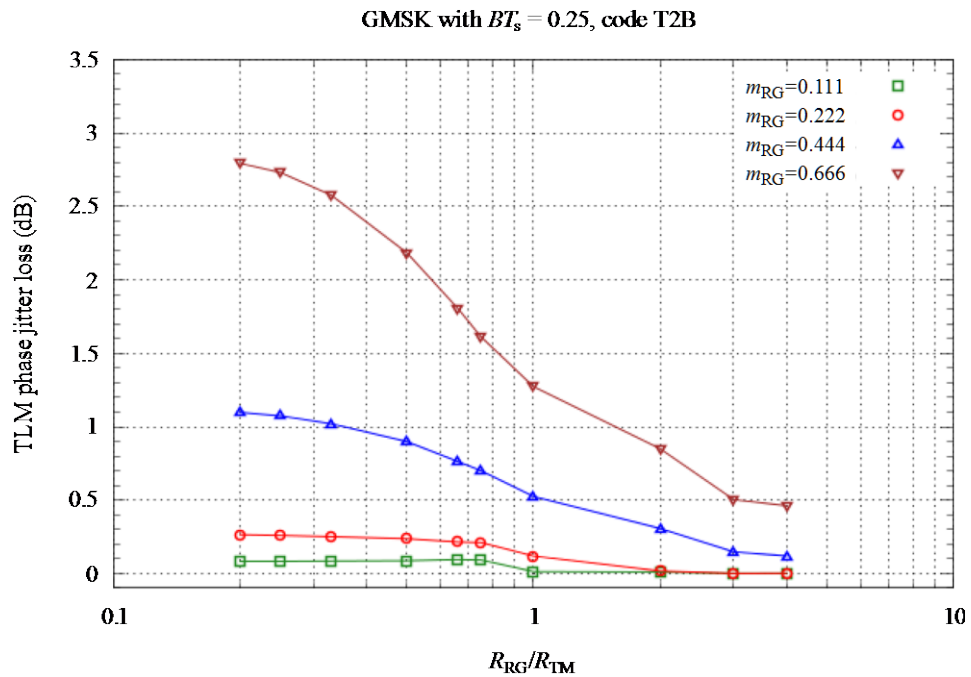


Figure A-23: Telemetry Carrier Phase Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

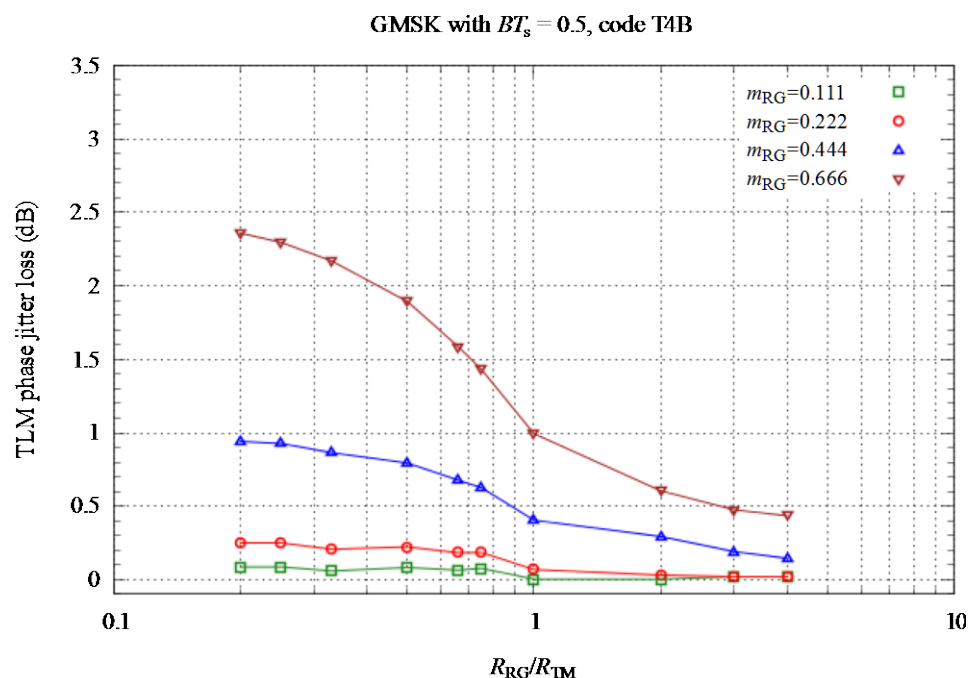


Figure A-24: Telemetry Carrier Phase Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

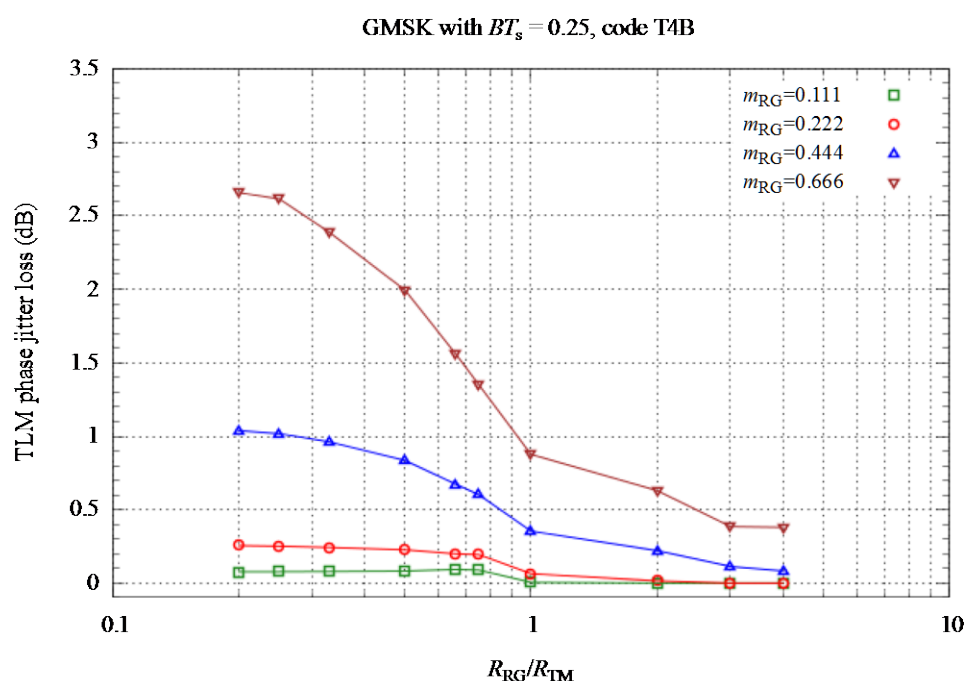


Figure A-25: Telemetry Carrier Phase Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

A3.3 TELEMETRY PERFORMANCE

A3.3.1 General

For the general unsynchronized case (defined by $R_{TM}=0.999$ instead of $R_{TM}=1$) and for various chip-rate/symbol-rate ratios, the telemetry losses versus m_{RG} due to sinewave-shaped ranging are reported in figures A-26 to A-29. In this case, to ease the overall system design, the telemetry loss is defined as the signal-to-noise ratio $P_T T_s / N_0$ (dB) necessary to obtain $SER_{TM}=0.1$ in the system with both telemetry and ranging minus the signal-to-noise ratio $P_T T_s / N_0 = -0.86$ dB necessary to obtain $SER_{TM}=0.1$, assuming an ideal BPSK modulation. Therefore the loss shown from figures A-26 to A-29 is the total loss, which includes a) the loss of the GMSK modulation with respect to BPSK, b) the loss due to the fact that part of the transmitted power P_T is actually used by ranging, and c) the loss due to the ranging interference. These losses can be compared with those listed in table A-3 with a little effort. As an example, consider the case of GMSK with $BT_s=0.25$, code T4B, $h_{sin}(t)$, $m_{RG}=0.444$: table A-3 gives a loss equal to 0.44 dB, but this loss does not include the GMSK loss, which, for GMSK with $BT_s=0.25$, is equal to $0.86-0.698=0.162$ dB, as given in table A-3; the overall loss with respect to BPSK is then $0.44+0.162=0.602$ dB. Figure A-29 shows a telemetry loss equal to 0.67 dB, which is in good agreement with the loss of table A-3, obtained with the simplified case of ideal synchronization (0.08 dB loss due to synchronization errors).

It can be seen from figures A-26 to A-29 that for $BT_s=0.5$ or 0.25 , PN code T2B or T4B, the loss is less than 0.4 dB for $m_{RG}<0.222$ for any of the considered ranging chip rates; but it is evident that values of R_{RG} smaller than 1 correspond to larger telemetry losses. For ratios $R_{RG}/R_{TM} \geq 1$ the loss is lower than 0.7 dB for $m_{RG} \leq 0.444$ and lower than 1.5 dB for the much higher value $m_{RG}=0.666$. The figures also show that in the cases $R_{RG}/R_{TM} > 1$ the telemetry loss is almost independent of the ratio R_{RG}/R_{TM} , especially for code T4B, while in the cases $R_{RG}/R_{TM} < 1$ the loss significantly increases as R_{RG} decreases. GMSK with $BT_s=0.25$ appears to be less robust to the ranging interference, but only for ratios $R_{RG}/R_{TM} < 1$, while in the other cases its loss is substantially equal to that of GMSK with $BT_s=0.5$ plus the inherent modulation loss of 0.15 dB resulting from the increased GMSK intersymbol interference.

It can be concluded that to prevent significant telemetry losses, the ratio R_{RG}/R_{TM} should be greater than or equal to 1 and the ranging peak modulation index m_{RG} should be kept in the range $[0.222, 0.444]$.

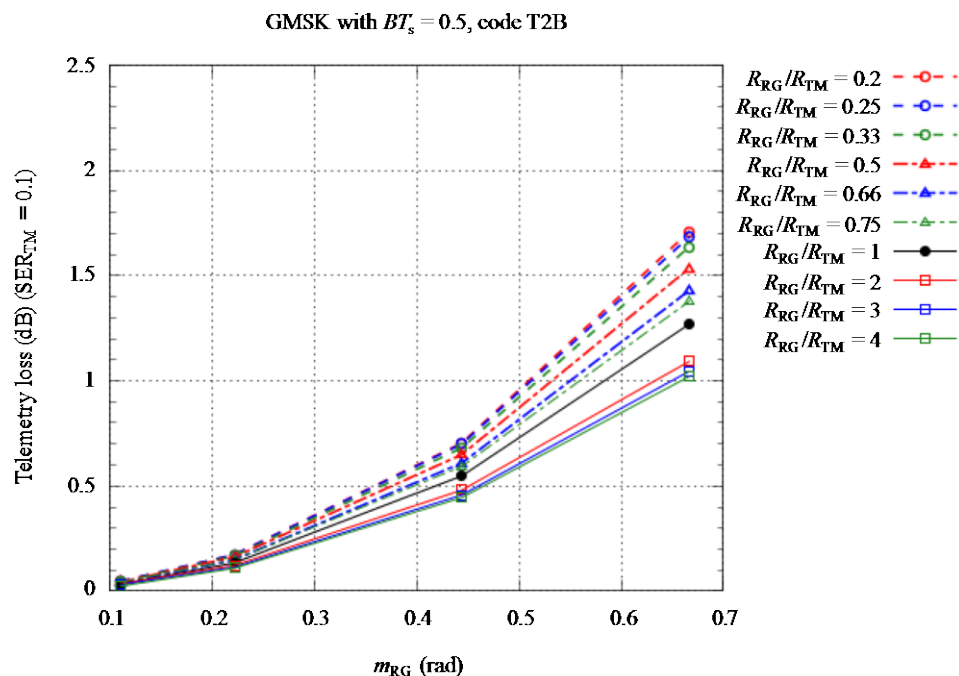


Figure A-26: Telemetry Degradation versus m_{RG} for GMSK ($BT_s=0.5$)/PN(T2B) Sinusoidal Pulse, Various R_{RG} Values

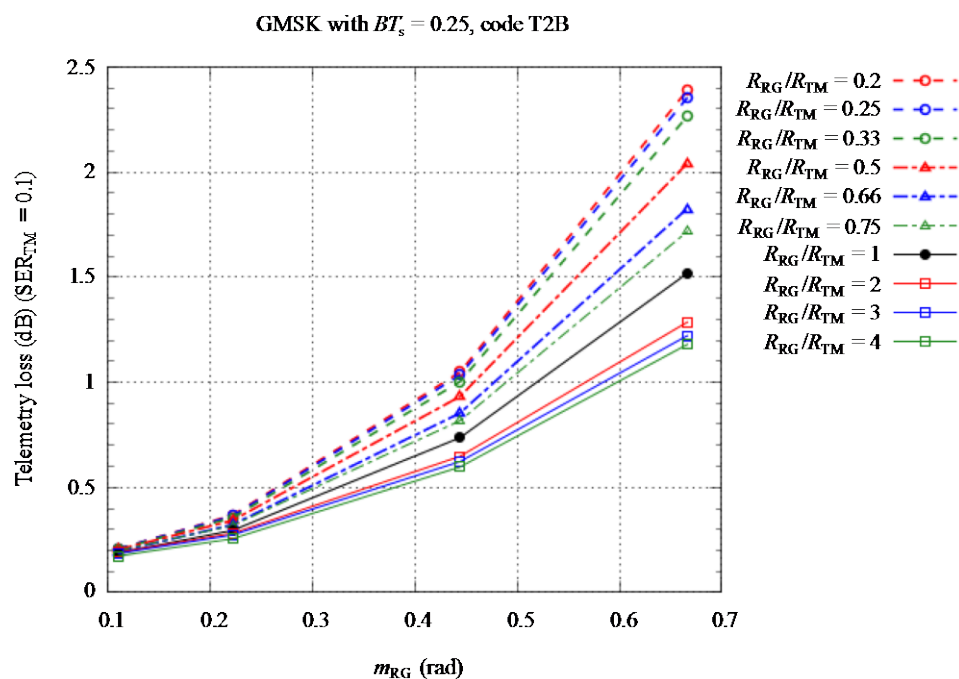


Figure A-27: Telemetry Degradation versus m_{RG} for GMSK ($BT_s=0.25$)/PN(T2B) Sinusoidal Pulse, Various R_{RG} Values

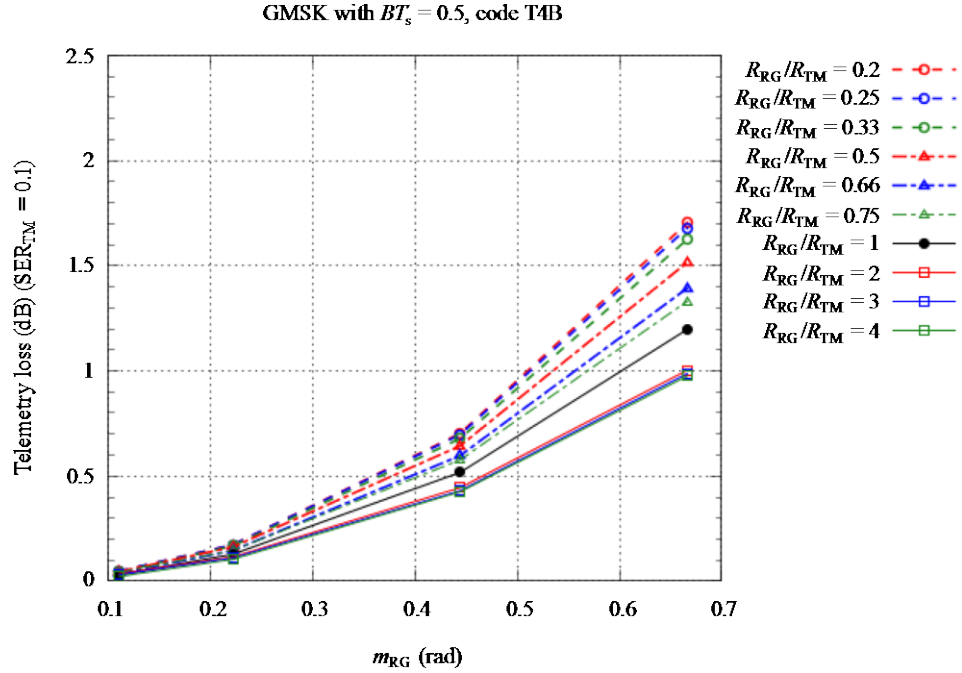


Figure A-28: Telemetry Degradation versus m_{RG} for GMSK ($BT_s=0.5$)/PN(T4B) Sinusoidal Pulse, Various R_{RG} Values

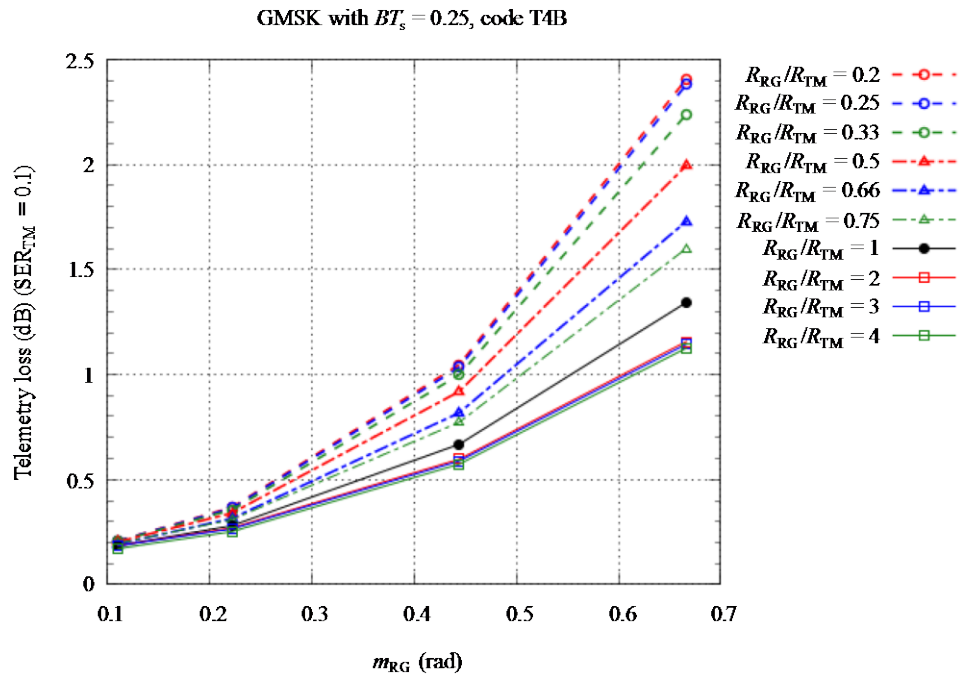


Figure A-29: Telemetry Degradation versus m_{RG} for GMSK ($BT_s=0.25$)/PN(T4B) Sinusoidal Pulse, Various R_{RG} Values

A3.3.2 Telemetry Clock Synchronization

The degradation figures given in A3.3 are the total telemetry losses to be used in link budgets. The losses described in the current subsection only encompass the telemetry clock synchronizer and as such are already included in the total telemetry loss. The purpose of this subsection is only to illustrate the ranging effect on the telemetry clock synchronization.

The theoretical analysis of the GMSK clock synchronizer (reference [7]) provides the formula given in equation (20) for the recovered clock jitter normalized standard deviation:

$$\sigma_\varepsilon = \sqrt{4B_L T_s \left[\gamma_1 + \gamma_2 \frac{N_0}{2P_T T_s} + \gamma_3 \left(\frac{N_0}{2P_T T_s} \right)^2 \right]}, \quad (20)$$

where the coefficients γ_1 , γ_2 , γ_3 (see reference [14]) are given in table A-6. Parameter B_L is the noise equivalent bandwidth of the phase synchronizer loop.

Table A-6: Parameters γ_1 , γ_2 , γ_3 for the Evaluation of the Recovered Telemetry Clock Jitter Standard Deviation

BT_s	γ_1	γ_2	γ_3
0.5	$1.420 \cdot 10^{-5}$	$7.351 \cdot 10^{-2}$	$7.718 \cdot 10^{-2}$
0.25	$2.824 \cdot 10^{-4}$	$1.187 \cdot 10^{-1}$	$1.470 \cdot 10^{-1}$

Because of the presence of the ranging component, the measured clock jitter standard deviation is higher than the theoretical one, for which an ideal AWGN channel is considered. Figures A-30 to A-33 show the clock normalized jitter standard deviation with symbols identifying the measured values and dashed lines for the theoretical values at the same signal-to-noise ratio $P_T T_s / N_0$; continuous lines in the figures correspond to values obtained using a best-fit first-order polynomial (best fitting in the plane $\log_{10}(\sigma_\varepsilon)$ versus $\log_{10}(P_T T_s / N_0)$).

As a general trend, it can be noticed that for $R_{RG}/R_{TM} > 1$ the measured standard deviation is almost independent of m_{RG} , but, because of the telemetry loss, the standard deviation should actually be lower, and a non-negligible loss thus exists.

When $R_{RG}/R_{TM} < 1$, the standard deviation increases as R_{RG}/R_{TM} decreases, and the increase is more evident for $m_{RG} = 0.444$ and 0.666 .

It is possible to measure the loss due to the presence of ranging as the difference between the value of $P_T T_s / N_0$ that provides $SER_{TM} = 0.1$ in the simulations with a given measured value of σ_ε and the value of $P_T T_s / N_0$ that, according to equation (20), gives the same value of σ_ε ; this clock jitter loss (dB) is shown in figures A-34 to A-37 for the considered cases.

Actually, in order to smooth out the measurement errors, the comparison is performed using the standard deviations obtained from the best-fit procedure. Part of the measured loss is due to the fact that a portion of the total power P_T is used for the ranging signal. Code T2B gives a higher loss than code T4B, whereas the difference between $BT_s=0.5$ and $BT_s=0.25$ is negligible; for m_{RG} in the range $[0.222-0.444]$ and $R_{RG}/R_{TM}>1$ the maximum clock jitter loss is 0.6 dB.

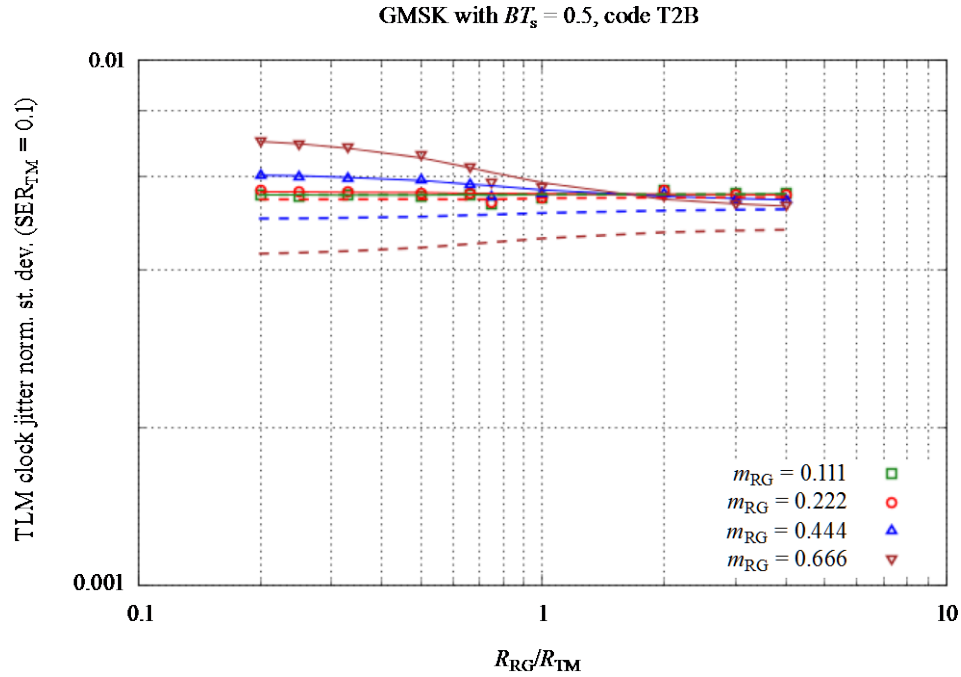


Figure A-30: Telemetry Clock Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

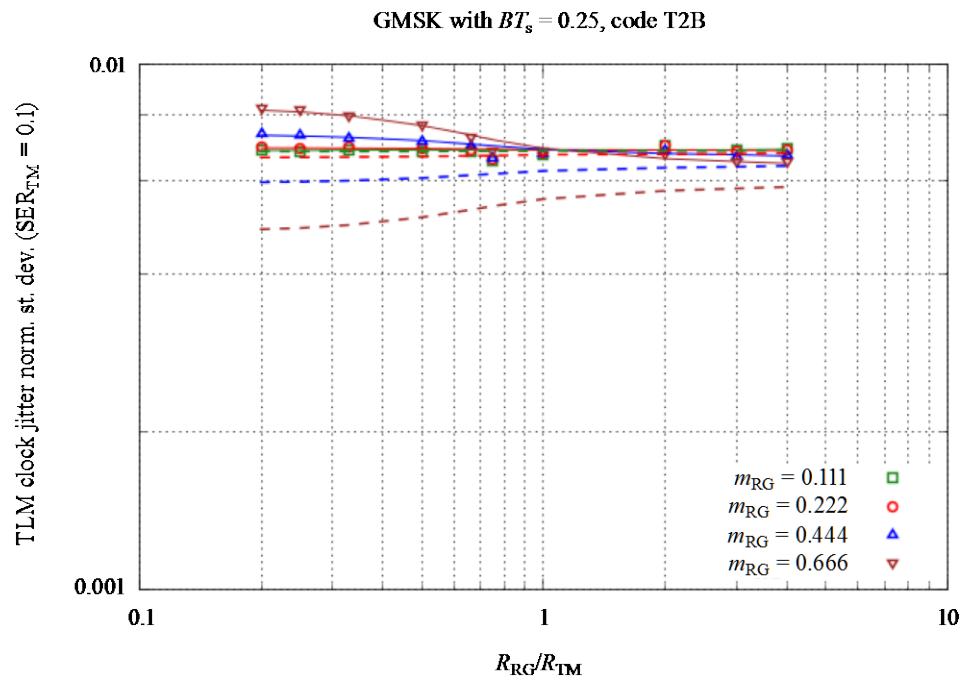


Figure A-31: Telemetry Clock Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

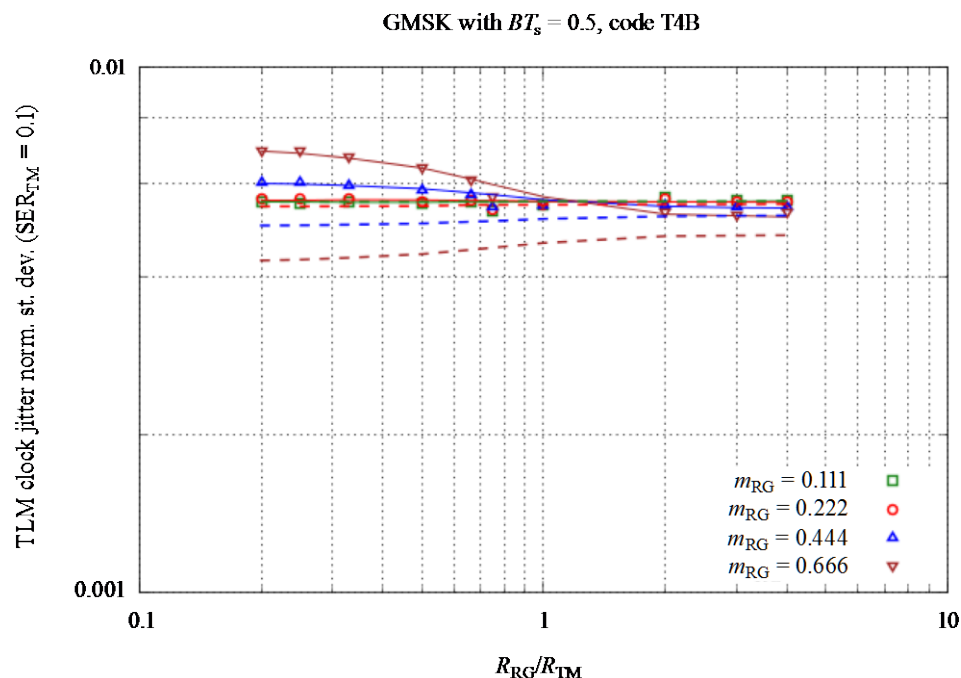


Figure A-32: Telemetry Clock Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

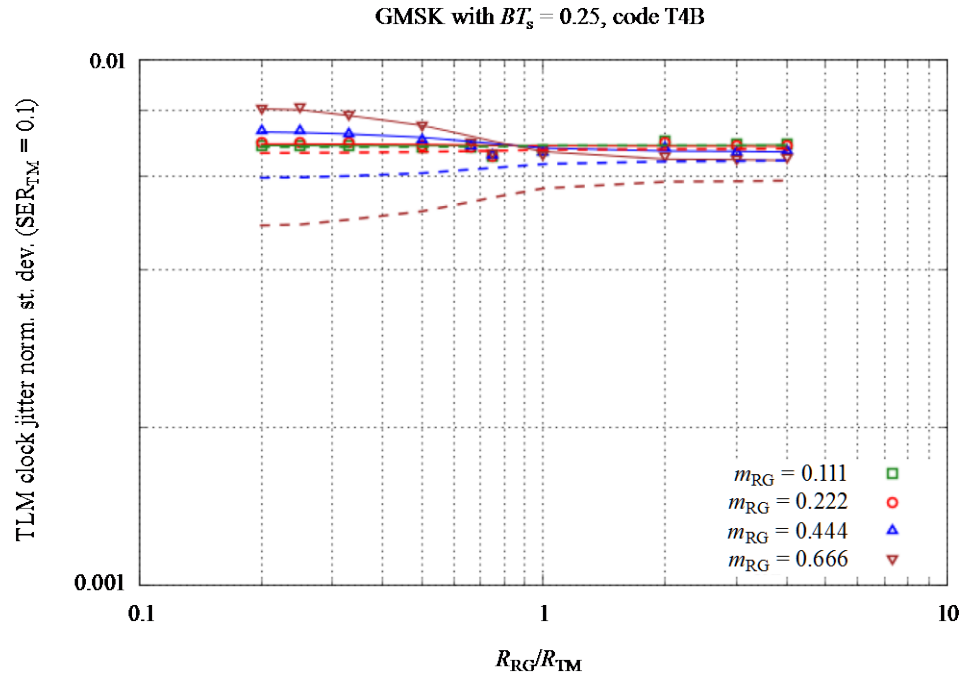


Figure A-33: Telemetry Clock Jitter Normalized Standard Deviation versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

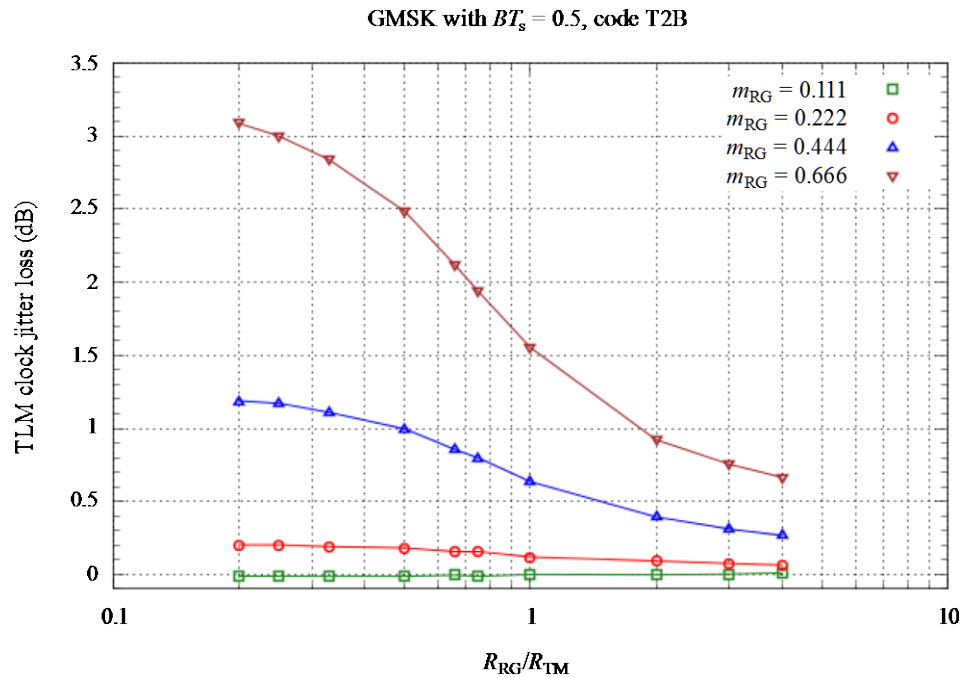


Figure A-34: Telemetry Clock Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

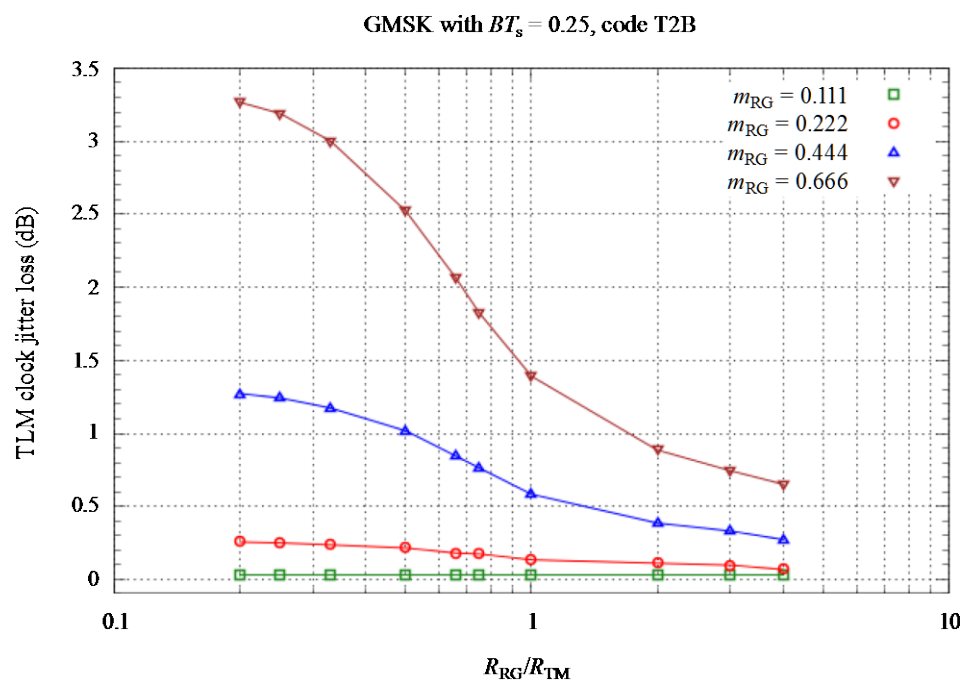


Figure A-35: Telemetry Clock Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T2B, Various m_{RG} Values, $B_L T_s=10^{-4}$

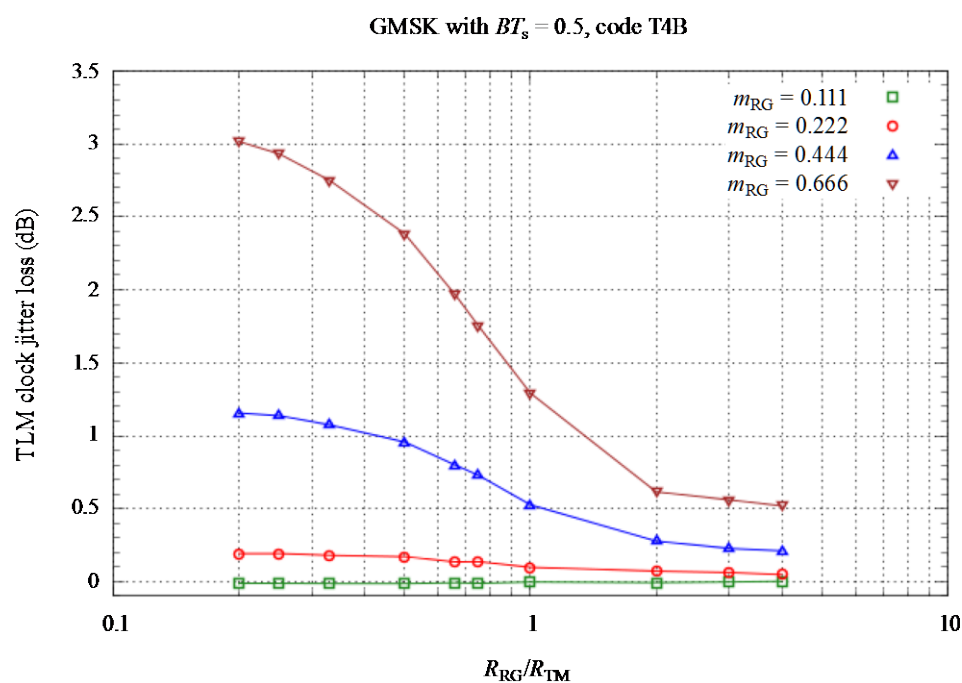


Figure A-36: Telemetry Clock Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.5$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

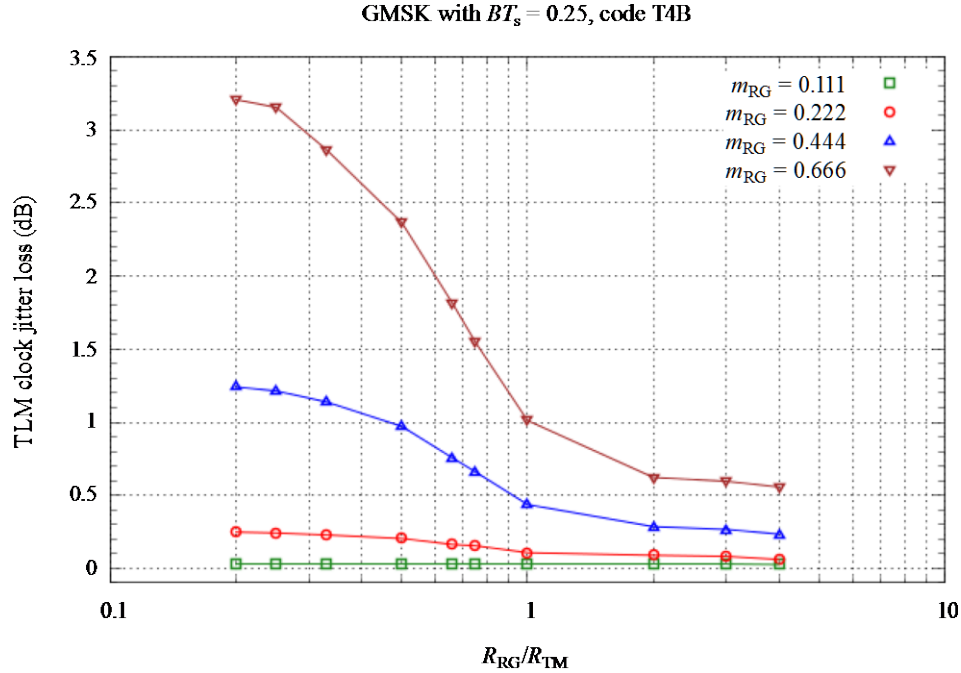


Figure A-37: Telemetry Clock Jitter Loss versus R_{RG}/R_{TM} for GMSK ($BT_s=0.25$), $R_{TM}=0.999$, Code T4B, Various m_{RG} Values, $B_L T_s=10^{-4}$

A3.4 RANGING PERFORMANCE

A3.4.1 Acquisition Time

For the general unsynchronized case (defined by $R_{TM}=0.999$ instead of $R_{TM}=1$) and for various chip-rate/symbol-rate ratios, the sinewave ranging acquisition losses versus m_{RG} due to telemetry can be seen in figures A-38 to A-41. They can be compared with table A-4, where the losses for the synchronized case with $R_{TM} = R_{RG}$ are given.

The figures give the difference in dB between the value of the acquisition time estimated through the simulations (taking into consideration the telemetry interference and the imperfect synchronization) and the theoretical value assuming no telemetry interference (see 3.4.4.2 for the formulas). The general trend already observed is again confirmed: values of R_{RG}/R_{TM} smaller than 1 give a larger ranging loss than values larger than 1, the loss is practically independent of m_{RG} for R_{RG}/R_{TM} larger than 1, being the loss equal to 2 dB for GMSK with $BT_s=0.5$ and code T4B and GMSK with $BT_s=0.25$ and code T2B, 2.2 dB for GMSK with $BT_s=0.5$ and code T2B, and 1.9 dB for GMSK with $BT_s=0.25$ and code T4B. It should be noted that the case $R_{RG} \approx R_{TM}$ has a larger loss than the cases R_{RG}/R_{TM} larger than 1 by approximately 0.3–0.4 dB.

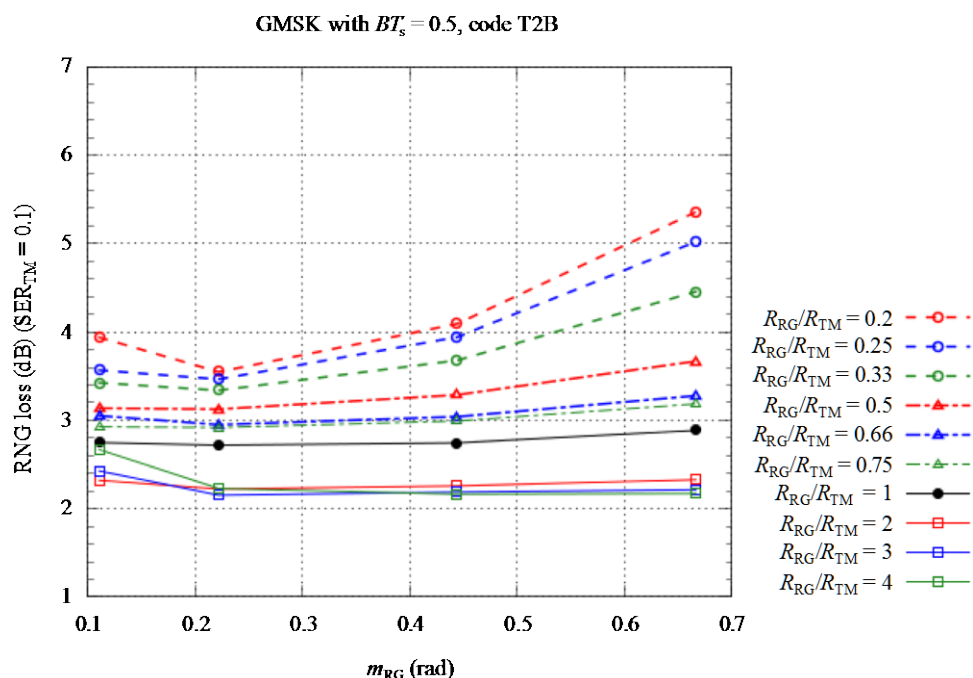


Figure A-38: Ranging (T2B) Acquisition Degradation Due to GMSK($BT_s=0.5$) versus m_{RG} , $R_{TM}=0.999$, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

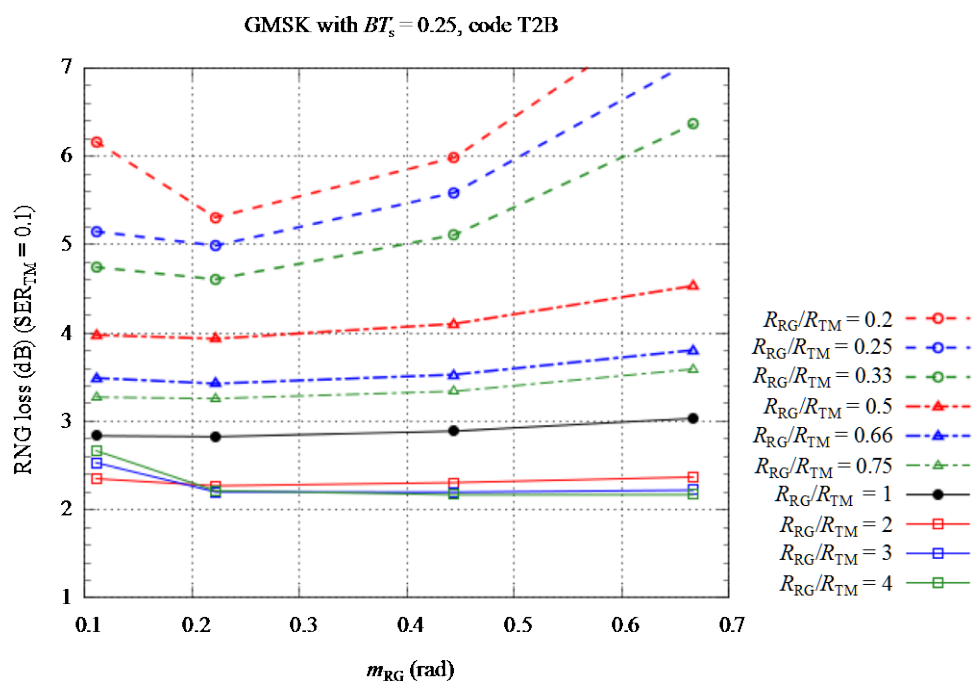


Figure A-39: Ranging (T2B) Acquisition Degradation Due to GMSK($BT_s=0.25$) versus m_{RG} , $R_{TM}=0.999$, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

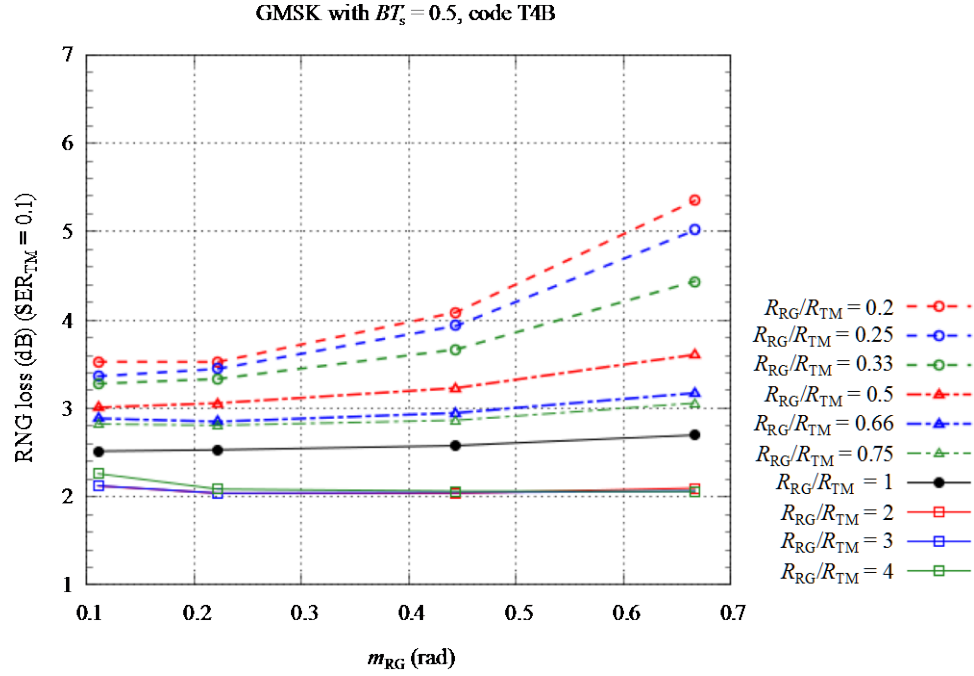


Figure A-40: Ranging (T4B) Acquisition Degradation Due to GMSK($BT_s=0.5$) versus m_{RG} , $R_{TM}=0.999$, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

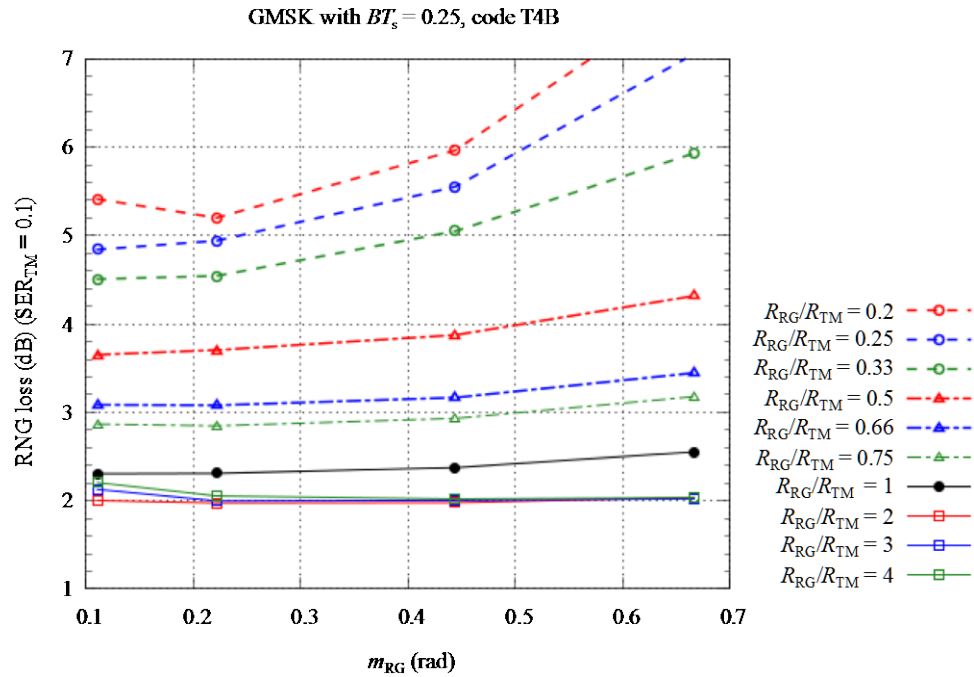


Figure A-41: Ranging (T4B) Acquisition Degradation Due to GMSK($BT_s=0.25$) versus m_{RG} , $R_{TM}=0.999$, Various R_{RG} Values, $B_{Lc} T_c=10^{-5}$

A3.4.2 Jitter

The actual ranging jitter variance taking into account the telemetry interference to the ranging signal is larger than its theoretical value. The ranging jitter standard deviation and the ratio (in dB) between the measured jitter variance and the theoretical one (see 3.4.4.3) are plotted from figures A-42 to A-49 for the selected ranging sinewave shaping. In particular, the filled squares in the figures showing the jitter standard deviation represent the simulation measurements, the dashed lines are the best fit first order polynomial approximation of the measurements, and the solid lines correspond to the theoretical performance (see equation (16)); the ranging jitter loss is obtained as the ratio in dB between the theoretical variance and the approximated variance. The best fit approximated variance has been used to remove the large variations due to measurement errors that otherwise would appear.

It should be noted that the jitter loss can be seen as an equivalent loss in the ranging clock energy-to-noise ratio $E_{cl,sin}/N_0$. The curves show that the jitter loss decreases as R_{RG}/R_{TM} increases and it is less than 1.5 dB for $R_{RG}/R_{TM} > 1$.

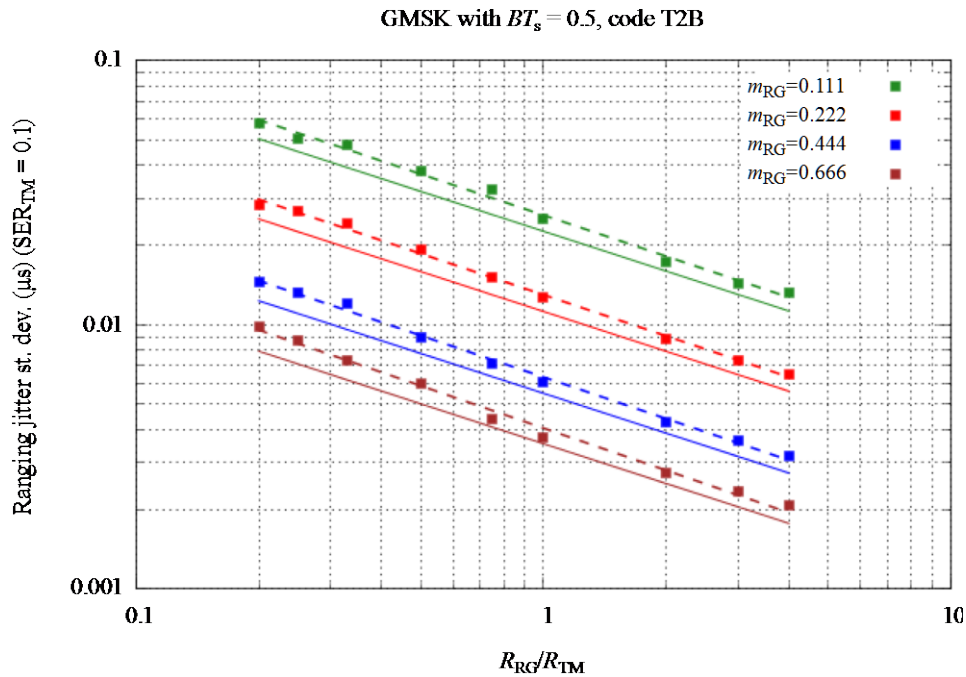


Figure A-42: Ranging (T2B) Jitter Standard Deviation versus R_{RG}/R_{TM} for GMSK($BT_s=0.5$), $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

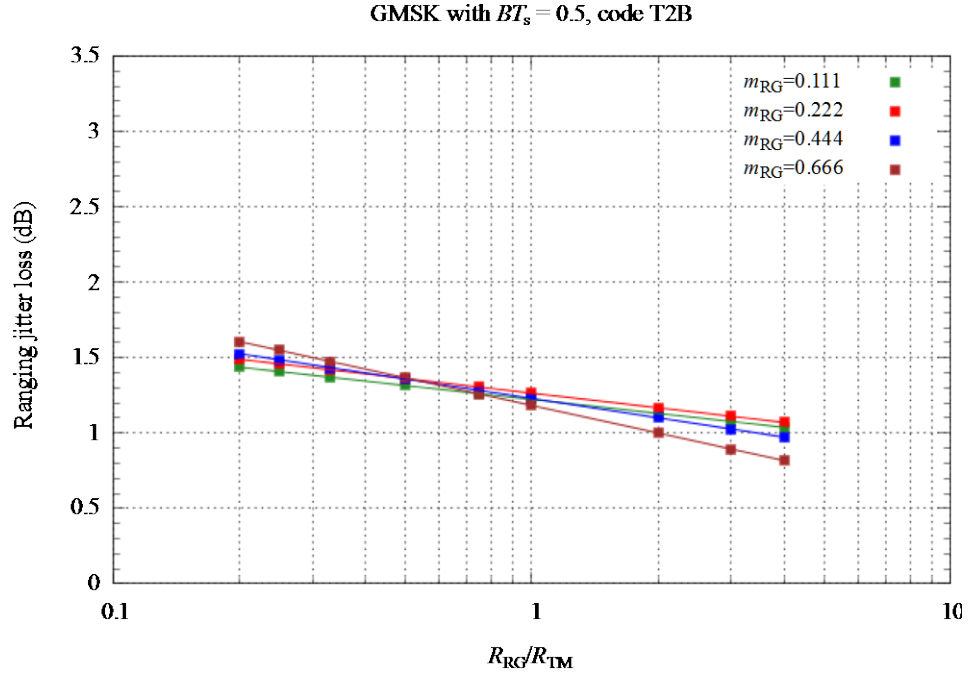


Figure A-43: Ranging (T2B) Jitter Degradation Due to GMSK($BT_s=0.5$) versus R_{RG}/R_{TM} , $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

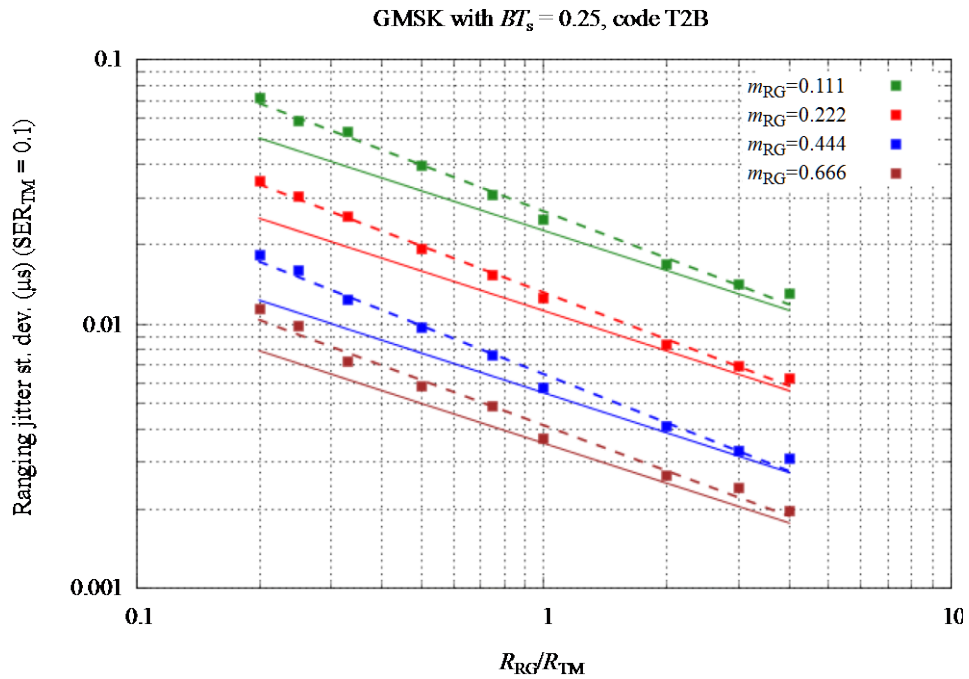


Figure A-44: Ranging (T2B) Jitter Standard Deviation versus R_{RG}/R_{TM} for GMSK($BT_s=0.25$), $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

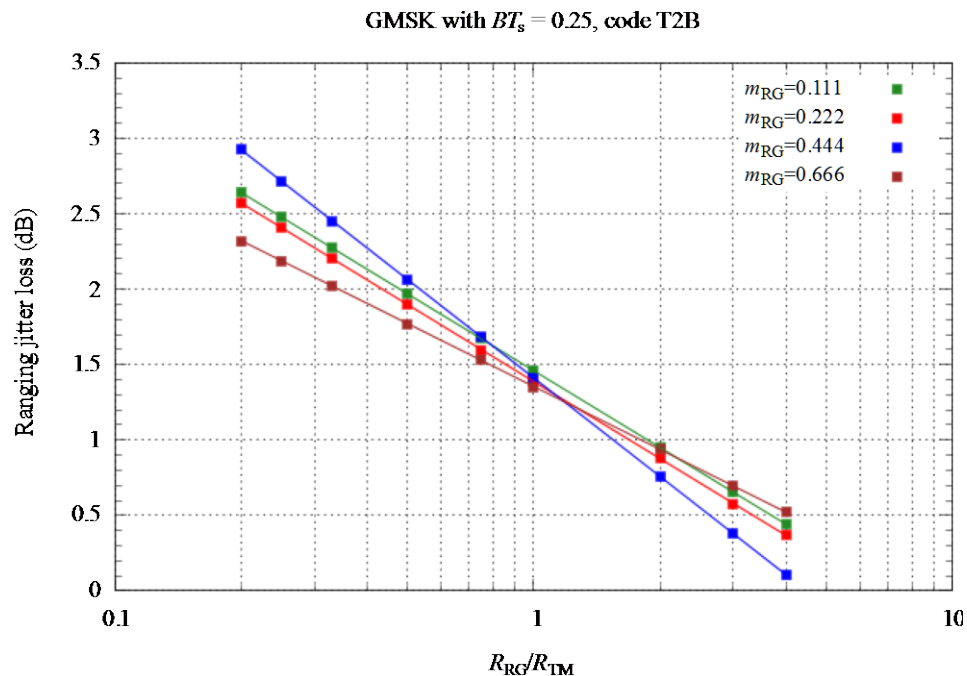


Figure A-45: Ranging (T2B) Jitter Degradation Due to GMSK($BT_s=0.25$) versus R_{RG}/R_{TM} , $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

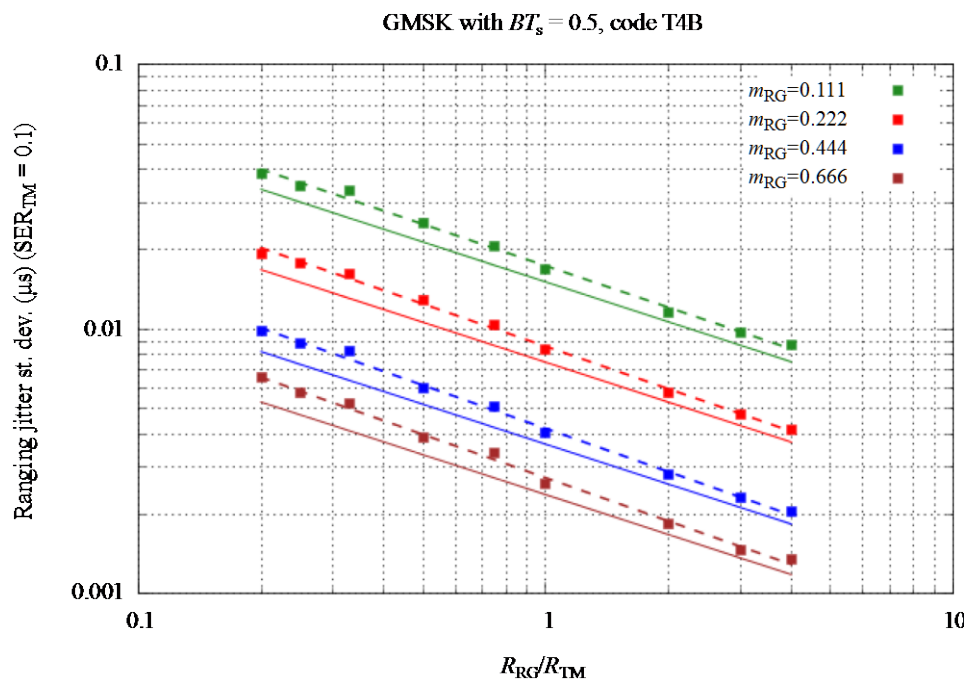


Figure A-46: Ranging (T4B) Jitter Standard Deviation versus R_{RG}/R_{TM} for GMSK($BT_s=0.5$), $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

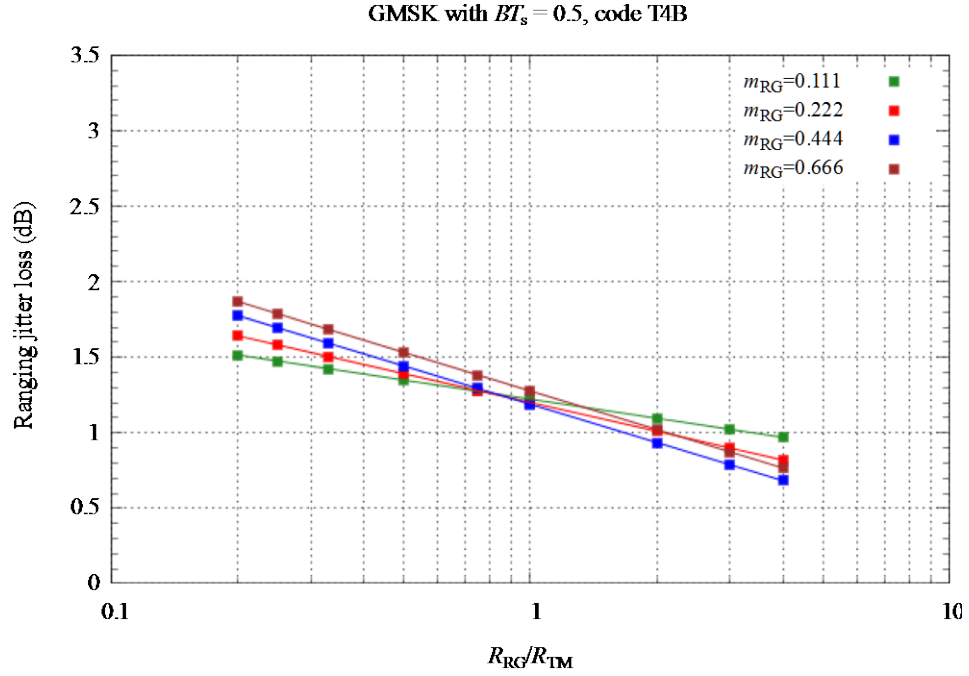


Figure A-47: Ranging (T4B) Jitter Degradation Due to GMSK($BT_s=0.5$) versus R_{RG}/R_{TM} , $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

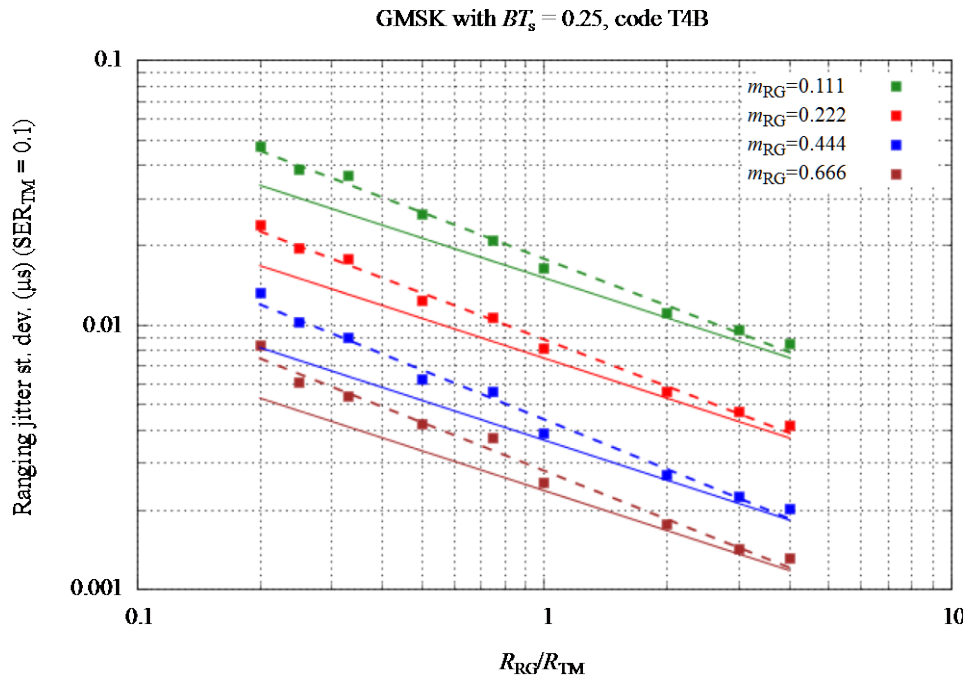


Figure A-48: Ranging (T4B) Jitter Standard Deviation versus R_{RG}/R_{TM} for GMSK($BT_s=0.25$), $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

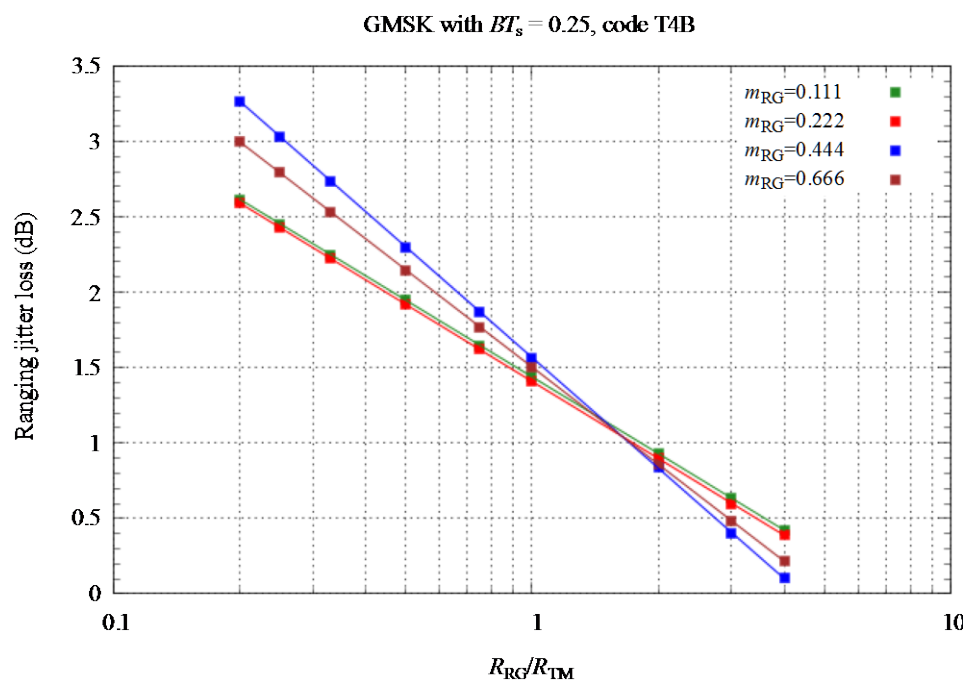


Figure A-49: Ranging (T4B) Jitter Degradation Due to GMSK($BT_s=0.25$) versus R_{RG}/R_{TM} , $R_{TM}=0.999$, Various m_{RG} Values, $B_{Lc} T_c=10^{-5}$

A4 PHASE ARRAY JUMPS AND NCO CONSIDERATIONS

The effects of phase jumps and phase noise have been studied and documented in reference [8], where it can be seen that under the assumed conditions (phase jumps less than 15 degrees with several time constants, and phase noise standard deviation up to 15 degrees) GMSK+PN ranging does not behave differently from conventional modulation.

ANNEX B**ABBREVIATIONS AND ACRONYMS**

<u>Term</u>	<u>Meaning</u>
AWGN	additive white Gaussian noise
BER	bit error rate
BPSK	binary phase-shift keying
CER	chip error rate
GMSK	Gaussian minimum shift keying
ITU	International Telecommunication Union
NCO	numerically controlled oscillator
OQPSK	offset quadrature-phase shift keying
PN	pseudo-random noise
RNG, RG	ranging
SER	symbol error rate
SFCG	Space Frequency Coordination Group
sin	sinewave
sq	squarewave
SRRC	square root raised cosine
SRS	Space Research Service
TLM, TM	telemetry